

Bit	Attr	Reset Value	Description
7	RW	0x0	WRCEIEN AXI write channel error interrupt enable. 1'b0: Disable 1'b1: Enable
6	RW	0x0	DISEIEN INFLATE huffman distance error interrupt enable. 1'b0: Disable 1'b1: Enable
5	RW	0x0	LENEIEN INFLATE huffman length error interrupt enable. 1'b0: Disable 1'b1: Enable
4	RW	0x0	LITEIEN INFLATE huffman literal error interrupt enable. 1'b0: Disable 1'b1: Enable
3	RW	0x0	SQMEIEN INFLATE SQ match error interrupt enable. 1'b0: Disable 1'b1: Enable
2	RW	0x0	SLCIEN INFLATE store block length check error interrupt enable. 1'b0: Disable 1'b1: Enable
1	RW	0x0	HDEIEN INFLATE file header error interrupt enable. 1'b0: Disable 1'b1: Enable
0	RW	0x0	DSIEN DECOM stop interrupt enable. 1'b0: Disable 1'b1: Enable

DECOM AXI STAT

Address: Operational Base + offset (0x0030)

Bit	Attr	Reset Value	Description
31:5	RO	0x0000000	reserved
4	RO	0x1	AXI_IDLE AXI master idle state register. 1'b0: AXI master is busy 1'b1: AXI master is idle When the decompression is aborted, DECOM will wait for AXI master's current read/write transfer to complete, that is, wait for AXI_IDLE=1, then reset DECOM, otherwise the uncompleted AXI transmission will cause AXI bus exception. It should be noted that after the decompression is abnormal, the AXI master's unfinished write operation will continue, but the output value of wstrb[15:0] will become 16'b0.
3:2	RO	0x0	RRESP AXI read channel response state.
1:0	RO	0x0	BRESP AXI write channel response state.

DECOM TSIZEL

Address: Operational Base + offset (0x0034)

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TSIZEL The total size of the data after decompression. TSIZE[63:0]. (Unit:byte)

DECOM_TSIZEH

Address: Operational Base + offset (0x0038)

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	TSIZEH The total size of the data after decompression.

DECOM_MGNUM

Address: Operational Base + offset (0x003C)

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	MGNUM Magic number of LZ4 is 0x184D2204. If the MGNUM is not 0x184D2204, a magic number error(mn_err) will be generated. And DECOM will stop decompressing.

DECOM_FRAME

Address: Operational Base + offset (0x0040)

Bit	Attr	Reset Value	Description
31:24	RO	0x00	reserved
23:16	RO	0x00	HCC Header checksum byte. One-byte checksum of combined descriptor fields, including optional ones. The value is the second byte of xxh32() using zero as a seed, and the full frame descriptor as an input (including optional fields when they are present). A wrong checksum indicates an error in the descriptor. Header checksum is informational and can be skipped. The header checksum check function can be enabled by controlling the HCEN enable signal in the CTRL register. After being turned on, DECOM calculates the hash32 check value based on the data of the frame descriptor part of the LZ4 compressed file, and performs the header checksum byte in the file. In contrast, if they are not equal, an HCC error will be generated and an interrupt will be generated.
15:8	RO	0x00	BD BD byte. Including Block Maximum Size information. This information is useful to help the decoder allocate memory. Size here refers to the original(uncompressed) data size.
7:6	RO	0x0	VN_NUM Version number. 2-bits field, must be set to 2'b01. Any other value cannot be decoded by this version of the specification. If the version number is error, Version number error(VNE) will be generated.
5	RO	0x0	BCC_FLG Block checksum flag. If this flag is set, each data block will be followed by a 4-bytes checksum, calculated by using the xxHash-32 algorithm on the raw (compressed) data block. The intention is to detect data corruption (storage or transmission errors) immediately, before decoding. Block checksum usage is optional.

Bit	Attr	Reset Value	Description
4	RO	0x0	BIND_FLG Block independence flag. If this flag is set to "1", blocks are independent. If this flag is set to "0", each block depends on previous ones (up to LZ4 window size, which is 64 KB). In such case, it's necessary to decode all blocks in sequence. Block dependency improves compression ratio, especially for small blocks. On the other hand, it makes random access or multi-threaded decoding impossible. For debugging.
3	RO	0x0	CS_FLG Content size flag. If this flag is set, the uncompressed size of data included within the frame will be present as an 8 bytes unsigned little endian value, after the flags. Content size usage is optional.
2	RO	0x0	CCC_FLG Content checksum flag. If this flag is set, a 32-bits content checksum will be appended after the EndMark.
1	RO	0x0	reserved
0	RO	0x0	DICT_FLG Dictionary ID flag. If this flag is set, a 4-bytes Dict-ID field will be present, after the descriptor flags and the Content Size. If this flag is set, Dictionary ID error will be generated. DECOM will stop decompressing.

DECOM_DICTID

Address: Operational Base + offset (0x0044)

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	DICTID Dictionary ID in the compressed file header. Dictionary ID is only present if the DID_FLG is set. It works as a kind of "known prefix" which is used by both the compressor and the decompressor to "warm-up" reference tables.

DECOM_CSL

Address: Operational Base + offset (0x0048)

Bit	Attr	Reset Value	Description
31:0	RO	0x00000000	CSL Content size(CS) lower 32-bits. (Unit:byte) CS[63:0] is the original(uncompressed) size. This information is optional in LZ4 file header, and only present if CS_FLG is set. CS[63:0]={CSH, CSL};

DECOM_CSH

Address: Operational Base + offset (0x004C)

Bit	Attr	Reset Value	Description
31:1	RO	0x00000000	reserved
0	RO	0x0	CSH Content size(CS) upper 32-bits.

DECOM_LMTSL

Address: Operational Base + offset (0x0050)

Bit	Attr	Reset Value	Description
31:0	RW	0xffffffff	LMTSL Limit size of decompressed data lower 32bits. (Unit:byte) When the amount of decompressed data is greater than the LMTS, DSOLI interrupt is generated and the decompression process is stopped.

DECOM_LMTSH

Address: Operational Base + offset (0x0054)

Bit	Attr	Reset Value	Description
31:0	RW	0xffffffff	LMTSH Limit size of decompressed data higher 32bits.

DECOM_GZFHD

Address: Operational Base + offset (0x0058)

Bit	Attr	Reset Value	Description
31:1	RO	0x00000000	reserved
0	RO	0x0	GZFHD GZIP/ZLIB file header information. When GZIPM = 1, GZFHD is GZIP file header information. When ZLIBM = 1, GZFHD is ZLIB file header information.

DECOM_VERSION

Address: Operational Base + offset (0x00F0)

Bit	Attr	Reset Value	Description
31:0	RO	0x00000926	VERSION Version number = 32'h0000_0926.

35.5 Application Notes

- After configuring the decompression mode and other decompression parameters, configure the decompression enable register ENR
- When the decompression is not completed, setting ENR from 1 to 0 will force the decompression process to stop and DECOM will return to the IDLE state. Need to reconfigure and enable DECOM to start the next decompression
- When an error is encountered in the decompression, DECOM will immediately stop the decompression process, set the relevant status register, and generate the corresponding interrupt
- After the decompression process is stopped, ENR will automatically change to 0
- After the decompression process is stopped, DSI will be set to 1. If DSIEN is 1, an interrupt will be generated; if DSIEN is 0, no interrupt will be generated
- After the decompression process is stopped, COMPLETE (STAT[0]) is not set to 1 if there is an error in the decompression process; COMPLETE is set to 1 only when the decompression is complete and there are no errors

Chapter 36 RKNN

36.1 Overview

RKNN is the process unit which is dedicated to neural network. It is designed to accelerate the neural network arithmetic in field of AI (artificial intelligence) such as machine vision and natural language processing. The variety of applications for AI is expanding, and currently provides functionality in a variety of areas, including face tracking as well as gesture and body tracking, image classification, video surveillance, automatic speech recognition (ASR) and advanced driver assistance systems (ADAS).

RKNN supports the following features:

- Include triple NPU CORE
- Support triple core co-work, dual core co-work, and work independently
- AHB interface used for configuration only support single
- AXI interface used to fetch data from memory
- Support integer 4, integer 8, integer 16, float 16, Bfloat 16 and tf32 operation
- 1024x3 integer 8 MAC operations per cycle
- 512x3 integer 16 MAC operations per cycle
- 512x3 float 16 MAC operations per cycle
- 512x3 bfloat 16 MAC operations per cycle
- 256x3 tf32 MAC operation per cycle
- 2048x3 integer 4 MAC operation per cycle
- 384KBx3 internal buffer
- Inference Engine: TensorFlow, Caffe, Tflite, Pytorch, Onnx NN, Android NN, etc.

36.2 Block Diagram

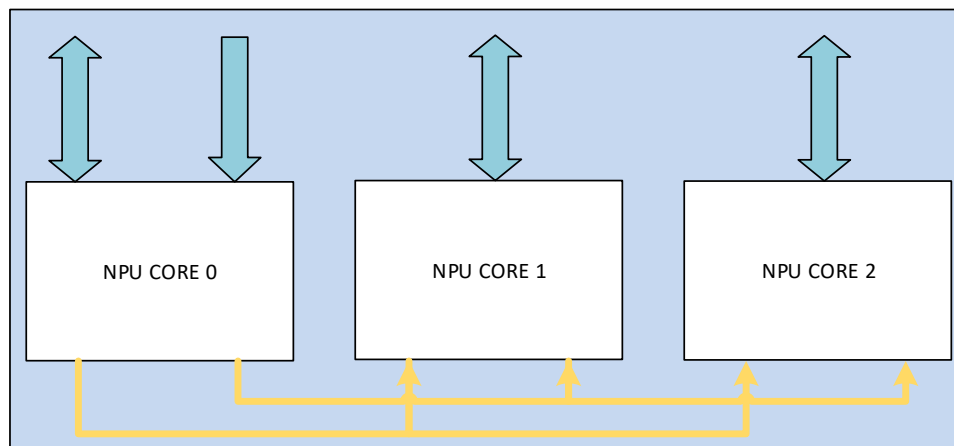


Fig. 36-1 RKNN Triple Core Architecture

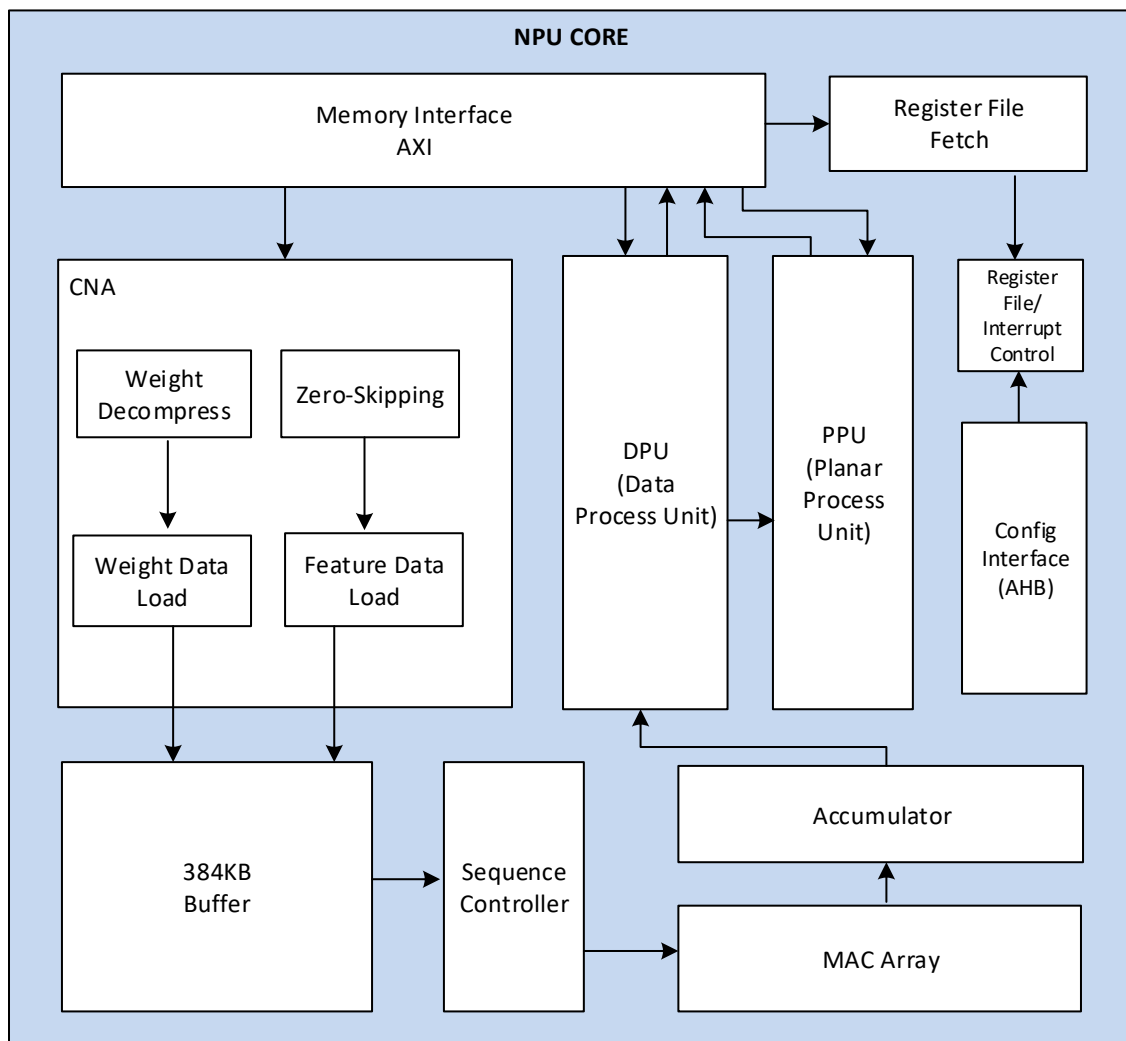


Fig. 36-2 RKNN Single Core Architecture

36.3 Function Description

36.3.1 1.3.1 AHB/AXI Interface

The AXI master interface is used to fetch data from memory that is attached to the Soc AXI interconnect. The AHB slave interface is used to access the registers for configuration, debug and test.

36.3.2 1.3.2 Neural Network Accelerating Engine

As the unit name, this engine is the main process unit for Neural Network arithmetic. This unit include convolution pre-process controller, internal buffer, mac array, accumulator. It provides parallel convolution MAC for recognition functions and int4, int8, int16, fp16, bfloat16 and tf16 are supported.

36.3.3 1.3.3 Data Processing Unit

Data Processing Unit mainly process the single data calculate, such as leaky_relu, relu, relux, sigmoid, tanh...etc. It also provides function: softmax, transpose, data format conversion, ...etc.

36.3.4 1.3.4 Planar Processing Unit

Planar Processing Unit mainly provide planar function followed by output data from Data Processing Unit, such as average pooling, max pooling, min pooling... are supported.

36.3.5 1. 3. 5 Register File Fetch Unit

Register File Fetch Unit fetch register configuration from external system memory through AXI interface.

36.4 Register Description

36.4.1 Internal Address Mapping

Slave address can be divided into different length for different usage, which is shown as follows.

Table 1- 1 RKNN Address Mapping

Base Address[15:12]	Device	Address Length	Offset Address Range
4'h0	PC	4K BYTE	0x0000 ~ 0x0fff
4'h1	CNA	4K BYTE	0x1000 ~ 0x1fff
4'h3	CORE	4K BYTE	0x3000 ~ 0x3fff
4'h4	DPU	4K BYTE	0x4000 ~ 0x4fff
4'h5	DPU_RDMA	4K BYTE	0x5000 ~ 0x5fff
4'h6	PPU	4K BYTE	0x6000 ~ 0x6fff
4'h7	PPU_RDMA	4K BYTE	0x7000 ~ 0x7fff
4'h8	DDMA	4K BYTE	0x8000 ~ 0x8fff
4'h9	SDMA	4K BYTE	0x9000 ~ 0x9fff
4'hf	GLOBAL	4 BYTE	0xf000 ~ 0xf004

36.4.2 Registers Summary

Name	Offset	Size	Reset Value	Description
RKNN_pc_operation_enable	0x0008	W	0x00000000	Operation Enable
RKNN_pc_base_address	0x0010	W	0x00000000	PC address register
RKNN_pc_register_amounts	0x0014	W	0x00000000	Register amount for each task
RKNN_pc_interrupt_mask	0x0020	W	0x0001FFFF	Interrupt Mask
RKNN_pc_interrupt_clear	0x0024	W	0x00000000	Interrupt clear
RKNN_pc_interrupt_status	0x0028	W	0x00000000	Interrupt status
RKNN_pc_interrupt_raw_status	0x002C	W	0x00000000	Interrupt raw status
RKNN_pc_task_control	0x0030	W	0x00000000	Task control register
RKNN_pc_task_dma_base_addr	0x0034	W	0x00000000	Task Base address
RKNN_pc_task_status	0x003C	W	0x00000000	Task status register
RKNN_cna_s_status	0x1000	W	0x00000000	Single register group status
RKNN_cna_s_pointer	0x1004	W	0x00000000	Single register group pointer
RKNN_cna_operation_enable	0x1008	W	0x00000000	Operation Enable
RKNN_cna_conv_control1	0x100C	W	0x00000000	Convolution control register1

Name	Offset	Size	Reset Value	Description
<u>RKNN cna conv con2</u>	0x1010	W	0x00000000	Convolution control register2
<u>RKNN cna conv con3</u>	0x1014	W	0x00000000	Convolution control register3
<u>RKNN cna data size0</u>	0x1020	W	0x00000000	Feature data size control register0
<u>RKNN cna data size1</u>	0x1024	W	0x00000000	Feature data size control register1
<u>RKNN cna data size2</u>	0x1028	W	0x00000000	Feature data size control register2
<u>RKNN cna data size3</u>	0x102C	W	0x00000000	Feature data size control register3
<u>RKNN cna weight size0</u>	0x1030	W	0x00000000	Weight size control 0
<u>RKNN cna weight size1</u>	0x1034	W	0x00000000	Weight size control 1
<u>RKNN cna weight size2</u>	0x1038	W	0x00000000	Weight size control 2
<u>RKNN cna cbuf con0</u>	0x1040	W	0x00000000	CBUF control register 0
<u>RKNN cna cbuf con1</u>	0x1044	W	0x00000000	CBUF control register 1
<u>RKNN cna cvt con0</u>	0x104C	W	0x00000000	Input convert control register0
<u>RKNN cna cvt con1</u>	0x1050	W	0x00000000	Input convert control register1
<u>RKNN cna cvt con2</u>	0x1054	W	0x00000000	Input convert control register2
<u>RKNN cna cvt con3</u>	0x1058	W	0x00000000	Input convert control register3
<u>RKNN cna cvt con4</u>	0x105C	W	0x00000000	Input convert control register4
<u>RKNN cna fc con0</u>	0x1060	W	0x00000000	Full connected control register0
<u>RKNN cna fc con1</u>	0x1064	W	0x00000000	Full connected control register1
<u>RKNN cna pad con0</u>	0x1068	W	0x00000000	Pad control register0
<u>RKNN cna feature data addr</u>	0x1070	W	0x00000000	Base address for input feature data
<u>RKNN cna fc con2</u>	0x1074	W	0x00000000	Full connected control register2
<u>RKNN cna dma con0</u>	0x1078	W	0x00000000	AXI control register 0
<u>RKNN cna dma con1</u>	0x107C	W	0x00000000	AXI control register 1
<u>RKNN cna dma con2</u>	0x1080	W	0x00000000	AXI control register 2
<u>RKNN cna fc data size0</u>	0x1084	W	0x00000000	Full connected data size control register0
<u>RKNN cna fc data size1</u>	0x1088	W	0x00000000	Full connected data size control register1
<u>RKNN cna clk gate</u>	0x1090	W	0x00000000	Clock gating control register
<u>RKNN cna dcomp ctrl</u>	0x1100	W	0x00000000	Weight decompress control register
<u>RKNN cna dcomp regnum</u>	0x1104	W	0x00000000	Weight decompress register number
<u>RKNN cna dcomp addr0</u>	0x1110	W	0x00000000	Base address of the weight
<u>RKNN cna dcomp amount0</u>	0x1140	W	0x00000000	Amount of the weight decompress for the 0 decompress
<u>RKNN cna dcomp amount1</u>	0x1144	W	0x00000000	Amount of the weight decompress for the 1 decompress
<u>RKNN cna dcomp amount2</u>	0x1148	W	0x00000000	Amount of the weight decompress for the 2 decompress
<u>RKNN cna dcomp amount3</u>	0x114C	W	0x00000000	Amount of the weight decompress for the 3 decompress

Name	Offset	Size	Reset Value	Description
<u>RKNN cna dcomp amount4</u>	0x1150	W	0x00000000	Amount of the weight decompress for the 4 decompress
<u>RKNN cna dcomp amount5</u>	0x1154	W	0x00000000	Amount of the weight decompress for the 5 decompress
<u>RKNN cna dcomp amount6</u>	0x1158	W	0x00000000	Amount of the weight decompress for the 6 decompress
<u>RKNN cna dcomp amount7</u>	0x115C	W	0x00000000	Amount of the weight decompress for the 7 decompress
<u>RKNN cna dcomp amount8</u>	0x1160	W	0x00000000	Amount of the weight decompress for the 8 decompress
<u>RKNN cna dcomp amount9</u>	0x1164	W	0x00000000	Amount of the weight decompress for the 9 decompress
<u>RKNN cna dcomp amount10</u>	0x1168	W	0x00000000	Amount of the weight decompress for the 10 decompress
<u>RKNN cna dcomp amount11</u>	0x116C	W	0x00000000	Amount of the weight decompress for the 11 decompress
<u>RKNN cna dcomp amount12</u>	0x1170	W	0x00000000	Amount of the weight decompress for the 12 decompress
<u>RKNN cna dcomp amount13</u>	0x1174	W	0x00000000	Amount of the weight decompress for the 13 decompress
<u>RKNN cna dcomp amount14</u>	0x1178	W	0x00000000	Amount of the weight decompress for the 14 decompress
<u>RKNN cna dcomp amount15</u>	0x117C	W	0x00000000	Amount of the weight decompress for the 15 decompress
<u>RKNN cna cvt con5</u>	0x1180	W	0x00000000	Input convert control register5
<u>RKNN cna pad con1</u>	0x1184	W	0x00000000	Pad controller register1
<u>RKNN core s status</u>	0x3000	W	0x00000000	Single register group status
<u>RKNN core s pointer</u>	0x3004	W	0x00000000	Single register group pointer
<u>RKNN core operation enable</u>	0x3008	W	0x00000000	Operation Enable
<u>RKNN core mac gating</u>	0x300C	W	0x07800800	MAC gating register
<u>RKNN core misc cfg</u>	0x3010	W	0x00000000	Misc configuration register
<u>RKNN core dataout size0</u>	0x3014	W	0x00000000	Feature size register 0 of output
<u>RKNN core dataout size1</u>	0x3018	W	0x00000000	Feature size register 1 of output
<u>RKNN core clip truncate</u>	0x301C	W	0x00000000	Shift value register
<u>RKNN dpu s status</u>	0x4000	W	0x00000000	Single register group status
<u>RKNN dpu s pointer</u>	0x4004	W	0x00000000	Single register group pointer
<u>RKNN dpu operation enable</u>	0x4008	W	0x00000000	Operation Enable
<u>RKNN dpu feature mode cfg</u>	0x400C	W	0x00000000	Configuration of the feature mode

Name	Offset	Size	Reset Value	Description
<u>RKNN_dpu_data_format</u>	0x4010	W	0x00000000	Configuration of the data format
<u>RKNN_dpu_offset_pend</u>	0x4014	W	0x00000000	Value of the offset pend
<u>RKNN_dpu_dst_base_addr</u>	0x4020	W	0x00000000	Destination base address
<u>RKNN_dpu_dst_surf_stride</u>	0x4024	W	0x00000000	Destination surface size
<u>RKNN_dpu_data_cube_width</u>	0x4030	W	0x00000000	Width of the input cube
<u>RKNN_dpu_data_cube_height</u>	0x4034	W	0x00000000	Height of the input cube
<u>RKNN_dpu_data_cube_notch_addr</u>	0x4038	W	0x00000000	Notch signal of the input cube
<u>RKNN_dpu_data_cube_channel</u>	0x403C	W	0x00000000	Channel of the input cube
<u>RKNN_dpu_bs_cfg</u>	0x4040	W	0x00000000	Configuration of the BS
<u>RKNN_dpu_bs_alu_cfg</u>	0x4044	W	0x00000000	Configuration of the BS ALU
<u>RKNN_dpu_bs_mul_cfg</u>	0x4048	W	0x00000000	Configuration of the BS MUL
<u>RKNN_dpu_bs_relux_compare_value</u>	0x404C	W	0x00000000	Value of the RELUX compare with
<u>RKNN_dpu_bs_ow_cfg</u>	0x4050	W	0x00000000	Configuration of the BS OW
<u>RKNN_dpu_bs_ow_op</u>	0x4054	W	0x00000000	Ow op of the BS OW
<u>RKNN_dpu_wdma_size_0</u>	0x4058	W	0x00000000	Size 0 of the WDMA
<u>RKNN_dpu_wdma_size_1</u>	0x405C	W	0x00000000	Size 1 of the WDMA
<u>RKNN_dpu_bn_cfg</u>	0x4060	W	0x00000000	Configuration of BN
<u>RKNN_dpu_bn_alu_cfg</u>	0x4064	W	0x00000000	Configuration of the BN ALU
<u>RKNN_dpu_bn_mul_cfg</u>	0x4068	W	0x00000000	Configuration of the BN MUL
<u>RKNN_dpu_bn_relux_compare_value</u>	0x406C	W	0x00000000	Value of the RELUX compare with
<u>RKNN_dpu_ew_cfg</u>	0x4070	W	0x00000000	Configuration of EW
<u>RKNN_dpu_ew_cvt_offset_value</u>	0x4074	W	0x00000000	Offset of the EW input convert
<u>RKNN_dpu_ew_cvt_scale_value</u>	0x4078	W	0x00000000	Scale of the EW input convert
<u>RKNN_dpu_ew_relux_compare_value</u>	0x407C	W	0x00000000	Value of the RELUX compare with
<u>RKNN_dpu_out_cvt_offset</u>	0x4080	W	0x00000000	Offset of the output converter
<u>RKNN_dpu_out_cvt_scale</u>	0x4084	W	0x00000000	Scale of the output converter
<u>RKNN_dpu_out_cvt_shift</u>	0x4088	W	0x00000000	Shift of the output converter
<u>RKNN_dpu_ew_op_value_0</u>	0x4090	W	0x00000000	Configure operand0 of the EW
<u>RKNN_dpu_ew_op_value_1</u>	0x4094	W	0x00000000	Configure operand1 of the EW

Name	Offset	Size	Reset Value	Description
<u>RKNN dpu ew op value 2</u>	0x4098	W	0x00000000	Configure operand2 of the EW
<u>RKNN dpu ew op value 3</u>	0x409C	W	0x00000000	Configure operand3 of the EW
<u>RKNN dpu ew op value 4</u>	0x40A0	W	0x00000000	Configure operand4 of the EW
<u>RKNN dpu ew op value 5</u>	0x40A4	W	0x00000000	Configure operand5 of the EW
<u>RKNN dpu ew op value 6</u>	0x40A8	W	0x00000000	Configure operand6 of the EW
<u>RKNN dpu ew op value 7</u>	0x40AC	W	0x00000000	Configure operand7 of the EW
<u>RKNN dpu surface add</u>	0x40C0	W	0x00000000	Value of the surface adder
<u>RKNN dpu lut access cfg</u>	0x4100	W	0x00000000	LUT access address and type
<u>RKNN dpu lut access data</u>	0x4104	W	0x00000000	Configuration of LUT access data
<u>RKNN dpu lut cfg</u>	0x4108	W	0x00000000	Configuration of the LUT
<u>RKNN dpu lut info</u>	0x410C	W	0x00000000	LUT information register
<u>RKNN dpu lut le start</u>	0x4110	W	0x00000000	LE LUT start point
<u>RKNN dpu lut le end</u>	0x4114	W	0x00000000	LE LUT end point
<u>RKNN dpu lut lo start</u>	0x4118	W	0x00000000	LO LUT start point
<u>RKNN dpu lut lo end</u>	0x411C	W	0x00000000	LO LUT end point
<u>RKNN dpu lut le slope scale</u>	0x4120	W	0x00000000	LE LUT slope scale
<u>RKNN dpu lut le slope shift</u>	0x4124	W	0x00000000	LE LUT slope shift
<u>RKNN dpu lut lo slope scale</u>	0x4128	W	0x00000000	LO LUT slope scale
<u>RKNN dpu lut lo slope shift</u>	0x412C	W	0x00000000	LO LUT slope shift
<u>RKNN dpu rdma s status</u>	0x5000	W	0x00000000	Single register group status
<u>RKNN dpu rdma s pointer</u>	0x5004	W	0x00000000	Single register group pointer
<u>RKNN dpu rdma operation enable</u>	0x5008	W	0x00000000	Operation Enable
<u>RKNN dpu rdma data cube width</u>	0x500C	W	0x00000000	Input cube width
<u>RKNN dpu rdma data cube height</u>	0x5010	W	0x00000000	Input cube height
<u>RKNN dpu rdma data cube channel</u>	0x5014	W	0x00000000	Input cube channel

Name	Offset	Size	Reset Value	Description
<u>RKNN_dpu_rdma_src_base_addr</u>	0x5018	W	0x00000000	Base address of the input cube
<u>RKNN_dpu_rdma_brdma_cfg</u>	0x501C	W	0x00000000	Configurations of BRDMA
<u>RKNN_dpu_rdma_bs_base_addr</u>	0x5020	W	0x00000000	Source base address of BRDMA
<u>RKNN_dpu_rdma_nrdma_cfg</u>	0x5028	W	0x00000000	Configurations of NRDMA
<u>RKNN_dpu_rdma_bn_base_addr</u>	0x502C	W	0x00000000	Source base address of NRDMA
<u>RKNN_dpu_rdma_erdma_cfg</u>	0x5034	W	0x00000000	Configurations of ERDMA
<u>RKNN_dpu_rdma_ew_base_addr</u>	0x5038	W	0x00000000	Source base address of ERDMA
<u>RKNN_dpu_rdma_ew_surf_stride</u>	0x5040	W	0x00000000	Surface size of the cube that the ERDMA read
<u>RKNN_dpu_rdma_feature_mode_cfg</u>	0x5044	W	0x00000000	Configuration of the feature mode
<u>RKNN_dpu_rdma_src_dma_cfg</u>	0x5048	W	0x00000000	Configuration of the source read DMA
<u>RKNN_dpu_rdma_surf_notch</u>	0x504C	W	0x00000000	Surface notch
<u>RKNN_dpu_rdma_pad_cfg</u>	0x5064	W	0x00000000	Configuration of the pad
<u>RKNN_dpu_rdma_weight</u>	0x5068	W	0x00000000	Weight of the arbiter
<u>RKNN_dpu_rdma_ew_surf_notch</u>	0x506C	W	0x00000000	Surface notch
<u>RKNN_ppu_s_status</u>	0x6000	W	0x00000000	Single register group status
<u>RKNN_ppu_s_pointer</u>	0x6004	W	0x00000000	Single register group pointer
<u>RKNN_ppu_operation_enable</u>	0x6008	W	0x00000000	Operation Enable
<u>RKNN_ppu_data_cube_in_width</u>	0x600C	W	0x00000000	Width of the input cube
<u>RKNN_ppu_data_cube_in_height</u>	0x6010	W	0x00000000	Height of the input cube
<u>RKNN_ppu_data_cube_in_channel</u>	0x6014	W	0x00000000	Channel of the input cube
<u>RKNN_ppu_data_cube_out_width</u>	0x6018	W	0x00000000	Width of the output cube
<u>RKNN_ppu_data_cube_out_height</u>	0x601C	W	0x00000000	Height of the output cube
<u>RKNN_ppu_data_cube_out_channel</u>	0x6020	W	0x00000000	Channel of the output cube

Name	Offset	Size	Reset Value	Description
<u>RKNN ppu operation mode_cfg</u>	0x6024	W	0x00000000	Configuration of the operation mode
<u>RKNN ppu pooling kernel_cfg</u>	0x6034	W	0x00000000	Configuration of the pooling kernel size
<u>RKNN ppu recip kernel width</u>	0x6038	W	0x00000000	The reciprocal of the kernel width
<u>RKNN ppu recip kernel height</u>	0x603C	W	0x00000000	The reciprocal of the kernel height
<u>RKNN ppu pooling padding_cfg</u>	0x6040	W	0x00000000	Configuration of the pooling pad
<u>RKNN ppu padding value_1_cfg</u>	0x6044	W	0x00000000	Pad Value register0
<u>RKNN ppu padding value_2_cfg</u>	0x6048	W	0x00000000	Pad Value register1
<u>RKNN ppu dst base_addr</u>	0x6070	W	0x00000000	Destination address of the output cube
<u>RKNN ppu dst surf stride</u>	0x607C	W	0x00000000	Destination surface size
<u>RKNN ppu data format</u>	0x6084	W	0x00000000	Configuration of the data format
<u>RKNN ppu misc_ctrl</u>	0x60DC	W	0x00000000	Misc configuration
<u>RKNN ppu rdma status</u>	0x7000	W	0x00000000	Single register group status
<u>RKNN ppu rdma s_pointer</u>	0x7004	W	0x00000000	Single register group pointer
<u>RKNN ppu rdma operation_enable</u>	0x7008	W	0x00000000	Operation Enable
<u>RKNN ppu rdma cube input width</u>	0x700C	W	0x00000000	Input cube width
<u>RKNN ppu rdma cube input height</u>	0x7010	W	0x00000000	Input cube height
<u>RKNN ppu rdma cube input channel</u>	0x7014	W	0x00000000	Input cube channel
<u>RKNN ppu rdma src base_addr</u>	0x701C	W	0x00000000	Source base address for input feature
<u>RKNN ppu rdma src line_stride</u>	0x7024	W	0x00000000	Source line number including Virtual box
<u>RKNN ppu rdma src surf_stride</u>	0x7028	W	0x00000000	Source surface size including Virtual box
<u>RKNN ppu rdma data format</u>	0x7030	W	0x00000000	Data format for the input feature
<u>RKNN ddma_cfg_outstanding</u>	0x8000	W	0x00000000	Outstanding config register
<u>RKNN ddma_rd_weight_0</u>	0x8004	W	0x00000000	Weight of read arbiter

Name	Offset	Size	Reset Value	Description
<u>RKNN_ddma_wr_weight_0</u>	0x8008	W	0x00000000	Weight of write arbiter
<u>RKNN_ddma_cfg_id_error</u>	0x800C	W	0x00000000	Id where error happened
<u>RKNN_ddma_rd_weight_1</u>	0x8010	W	0x00000000	Weight of read arbiter register1
<u>RKNN_ddma_cfg_dma_fifo_clr</u>	0x8014	W	0x00000000	Clear DMA FIFO
<u>RKNN_ddma_cfg_dma_arb</u>	0x8018	W	0x00000000	DMA arbiter configuration register
<u>RKNN_ddma_cfg_dma_rd_qos</u>	0x8020	W	0x00000000	Read Qos for DMA
<u>RKNN_ddma_cfg_dma_rd_cfg</u>	0x8024	W	0x00000000	Read configuration for AXI signals
<u>RKNN_ddma_cfg_dma_wr_cfg</u>	0x8028	W	0x00000000	Write configuration for AXI signals
<u>RKNN_ddma_cfg_dma_wstrb</u>	0x802C	W	0x00000000	Write strobe signal for AXI
<u>RKNN_ddma_cfg_status</u>	0x8030	W	0x00000000	AXI status signal
<u>RKNN_sdma_cfg_outstanding</u>	0x9000	W	0x00000000	Outstanding configuration register
<u>RKNN_sdma_rd_weight_0</u>	0x9004	W	0x00000000	Weight of read arbiter
<u>RKNN_sdma_wr_weight_0</u>	0x9008	W	0x00000000	Weight of write arbiter
<u>RKNN_sdma_cfg_id_error</u>	0x900C	W	0x00000000	Id where error happened
<u>RKNN_sdma_rd_weight_1</u>	0x9010	W	0x00000000	Weight of read arbiter register1
<u>RKNN_sdma_cfg_dma_fifo_clr</u>	0x9014	W	0x00000000	Clear DMA FIFO
<u>RKNN_sdma_cfg_dma_arb</u>	0x9018	W	0x00000000	DMA arbiter configuration register
<u>RKNN_sdma_cfg_dma_rd_qos</u>	0x9020	W	0x00000000	Read Qos for DMA
<u>RKNN_sdma_cfg_dma_rd_cfg</u>	0x9024	W	0x00000000	Read configuration for AXI signals
<u>RKNN_sdma_cfg_dma_wr_cfg</u>	0x9028	W	0x00000000	Write configuration for AXI signals
<u>RKNN_sdma_cfg_dma_wstrb</u>	0x902C	W	0x00000000	Write strobe signal for AXI
<u>RKNN_sdma_cfg_status</u>	0x9030	W	0x00000000	AXI status signal
<u>RKNN_global_operation_enable</u>	0xF008	W	0x00000000	Combine Operation Enable

Notes: **Size:** **B**- Byte (8 bits) access, **HW**- Half WORD (16 bits) access, **W**-WORD (32 bits) access, **DW**- Double WORD (64 bits) access

36.4.3 Detail Registers Description

RKNN_pc_operation_enable

Address: Operational Base + offset (0x0008)

Bit	Attr	Reset Value	Description
31:1	RO	0x00000000	reserved
0	RW	0x0	op_en PC operation enable. 1'd0: Disable PC module; 1'd1: Enable PC module to fetch register for each task.

RKNN_pc_base_address

Address: Operational Base + offset (0x0010)

Bit	Attr	Reset Value	Description
31:4	RW	0x00000000	pc_source_addr PC base address. This is the address of DMA instruction where it located.
3:1	RO	0x0	reserved
0	RW	0x0	pc_sel PC mode enable. 1'd0: PC mode, use AXI DMA to fetch register config; 1'd1: Slave mode, use AHB to set register.

RKNN_pc_register_amounts

Address: Operational Base + offset (0x0014)

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	reserved
15:0	RW	0x0000	pc_data_amount Data amount. The register number need to be fetched of one task. Each register takes 64 bits, it is combined as following: bit[63:48] indicates which block the register forward to. bit[47:16] the register's value bit[15: 0] the register's offset address in each block. bit[56]= 1 means this register is for pc block. bit[57]= 1 CNA bit[59]= 1 CORE bit[60]= 1 DPU bit[61]= 1 DPU_RDMA bit[62]= 1 PPU bit[63]= 1 PPU_RDMA bit[55]= 1 to set each block's op_en eg. 64'h0081_0000_007f_0008 will set each block's op_en(CNA, CORE, ..., PPU_RDMA). note: op_en is strongly recommended set at the end of register list. before op_en, 64'h0041_xxxx_xxxx_xxxx must be set.

RKNN_pc_interrupt_mask

Address: Operational Base + offset (0x0020)

Bit	Attr	Reset Value	Description
31:17	RO	0x0000	reserved
16:0	RW	0x1ffff	int_mask Interrupt mask. mask[0]: CNA feature group 0 interrupt mask. set 1 to enable interrupt; mask[1]: CNA feature group 1; mask[2]: CNA weight group 0; mask[3]: CNA weight group 1; mask[4]: CNA csc group 0; mask[5]: CNA csc group 1; mask[6]: CORE group 0; mask[7]: CORE group 1; mask[8]: DPU group 0; mask[9]: DPU group 1; mask[10]: PPU group 0; mask[11]: PPU group 1; mask[12]: DMA read error; mask[13]: DMA write error; Note: In pc mode, int mask set the last one task's interrupt masking.

RKNN_pc interrupt clear

Address: Operational Base + offset (0x0024)

Bit	Attr	Reset Value	Description
31:17	RO	0x0000	reserved
16:0	W1C	0x00000	int_clr Interrupt clear. done_clr[0]: CNA feature group 0 interrupt clear; done_clr[1]: CNA feature group 1; done_clr[2]: CNA weight group 0; done_clr[3]: CNA weight group 1; done_clr[4]: CNA csc group 0; done_clr[5]: CNA csc group 1; done_clr[6]: CORE group 0; done_clr[7]: CORE group 1; done_clr[8]: DPU group 0; done_clr[9]: DPU group 1; done_clr[10]: PPU group 0; done_clr[11]: PPU group 1; done_clr[12]: DMA read error; done_clr[13]: DMA write error.

RKNN_pc interrupt status

Address: Operational Base + offset (0x0028)

Bit	Attr	Reset Value	Description
31:17	RO	0x0000	reserved
16:0	W1 C	0x00000	int_st Interrupt status. int_st[0]: CNA feature group 0 interrupt status, which and with mask bit; int_st[1]: CNA feature group 1; int_st[2]: CNA weight group 0; int_st[3]: CNA weight group 1; int_st[4]: CNA csc group 0; int_st[5]: CNA csc group 1; int_st[6]: CORE group 0; int_st[7]: CORE group 1; int_st[8]: DPU group 0; int_st[9]: DPU group 1; int_st[10]: PPU group 0; int_st[11]: PPU group 1; int_st[12]: DMA read error; int_st[13]: DMA write error.

RKNN_pc interrupt raw status

Address: Operational Base + offset (0x002C)

Bit	Attr	Reset Value	Description
31:17	RO	0x0000	reserved
16:0	W1 C	0x00000	int_raw_st Interrupt raw status. int_st[0]: CNA feature group 0 interrupt raw status; int_st[1]: CNA feature group 1; int_st[2]: CNA weight group 0; int_st[3]: CNA weight group 1; int_st[4]: CNA csc group 0; int_st[5]: CNA csc group 1; int_st[6]: CORE group 0; int_st[7]: CORE group 1; int_st[8]: DPU group 0; int_st[9]: DPU group 1; int_st[10]: PPU group 0; int_st[11]: PPU group 1; int_st[12]: DMA read error; int_st[13]: DMA write error.

RKNN_pc task con

Address: Operational Base + offset (0x0030)

Bit	Attr	Reset Value	Description
31:14	RO	0x00000	reserved

Bit	Attr	Reset Value	Description
13	W1 C	0x0	task_count_clear Task counter clear register. Clear the counter that counting current task. Before task started, it is suggested to clear.
12	RW	0x0	task_pp_en PC task ping-pong mode enable. 1'd0: Ping-pong mode off. The second group register setting is fetched after first group task operation is finished; 1'd1: Tasks' registers are fetched in ping-pong mode. The second group register setting is fetched immediately after first group's register fetching is finished.
11:0	RW	0x000	task_number PC task number. Set the total task number to be executed.

RKNN_pc_task_dma_base_addr

Address: Operational Base + offset (0x0034)

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	dma_base_addr Task base address. This is for each DMA's base address. For feature DMA, weight DMA, DPU DMA, PPU DMA, the address is set as offset address. Final address appear on AXI bus is base address + offset address.
3:0	RO	0x0	reserved

RKNN_pc_task_status

Address: Operational Base + offset (0x003C)

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:0	RW	0x0000000	task_status Task status. [11:0]: Current task counter value; [12]: Indicate the first task is operating; [13]: Indicate the last task is operating; [12]: Indicate the first task's register is fetching; [13]: Indicate the last task's register is fetching.

RKNN_cna_s_status

Address: Operational Base + offset (0x1000)

Bit	Attr	Reset Value	Description
31:18	RO	0x0000	reserved

Bit	Attr	Reset Value	Description
17:16	RO	0x0	status_1 Executer 1 status. 2'd0: Executer 1 is in idle state; 2'd1: Executer 1 is operating; 2'd2: Executer 1 is operating, executer 1 is waiting to operate; 2'd3: Reserved.
15:2	RO	0x0000	reserved
1:0	RO	0x0	status_0 Executer 0 status. 2'd0: Executer 0 is in idle state; 2'd1: Executer 0 is operating; 2'd2: Executer 0 is operating, executer 1 is waiting to operate; 2'd3: Reserved.

RKNN_cna_s_pointer

Address: Operational Base + offset (0x1004)

Bit	Attr	Reset Value	Description
31:17	RO	0x0000	reserved
16	RO	0x0	executer Which register group to be used. 1'd0: Executer group 0; 1'd1: Executer group 1.
15:6	RO	0x000	reserved
5	W1C	0x0	executer_pp_clear Clear executer group pointer. Set this bit to 1 to clear pointer to 0.
4	W1C	0x0	pointer_pp_clear Clear register group pointer. Set this bit to 1 to clear pointer to 0.
3	RW	0x0	pointer_pp_mode Register group ping-pong mode. 1'd0: Pointer ping-pong by executer; eg. if current executer is 0, next pointer will toggle to 1; 1'd1: Pointer ping-pong by pointer; eg. if current pointer is 0, next pointer will toggle to 1.
2	RW	0x0	executer_pp_en Executer group ping-pong enable. 1'd0: Disable; 1'd1: Enable.
1	RW	0x0	pointer_pp_en Register group ping-pong enable. 1'd0: Disable; 1'd1: Enable.

Bit	Attr	Reset Value	Description
0	RW	0x0	pointer Which register group ready to be set. 1'd0: Register group 0; 1'd1: Register group 1.

RKNN_cna_operation_enable

Address: Operational Base + offset (0x1008)

Bit	Attr	Reset Value	Description
31:1	RO	0x00000000	reserved
0	RW	0x0	op_en Set this register will trigger CNA block operate. This register and after this are all shadowed for ping-pong operation. 1'd0: Disable; 1'd1: Enable.

RKNN_cna_conv_con1

Address: Operational Base + offset (0x100C)

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30	RW	0x0	nonalign_dma CNA DMA non-align mode. 1'd0: Disable; 1'd1: Enable, please enable this bit under ARGB mode. Enable this bit will enable DMA fetching feature data continuously.
29	RW	0x0	group_line_off Group line fetch off. 1'd0: Enable group line fetch; 1'd1: Disable. This setting only influence line fetch efficiency.
28:17	RO	0x000	reserved
16	RW	0x0	deconv Enable deconvolution function. 1'd0: Disable; 1'd1: Enable.
15:12	RW	0x0	argb_in Non-align channel layer control register. 4'd8: 1 channel input mode; 4'd9: 2 channel input mode; 4'd10: 3 channel input mode; 4'd11: 4 channel input mode.
11:10	RO	0x0	reserved

Bit	Attr	Reset Value	Description
9:7	RW	0x0	proc_precision Process precision. 3'd0: Input precision is int 8; 3'd1: Input data precision is int 16; 3'd2: Input data precision is float 16; 3'd3: Input data precision is bfloat 16; 3'd4: Reserved; 3'd5: Reserved; 3'd6: Input data precision is int4; 3'd7: Input data precision is tf32.
6:4	RW	0x0	in_precision Input precision. 3'd0: Input precision is int 8; 3'd1: Input data precision is int 16; 3'd2: Input data precision is float 16; 3'd3: Input data precision is bfloat 16; 3'd4: Reserved; 3'd5: Reserved; 3'd6: Input data precision is int4; 3'd7: Input data precision is tf32.
3:0	RW	0x0	conv_mode Convolution mode. 2'd0: Direct convolution; 2'd1: Reserved; 2'd2: Reserved; 2'd3: Depthwise convolution.

RKNN_cna_conv_con2

Address: Operational Base + offset (0x1010)

Bit	Attr	Reset Value	Description
31:24	RO	0x00	reserved
23:16	RW	0x00	kernel_group Kernels group. In int8, 32 kernels form 1 group, in int16 or fp16, 16 kernels form 1 group. eg, weight kernel is 256, in int8, you can set this register to be 256/32 - 1 = 15.
15:14	RO	0x0	reserved
13:4	RW	0x000	feature_grains Feature data rows needs to be buffered before convolution start. It's suggested to set this field as y_stride+weight_height+1.
3	RO	0x0	reserved

Bit	Attr	Reset Value	Description
2	RW	0x0	csc_wo_en Do weight scan. 1'd0: Enable csc output weight data to core; 1'd1: Disable.
1	RW	0x0	csc_do_en Do data scan. 1'd0: Enable csc output feature data to core; 1'd1: Disable.
0	RW	0x0	cmd_fifo_srst Command FIFO soft reset. Reserved for debug purpose.

RKNN_cna_conv_con3

Address: Operational Base + offset (0x1014)

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved
30:28	RW	0x0	nn_mode co-work mode. 3'd0: Int8 mac array 32x32 mode; 3'd1: 64x32 mode; 3'd2: 96x32 mode; 3'd3: Reserved; 3'd4: 32x64 mode; 3'd5: 32x96 mode; 3'd6: Reserved; 3'd7: Reserved. This register is target for multicore mode. By single core mode keep it at 3'd0.
27:26	RO	0x0	reserved
25:21	RW	0x00	atrous_y_dilation Atrous x dilation. Pad numbers inserted in feature map column between 2 pixels.
20:16	RW	0x00	atrous_x_dilation Atrous x dilation. Pad numbers inserted in feature map row between 2 pixels. Set this register value >0 will enable atrous convolution.
15:14	RO	0x0	reserved
13:11	RW	0x0	deconv_y_stride Deconvolution y stride. Pad numbers inserted in feature map column between 2 pixels.
10:8	RW	0x0	deconv_x_stride Deconvolution x stride. Pad numbers inserted in feature map row between 2 pixels.
7:6	RO	0x0	reserved

Bit	Attr	Reset Value	Description
5:3	RW	0x0	conv_y_stride Convolution y stride. Stride value in y direction.
2:0	RW	0x0	conv_x_stride Convolution x stride. Stride value in x direction.

RKNN_cna_data_size0

Address: Operational Base + offset (0x1020)

Bit	Attr	Reset Value	Description
31:27	RO	0x00	reserved
26:16	RW	0x000	datain_width Input feature data width.
15:11	RO	0x00	reserved
10:0	RW	0x000	datain_height Input feature data height.

RKNN_cna_data_size1

Address: Operational Base + offset (0x1024)

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	datain_channel_real Real channel number. If the input channel is not integer times of 8(int8) or 4(int16/float16), set the real channel number in this field.
15:0	RW	0x0000	datain_channel Input feature data channel number; Int 8 mode, this number should be integer times of 8; Int 16/float 16 mode, should be integer times of 4.

RKNN_cna_data_size2

Address: Operational Base + offset (0x1028)

Bit	Attr	Reset Value	Description
31:11	RO	0x000000	reserved
10:0	RW	0x000	dataout_width Data width after convolution.

RKNN_cna_data_size3

Address: Operational Base + offset (0x102C)

Bit	Attr	Reset Value	Description
31:24	RO	0x00	reserved

Bit	Attr	Reset Value	Description
23:22	RW	0x0	surf_mode Surface serial mode. 2'd0: 1surf series; 2'd1: 1surf series; 2'd2: 2 surf series; 2'd3: 4 surf series.
21:0	RW	0x000000	dataout_atomics Data atoms after convolution which is data out total pixels number.

RKNN_cna_weight_size0

Address: Operational Base + offset (0x1030)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	weight_bytes Weight bytes in total for this convolution.

RKNN_cna_weight_size1

Address: Operational Base + offset (0x1034)

Bit	Attr	Reset Value	Description
31:19	RO	0x0000	reserved
18:0	RW	0x00000	weight_bytes_per_kernel Weight bytes for one kernel.

RKNN_cna_weight_size2

Address: Operational Base + offset (0x1038)

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:24	RW	0x00	weight_width Kernel width.
23:21	RO	0x0	reserved
20:16	RW	0x00	weight_height Kernel height.
15:14	RO	0x0	reserved
13:0	RW	0x0000	weight_kernels Weight kernels.

RKNN_cna_cbuf_con0

Address: Operational Base + offset (0x1040)

Bit	Attr	Reset Value	Description
31:14	RO	0x00000	reserved

Bit	Attr	Reset Value	Description
13	RW	0x0	weight_reuse Weight data reuse enable. 1'd0: Disable; 1'd1: Enable data reuse. fetching weight directly from internal buffer.
12	RW	0x0	data_reuse Feature data reuse enable. 1'd0: Disable; 1'd1: Enable data reuse. fetching data directly from internal buffer.
11	RO	0x0	reserved
10:8	RW	0x0	fc_data_bank Bank numbers for fc zero-skipping feature data. In FC zero-skipping mode, set to be 1, Otherwise, must set to be 0.
7:4	RW	0x0	weight_bank Bank numbers for weight data. 4'd1: Bank 7 occupied by weight data; 4'd2: Bank 6/7 occupied by weight data; ... 4'd7: Bank 1-7 occupied by weight data.
3:0	RW	0x0	data_bank Bank numbers for feature data. 4'd0: Bank 0 occupied by feature data; 4'd1: Bank 0 and bank 1 occupied by feature data; 4'd2: Bank 0/1/2 occupied by feature data; ... 4'd6: Bank 0-6 occupied by feature data.

RKNN_cna_cbuf_con1

Address: Operational Base + offset (0x1044)

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	reserved
12:0	RW	0x0000	data_entries How many banks space needed to store one feature map row.

RKNN_cna_cvt_con0

Address: Operational Base + offset (0x104C)

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:22	RW	0x00	cvt_truncate_3 CVT truncate value 3.
21:16	RW	0x00	cvt_truncate_2 CVT truncate value 2.
15:10	RW	0x00	cvt_truncate_1 CVT truncate value 1.

Bit	Attr	Reset Value	Description
9:4	RW	0x00	cvt_truncate_0 CVT truncate value 0.
3	RW	0x0	data_sign Feature data is signed or unsigned. 1'd0: Unsigned; 1'd1: Signed.
2	RW	0x0	round_type Rounding type of the input convert. 1'd0: Odd in, even not; 1'd1: Round-up 0.5 to 1.
1	RW	0x0	cvt_type Cal type of the input convert. 1'd0: Multiply first, then add; 1'd1: CVT function will do add first, then multiply.
0	RW	0x0	cvt_bypass Bypass input convert. 1'd0: Enable CVT function; 1'd1: Disable CVT function.

RKNN cna_cvt_con1

Address: Operational Base + offset (0x1050)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	cvt_scale0 CVT scale 0. Multiplier operand for 1st channel.
15:0	RW	0x0000	cvt_offset0 CVT offset 0. Adder operand for 1st channel.

RKNN cna_cvt_con2

Address: Operational Base + offset (0x1054)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	cvt_scale1 CVT scale 1 Multiplier operand for 2nd channel.
15:0	RW	0x0000	cvt_offset1 CVT offset 1. Adder operand for 2nd channel.

RKNN cna_cvt_con3

Address: Operational Base + offset (0x1058)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	cvt_scale2 CVT scale 2. Multiplier operand for 3rd channel.

Bit	Attr	Reset Value	Description
15:0	RW	0x0000	cvt_offset2 CVT offset 2. Adder operand for 3rd channel.

RKNN_cna_cvt_con4

Address: Operational Base + offset (0x105C)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	cvt_scale3 CVT scale 3. Multiplier operand for 4th channel.
15:0	RW	0x0000	cvt_offset3 CVT offset 3. Adder operand for 4th channel.

RKNN_cna_fc_con0

Address: Operational Base + offset (0x1060)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	fc_skip_data FC zero skipping data Skipped feature data value, normally set to 0.
15:1	RO	0x0000	reserved
0	RW	0x0	fc_skip_en FC zero skipping enable 1'd0: Disable; 1'd1: Enable skip some feature data value, normally skip zero. When one pixel feature data is 0, the corresponding weight data is not fetched from system memory.

RKNN_cna_fc_con1

Address: Operational Base + offset (0x1064)

Bit	Attr	Reset Value	Description
31:17	RO	0x0000	reserved
16:0	RW	0x00000	data_offset FC zero skipping data offset. Feature data offset in fc skip mode.

RKNN_cna_pad_con0

Address: Operational Base + offset (0x1068)

Bit	Attr	Reset Value	Description
31:8	RO	0x000000	reserved
7:4	RW	0x0	pad_left Pad left Pad numbers in left of the feature map.

Bit	Attr	Reset Value	Description
3:0	RW	0x0	pad_top PAD top Pad numbers in top of the feature map.

RKNN_cna_feature_data_addr

Address: Operational Base + offset (0x1070)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	feature_base_addr Feature data address.

RKNN_cna_fc_con2

Address: Operational Base + offset (0x1074)

Bit	Attr	Reset Value	Description
31:17	RO	0x0000	reserved
16:0	RW	0x00000	weight_offset Weight data address.

RKNN_cna_dma_con0

Address: Operational Base + offset (0x1078)

Bit	Attr	Reset Value	Description
31	RW	0x0	ov4k_bypass Separate the burst command of over 4k to 2 independent burst commands. 1'd0: Enable this feature; 1'd1: Bypass this feature.
30:20	RO	0x000	reserved
19:16	RW	0x0	weight_burst_len AXI burst length for weight data DMA. 4'd3: Burst length is 4; 4'd7: Burst length is 8; 4'd15: Burst length is 16.
15:4	RO	0x000	reserved
3:0	RW	0x0	data_burst_len AXI burst length for feature data DMA. 4'd3: Burst length is 4; 4'd7: Burst length is 8; 4'd15: Burst length is 16.

RKNN_cna_dma_con1

Address: Operational Base + offset (0x107C)

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:0	RW	0x0000000	line_stride Line stride Feature width with Virtual box.

RKNN_cna_dma_con2

Address: Operational Base + offset (0x1080)

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27:0	RW	0x0000000	surf_stride Surface stride Feature map actual surface area.

RKNN_cna_fc_data_size0

Address: Operational Base + offset (0x1084)

Bit	Attr	Reset Value	Description
31:30	RO	0x0	reserved
29:16	RW	0x0000	dma_width Feature input width for AXI DMA.
15:11	RO	0x00	reserved
10:0	RW	0x000	dma_height Feature input height for AXI DMA.

RKNN_cna_fc_data_size1

Address: Operational Base + offset (0x1088)

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	reserved
15:0	RW	0x0000	dma_channel Feature input channel for AXI DMA.

RKNN_cna_clk_gate

Address: Operational Base + offset (0x1090)

Bit	Attr	Reset Value	Description
31:5	RO	0x0000000	reserved
4	RW	0x0	cbuf_cs_disable_clkgate Cache auto gating. 1'd0: Auto clock gate is enabled; 1'd1: Disable CBUF clock auto gate.
3	RO	0x0	reserved
2	RW	0x0	csc_disable_clkgate Sequence scan auto gating. 1'd0: Auto clock gate is enabled; 1'd1: Disable csc block clock gate.
1	RW	0x0	cna_weight_disable_clkgate Weight fetch auto gating. 1'd0: Auto clock gate is enabled; 1'd1: Disable weight block clock gate.

Bit	Attr	Reset Value	Description
0	RW	0x0	cna_feature_disable_clkgate Feature fetch auto gating. 1'd0: Auto clock gate is enabled; 1'd1: Disable feature block clock gate.

RKNN_cna_dcomp_ctrl

Address: Operational Base + offset (0x1100)

Bit	Attr	Reset Value	Description
31:4	RO	0x00000000	reserved
3	RW	0x0	wt_dec_bypass Bypass weight decompress.
2:0	RW	0x0	decomp_control Control register for weight decompress.

RKNN_cna_dcomp_regnum

Address: Operational Base + offset (0x1104)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_regnum Weight decompress register number.

RKNN_cna_dcomp_addr0

Address: Operational Base + offset (0x1110)

Bit	Attr	Reset Value	Description
31:4	RW	0x00000000	decompress_addr0 Base address of the weight.
3:0	RO	0x0	reserved

RKNN_cna_dcomp_amount0

Address: Operational Base + offset (0x1140)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount0 Amount of the weight decompress for the 0 decompress.

RKNN_cna_dcomp_amount1

Address: Operational Base + offset (0x1144)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount1 Amount of the weight decompress for the 1 decompress.

RKNN_cna_dcomp_amount2

Address: Operational Base + offset (0x1148)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount2 Amount of the weight decompress for the 2 decompress.

RKNN_cna_dcomp_amount3

Address: Operational Base + offset (0x114C)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount3 Amount of the weight decompress for the 3 decompress.

RKNN_cna_dcomp_amount4

Address: Operational Base + offset (0x1150)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount4 Amount of the weight decompress for the 4 decompress.

RKNN_cna_dcomp_amount5

Address: Operational Base + offset (0x1154)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount5 Amount of the weight decompress for the 5 decompress.

RKNN_cna_dcomp_amount6

Address: Operational Base + offset (0x1158)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount6 Amount of the weight decompress for the 6 decompress.

RKNN_cna_dcomp_amount7

Address: Operational Base + offset (0x115C)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount7 Amount of the weight decompress for the 7 decompress.

RKNN_cna_dcomp_amount8

Address: Operational Base + offset (0x1160)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount8 Amount of the weight decompress for the 8 decompress.

RKNN_cna_dcomp_amount9

Address: Operational Base + offset (0x1164)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount9 Amount of the weight decompress for the 9 decompress.

RKNN_cna_dcomp_amount10

Address: Operational Base + offset (0x1168)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount10 Amount of the weight decompress for the 10 decompress.

RKNN_cna_dcomp_amount11

Address: Operational Base + offset (0x116C)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount11 Amount of the weight decompress for the 11 decompress.

RKNN_cna_dcomp_amount12

Address: Operational Base + offset (0x1170)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount12 Amount of the weight decompress for the 12 decompress.

RKNN_cna_dcomp_amount13

Address: Operational Base + offset (0x1174)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount13 Amount of the weight decompress for the 13 decompress.

RKNN_cna_dcomp_amount14

Address: Operational Base + offset (0x1178)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount14 Amount of the weight decompress for the 14 decompress.

RKNN_cna_dcomp_amount15

Address: Operational Base + offset (0x117C)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	dcomp_amount15 Amount of the weight decompress for the 15 decompress.

RKNN_cna_cvt_con5

Address: Operational Base + offset (0x1180)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	per_channel_cvt_en convert enable. Per channel enable CVT function. Int 4 has 32 channels in total for 128 bits. Int 8 16 channel...

RKNN_cna_pad_con1

Address: Operational Base + offset (0x1184)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pad_value Pad value.

RKNN core s status

Address: Operational Base + offset (0x3000)

Bit	Attr	Reset Value	Description
31:18	RO	0x0000	reserved
17:16	RO	0x0	status_1 Executer 1 status. 2'd0: Executer 1 is in idle state; 2'd1: Executer 1 is operating; 2'd2: Executer 1 is operating, executer 1 is waiting to operate; 2'd3: Reserved.
15:2	RO	0x0000	reserved
1:0	RO	0x0	status_0 Executer 0 status. 2'd0: Executer 0 is in idle state; 2'd1: Executer 0 is operating; 2'd2: Executer 0 is operating, executer 1 is waiting to operate; 2'd3: Reserved.

RKNN core s pointer

Address: Operational Base + offset (0x3004)

Bit	Attr	Reset Value	Description
31:17	RO	0x0000	reserved
16	RO	0x0	executer Which register group to be used. 1'd0: Executer group 0; 1'd1: Executer group 1.
15:6	RO	0x000	reserved
5	W1C	0x0	executer_pp_clear Clear executer group pointer. Set this bit to 1 to clear pointer to 0.
4	W1C	0x0	pointer_pp_clear Clear register group pointer. Set this bit to 1 to clear pointer to 0.
3	RW	0x0	pointer_pp_mode Register group ping-pong mode. 1'd0: Pointer ping-pong by executer; eg. if current executer is 0, next pointer will toggle to 1; 1'd1: Pointer ping-pong by pointer; eg. if current pointer is 0, next pointer will toggle to 1.

Bit	Attr	Reset Value	Description
2	RW	0x0	executer_pp_en Executer group ping-pong enable. 1'd0: Disable; 1'd1: Enable.
1	RW	0x0	pointer_pp_en Register group ping-pong enable. 1'd0: Disable; 1'd1: Enable.
0	RW	0x0	pointer Which register group ready to be set. 1'd0: Register group 0; 1'd1: Register group 1.

RKNN core operation enable

Address: Operational Base + offset (0x3008)

Bit	Attr	Reset Value	Description
31:1	RO	0x00000000	reserved
0	RW	0x0	op_en Set this register will trigger core block operate. This register and after this are all shadowed for ping-pong operation. 1'd0: Disable; 1'd1: Enable.

RKNN core mac gating

Address: Operational Base + offset (0x300C)

Bit	Attr	Reset Value	Description
31:27	RO	0x00	reserved
26:0	RW	0x7800800	slcg_op_en Soft clock gating signals.

RKNN core misc cfg

Address: Operational Base + offset (0x3010)

Bit	Attr	Reset Value	Description
31:20	RO	0x000	reserved
19:14	RW	0x00	soft_gating Accumulate soft gating signal.
13:11	RO	0x0	reserved

Bit	Attr	Reset Value	Description
10:8	RW	0x0	proc_precision Process precision 3'd0: Input precision is int 8; 3'd1: Input data precision is int 16; 3'd2: Input data precision is float 16; 3'd3: Input data precision is bfloat 16; 3'd4: Reserved; 3'd5: Reserved; 3'd6: Input data precision is int4; 3'd7: Input data precision is tf32.
7:2	RO	0x00	reserved
1	RW	0x0	dw_en Depthwise enable 1'd0: Disable; 1'd1: Depthwise mode enable.
0	RW	0x0	qd_en Quantify feature data calculate enable 1'd0: Disable; 1'd1: Enable.

RKNN core dataout size 0

Address: Operational Base + offset (0x3014)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	dataout_height Data out height Data height after activation.
15:0	RW	0x0000	dataout_width Data out width. Data width after activation.

RKNN core dataout size 1

Address: Operational Base + offset (0x3018)

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	reserved
15:0	RW	0x0000	dataout_channel Data out channel number Data channel number after activation.

RKNN core clip truncate

Address: Operational Base + offset (0x301C)

Bit	Attr	Reset Value	Description
31:7	RO	0x00000000	reserved

Bit	Attr	Reset Value	Description
6	RW	0x0	round_type Rounding type 1'b0: Odd in, even not; 1'b1: Round-up 0.5 to 1.
5	RO	0x0	reserved
4:0	RW	0x00	clip_truncate Truncate bits number.

RKNN_dpu_s_status

Address: Operational Base + offset (0x4000)

Bit	Attr	Reset Value	Description
31:18	RO	0x0000	reserved
17:16	RO	0x0	status_1 Executer 1 status. 2'd0: Executer 1 is in idle state; 2'd1: Executer 1 is operating; 2'd2: Executer 1 is operating, executer 1 is waiting to operate; 2'd3: Reserved.
15:2	RO	0x0000	reserved
1:0	RO	0x0	status_0 Executer 0 status. 2'd0: Executer 0 is in idle state; 2'd1: Executer 0 is operating; 2'd2: Executer 0 is operating, executer 1 is waiting to operate; 2'd3: Reserved.

RKNN_dpu_s_pointer

Address: Operational Base + offset (0x4004)

Bit	Attr	Reset Value	Description
31:17	RO	0x0000	reserved
16	RO	0x0	executer Which register group to be used. 1'd0: Executer group 0; 1'd1: Executer group 1.
15:6	RO	0x000	reserved
5	W1C	0x0	executer_pp_clear Clear executer group pointer. Set this bit to 1 to clear pointer to 0.
4	W1C	0x0	pointer_pp_clear Clear register group pointer. Set this bit to 1 to clear pointer to 0.

Bit	Attr	Reset Value	Description
3	RW	0x0	pointer_pp_mode Register group ping-pong mode. 1'd0: Pointer ping-pong by executer; eg. if current executer is 0, next pointer will toggle to 1; 1'd1: Pointer ping-pong by pointer; eg. if current pointer is 0, next pointer will toggle to 1.
2	RW	0x0	executer_pp_en Executer group ping-pong enable. 1'd0: Disable; 1'd1: Enable.
1	RW	0x0	pointer_pp_en Register group ping-pong enable. 1'd0: Disable; 1'd1: Enable.
0	RW	0x0	pointer Which register group ready to be set. 1'd0: Register group 0; 1'd1: Register group 1.

RKNN dpu operation enable

Address: Operational Base + offset (0x4008)

Bit	Attr	Reset Value	Description
31:1	RO	0x00000000	reserved
0	RW	0x0	op_en Set this register will trigger DPU block operate. This register and after this are all shadowed for ping-pong operation. 1'd0: Disable; 1'd1: Enable.

RKNN dpu feature mode cfg

Address: Operational Base + offset (0x400C)

Bit	Attr	Reset Value	Description
31	RW	0x0	comb_use Combine use, same as DPU_RDMA comb_use[0].
30	RW	0x0	tp_en If enable transpose.
29:26	RW	0x0	rgp_type Regroup type. 4'd0: Cut all input (128bit); 4'd1: Cut 4bit; 4'd2: Cut 8bit; 4'd3: Cut 16bit; 4'd4: Cut 32bit; 4'd5: Cut 64bit.

Bit	Attr	Reset Value	Description
25	RW	0x0	nonalign If non-align mode is enabled. If the output data flow is the same as the input data flow, this mode can be used.
24:9	RW	0x0000	surf_len In non-align mode, how many 8bytes to be stored.
8:5	RW	0x0	burst_len Burst length. 4'd3: Burst4; 4'd7: Burst8; 4'd15: Burst16.
4:3	RW	0x0	conv_mode Convolution mode. 2'd0: Normal convolution mode; 2'd1: Reserved; 2'd2: Reserved; 2'd3: Depthwise convolution mode.
2:1	RW	0x0	output_mode Where the DPU core output goes. [0]: If the output goes to PPU, high is active; [1]: If the output goes to outside, high is active.
0	RW	0x0	flying_mode Flying mode enable. 1'd0: DPU core main data is from convolution output; 1'd1: DPU core main data is from MRDMA;

RKNN_dpu_data_format

Address: Operational Base + offset (0x4010)

Bit	Attr	Reset Value	Description
31:29	RW	0x0	out_precision Output precision. 3'd0: Integer 8bit; 3'd1: Integer 16bit; 3'd2: Float point 16bit; 3'd3: Bfloat 16bit; 3'd4: Integer 32bit; 3'd5: Float point 32bit; 3'd6: Integer 4bit.

Bit	Attr	Reset Value	Description
28:26	RW	0x0	in_precision Input precision same with DPU_RDMA. 3'd0: Integer 8bit; 3'd1: Integer 16bit; 3'd2: Float point 16bit; 3'd3: Bfloat 16bit; 3'd4: Integer 32bit; 3'd5: Float point 32bit; 3'd6: Integer 4bit.
25:16	RW	0x000	ew_truncate_neg Shift value in EW core for negative data.
15:10	RW	0x00	bn_mul_shift_value_neg Shift value in BN core for negative data.
9:4	RW	0x00	bs_mul_shift_value_neg Shift value in BS core for negative data.
3	RW	0x0	mc_surf_out How many surfaces serial the DPU output. 1'd0: Output feature obey the rule of 16byte align for one pixel; 1'd1: Output feature can output 2 surface serial or 4 surf serials.
2:0	RW	0x0	proc_precision Proc precision. 3'd0: Integer 8bit; 3'd1: Integer 16bit; 3'd2: Float point 16bit; 3'd3: Bfloat 16bit; 3'd4: Integer 32bit; 3'd5: Float point 32bit; 3'd6: Integer 4bit.

RKNN_dpu_offset_pend

Address: Operational Base + offset (0x4014)

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	reserved
15:0	RW	0x0000	offset_pend What value the extra channel be set.

RKNN_dpu_dst_base_addr

Address: Operational Base + offset (0x4020)

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	dst_base_addr Destination base address.
3:0	RO	0x0	reserved

RKNN_dpu_dst_surf_stride

Address: Operational Base + offset (0x4024)

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	dst_surf_stride Output shape surface stride.
3:0	RO	0x0	reserved

RKNN dpu data cube width

Address: Operational Base + offset (0x4030)

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	reserved
12:0	RW	0x0000	width Width of the input cube.

RKNN dpu data cube height

Address: Operational Base + offset (0x4034)

Bit	Attr	Reset Value	Description
31:25	RO	0x00	reserved
24:22	RW	0x0	minmax_ctl Configuration of the minmax op. [0]: Enable minmax op; [1]: Minmax type; [2]: Probability only.
21:13	RO	0x000	reserved
12:0	RW	0x0000	height Height of the input cube.

RKNN dpu data cube notch addr

Address: Operational Base + offset (0x4038)

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	notch_addr_1 How many pixels from the end of width to the end of the shape line end.
15:13	RO	0x0	reserved
12:0	RW	0x0000	notch_addr_0 How many pixels from the end of width to the end of the shape line end.

RKNN dpu data cube channel

Address: Operational Base + offset (0x403C)

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	orig_channel Original output channel.
15:13	RO	0x0	reserved

Bit	Attr	Reset Value	Description
12:0	RW	0x0000	channel Cube channel.

RKNN_dpu_bs_cfg

Address: Operational Base + offset (0x4040)

Bit	Attr	Reset Value	Description
31:20	RO	0x000	reserved
19:16	RW	0x0	bs_alu_algo BS core ALU op type. 4'd0: Reserved; 4'd1: Reserved; 4'd2: Add 4'd3: Reserved; 4'd4: Minus; 4'd5: Reserved; 4'd6: Reserved; 4'd7: Reserved; 4'd8: Reserved.
15:9	RO	0x00	reserved
8	RW	0x0	bs_alu_src Where the ALU operand from. 1'd0: From configure register; 1'd1: From outside.
7	RW	0x0	bs_relux_en If enable RELUX. 1'd0: Disable; 1'd1: Enable.
6	RW	0x0	bs_relu_bypass If bypass BS core RELU op. 1'd0: Do not bypass; 1'd1: Bypass.
5	RW	0x0	bs_mul_prelu If enable MUL PRELU. 1'd0: Disable; 1'd1: Enable.
4	RW	0x0	bs_mul_bypass If bypass BS core MUL op. 1'd0: Do not bypass; 1'd1: Bypass.
3:2	RO	0x0	reserved
1	RW	0x0	bs_alu_bypass If bypass BS core ALU op. 1'd0: Do not bypass; 1'd1: Bypass.

Bit	Attr	Reset Value	Description
0	RW	0x0	bs_bypass If bypass BS core. 1'd0: Do not bypass bs core; 1'd1: Bypass bs core.

RKNN_dpu_bs_alu_cfg

Address: Operational Base + offset (0x4044)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	bs_alu_operand Bs core ALU operand.

RKNN_dpu_bs_mul_cfg

Address: Operational Base + offset (0x4048)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	bs_mul_operand BS core MUL operand.
15:14	RO	0x0	reserved
13:8	RW	0x00	bs_mul_shift_value Shift value in BS core for positive data.
7:2	RO	0x00	reserved
1	RW	0x0	bs_truncate_src Where the shift value from. 1'd0: From configure register; 1'd1: From outside.
0	RW	0x0	bs_mul_src Where the MUL operand from. 1'd0: From configure register; 1'd1: From outside.

RKNN_dpu_bs_relux_cmp_value

Address: Operational Base + offset (0x404C)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	bs_relux_cmp_dat Value of the RELUX compare with.

RKNN_dpu_bs_ow_cfg

Address: Operational Base + offset (0x4050)

Bit	Attr	Reset Value	Description
31:28	RW	0x0	rgp_cnter Regroup counter. 4'd0: Select all data; 4'd1: Select 1 from every 2; 4'd2: Select 1 from every 4; 4'd3: Select 1 from every 8; Else: Reserved.

Bit	Attr	Reset Value	Description
27	RW	0x0	tp_org_en If enable original transpose. 1'd0: Disable; 1'd1: Enable.
26:11	RO	0x0000	reserved
10:8	RW	0x0	size_e_2 How many 8 channels in a row the last output line (minus 1).
7:5	RW	0x0	size_e_1 How many 8 channels in a row the middle output line (minus 1).
4:2	RW	0x0	size_e_0 How many 8 channels in a row the first output line (minus 1).
1	RW	0x0	od_bypass If bypass CPEND. 1'd0: Do not bypass; 1'd1: Bypass.
0	RW	0x0	ow_src Where the CPEND operand from. 1'd0: From configure register; 1'd1: From outside.

RKNN_dpu_bs_ow_op

Address: Operational Base + offset (0x4054)

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	reserved
15:0	RW	0x0000	ow_op CPEND operand.

RKNN_dpu_wdma_size_0

Address: Operational Base + offset (0x4058)

Bit	Attr	Reset Value	Description
31:28	RO	0x0	reserved
27	RW	0x0	tp_precision Transpose precision. 1'd0: 8bit; 1'd1: 16bit.
26:16	RW	0x000	size_c_wdma Size_c for DPU_WDMA.
15:13	RO	0x0	reserved
12:0	RW	0x0000	channel_wdma Channel for DPU_WDMA.

RKNN_dpu_wdma_size_1

Address: Operational Base + offset (0x405C)

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	height_wdma Height for DPU_WDMA.
15:13	RO	0x0	reserved
12:0	RW	0x0000	width_wdma Width for DPU_WDMA.

RKNN_dpu_bn_cfg

Address: Operational Base + offset (0x4060)

Bit	Attr	Reset Value	Description
31:20	RO	0x000	reserved
19:16	RW	0x0	bn_alu_algo BS core ALU op type. 4'd0: Reserved; 4'd1: Reserved; 4'd2: Add 4'd3: Reserved; 4'd4: Minus; 4'd5: Reserved; 4'd6: Reserved; 4'd7: Reserved; 4'd8: Reserved.
15:9	RO	0x00	reserved
8	RW	0x0	bn_alu_src Where the ALU operand from 1'd0: From configure register; 1'd1: From outside.
7	RW	0x0	bn_relux_en If enable RELUX. 1'd0: Disable; 1'd1: Enable.
6	RW	0x0	bn_relu_bypass If bypass BN core RELU op. 1'd0: Do not bypass; 1'd1: Bypass.
5	RW	0x0	bn_mul_prelu If enable MUL PRELU. 1'd0: Disable; 1'd1: Enable.
4	RW	0x0	bn_mul_bypass If bypass BN core MUL op. 1'd0: Do not bypass; 1'd1: Bypass.
3:2	RO	0x0	reserved

Bit	Attr	Reset Value	Description
1	RW	0x0	bn_alu_bypass If bypass BN core ALU op. 1'd0: Do not bypass; 1'd1: Bypass.
0	RW	0x0	bn_bypass If bypass BN core. 1'd0: Do not bypass BN core; 1'd1: Bypass BN core.

RKNN_dpu_bn_alu_cfg

Address: Operational Base + offset (0x4064)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	bn_alu_operand BN core ALU operand.

RKNN_dpu_bn_mul_cfg

Address: Operational Base + offset (0x4068)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	bn_mul_operand BN core MUL operand.
15:14	RO	0x0	reserved
13:8	RW	0x00	bn_mul_shift_value Shift value in BN core for positive data.
7:2	RO	0x00	reserved
1	RW	0x0	bn_truncate_src Where the shift value from 1'd0: From configure register; 1'd1: From outside.
0	RW	0x0	bn_mul_src Where the MUL operand from. 1'd0: From configure register; 1'd1: From outside.

RKNN_dpu_bn_relux_cmp_value

Address: Operational Base + offset (0x406C)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	bn_relux_cmp_dat RELUX compare data in BN core.

RKNN_dpu_ew_cfg

Address: Operational Base + offset (0x4070)

Bit	Attr	Reset Value	Description
31	RW	0x0	ew_cvt_type Convert type of EW input convert when if 0.5 1'd0: Mul first; 1'd1: Add first.
30	RW	0x0	ew_cvt_round Rounding type of EW input convert when if 0.5 1'd0: If the integer is odd, carry 1; 1'd1: Carry 1 no matter what the integer is.
29:28	RW	0x0	ew_data_mode Data mode of the data from ERDMA.
27:24	RO	0x0	reserved
23:22	RW	0x0	edata_size Data size of the cube from ERDMA. 2'd0: 4bit; 2'd1: 8bit; 2'd2: 16bit; 2'd3: 32bit.
21	RW	0x0	ew_equal_en Min max equal enable. 1'd0: Disable; 1'd1: Enable.
20	RW	0x0	ew_binary_en Min max binary enable. 1'd0: Disable; 1'd1: Enable.
19:16	RW	0x0	ew_alu_algo EW core ALU op type. 4'd0: Max; 4'd1: Min; 4'd2: Add; 4'd3: Div; 4'd4: Minus; 4'd5: Abs; 4'd6: Neg; 4'd7: Floor; 4'd8: Ceil.
15:11	RO	0x00	reserved
10	RW	0x0	ew_relux_en If enable RELUX. 1'd0: Disable; 1'd1: Enable.
9	RW	0x0	ew_relu_bypass If bypass EW core RELU op. 1'd0: Do not bypass; 1'd1: Bypass.

Bit	Attr	Reset Value	Description
8	RW	0x0	ew_op_cvt_bypass If bypass EW input converter. 1'd0: Do not bypass; 1'd1: Bypass.
7	RW	0x0	ew_lut_bypass If bypass LUT. 1'd0: Do not bypass; 1'd1: Bypass.
6	RW	0x0	ew_op_src Where the operand from 1'd0: From configure register; 1'd1: From outside.
5	RW	0x0	ew_mul_prelu If enable MUL PRELU. 1'd0: Disable; 1'd1: Enable.
4:3	RO	0x0	reserved
2	RW	0x0	ew_op_type Operator type. 1'd0: ALU; 1'd1: MUL.
1	RW	0x0	ew_op_bypass If bypass EW core ALU and MUL op. 1'd0: Do not bypass; 1'd1: Bypass.
0	RW	0x0	ew_bypass If bypass EW core. 1'd0: Do not bypass EW core; 1'd1: Bypass EW core.

RKNN_dpu_ew_cvt_offset_value

Address: Operational Base + offset (0x4074)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ew_op_cvt_offset EW convert offset.

RKNN_dpu_ew_cvt_scale_value

Address: Operational Base + offset (0x4078)

Bit	Attr	Reset Value	Description
31:22	RW	0x000	ew_truncate EW core shift value.
21:16	RW	0x00	ew_op_cvt_shift EW convert shift value.
15:0	RW	0x0000	ew_op_cvt_scale EW convert scale.

RKNN dpu ew relax cmp value

Address: Operational Base + offset (0x407C)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ew_relux_cmp_dat EW RELUX compare data.

RKNN dpu out cvt offset

Address: Operational Base + offset (0x4080)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	out_cvt_offset Offset of the output converter.

RKNN dpu out cvt scale

Address: Operational Base + offset (0x4084)

Bit	Attr	Reset Value	Description
31:17	RO	0x0000	reserved
16	RW	0x0	fp32tofp16_en If enable output from fp32 to fp16. 1'd0: Disable; 1'd1: Enable.
15:0	RW	0x0000	out_cvt_scale Scale of the output converter.

RKNN dpu out cvt shift

Address: Operational Base + offset (0x4088)

Bit	Attr	Reset Value	Description
31	RW	0x0	cvt_type Convert type of out convert when if 0.5. 1'd0: MUL first; 1'd1: ALU first.
30	RW	0x0	cvt_round Rounding type of out convert when if 0.5. 1'd0: If the integer is odd, carry 1; 1'd1: Carry 1 no matter what the integer is.
29:20	RO	0x000	reserved
19:12	RW	0x00	minus_exp Minus exp of out CVT.
11:0	RW	0x000	out_cvt_shift Shift of the output converter.

RKNN dpu ew op value 0

Address: Operational Base + offset (0x4090)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ew_operand_0 The 1st EW operand for EW core op.

RKNN dpu ew op value 1

Address: Operational Base + offset (0x4094)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ew_operand_1 The 2nd EW operand for EW core op.

RKNN dpu ew op value 2

Address: Operational Base + offset (0x4098)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ew_operand_2 the 3thd EW operand for EW core op.

RKNN dpu ew op value 3

Address: Operational Base + offset (0x409C)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ew_operand_3 the 4th EW operand for EW core op.

RKNN dpu ew op value 4

Address: Operational Base + offset (0x40A0)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ew_operand_4 the 5th EW operand for EW core op.

RKNN dpu ew op value 5

Address: Operational Base + offset (0x40A4)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ew_operand_5 The 6th EW operand for EW core op.

RKNN dpu ew op value 6

Address: Operational Base + offset (0x40A8)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ew_operand_6 The 7th EW operand for EW core op.

RKNN dpu ew op value 7

Address: Operational Base + offset (0x40AC)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ew_operand_7 The 8th EW operand for EW core op.

RKNN dpu surface add

Address: Operational Base + offset (0x40C0)

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	surf_add How many surfaces in a row.
3:0	RO	0x0	reserved

RKNN_dpu_lut_access_cfg

Address: Operational Base + offset (0x4100)

Bit	Attr	Reset Value	Description
31:18	RO	0x0000	reserved
17	RW	0x0	lut_access_type Access type. 1'd0: Read; 1'd1: Write.
16	RW	0x0	lut_table_id Access Id. 1'd0: LE LUT; 1'd1: LO LUT.
15:10	RO	0x00	reserved
9:0	RW	0x000	lut_addr Access address.

RKNN_dpu_lut_access_data

Address: Operational Base + offset (0x4104)

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	reserved
15:0	RW	0x0000	lut_access_data Configuration of LUT access data.

RKNN_dpu_lut_cfg

Address: Operational Base + offset (0x4108)

Bit	Attr	Reset Value	Description
31:8	RO	0x000000	reserved
7	RW	0x0	lut_cal_sel LUT calculate sel. Only useful when lut_expand_en is 1.
6	RW	0x0	lut_hybrid_priority LUT hybrid flow priority. 1'd0: LE LUT; 1'd1: LO LUT.
5	RW	0x0	lut_oflow_priority Priority when over flow happened. 1'd0: LE LUT; 1'd1: LO LUT.

Bit	Attr	Reset Value	Description
4	RW	0x0	lut_uflow_priority Priority when under flow happened. 1'd0: LE LUT; 1'd1: LO LUT.
3:2	RW	0x0	lut_lo_le_mux LO LUT and LE LUT mux.
1	RW	0x0	lut_expand_en If expand two small LUT to a larger LUT. 1'd0: Disable; 1'd1: Enable.
0	RW	0x0	lut_road_sel LUT road sel. 1'd0: 1st; 1'd1: 2nd.

RKNN_dpu_lut_info

Address: Operational Base + offset (0x410C)

Bit	Attr	Reset Value	Description
31:24	RO	0x00	reserved
23:16	RW	0x00	lut_lo_index_select LUT LO index selected. Used in index generator, choose some bits to be the index.
15:8	RW	0x00	lut_le_index_select LUT LE index selected. Used in index generator, choose some bits to be the index.
7:0	RO	0x00	reserved

RKNN_dpu_lut_le_start

Address: Operational Base + offset (0x4110)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	lut_le_start LE LUT start point.

RKNN_dpu_lut_le_end

Address: Operational Base + offset (0x4114)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	lut_le_end LE LUT end point.

RKNN_dpu_lut_lo_start

Address: Operational Base + offset (0x4118)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	lut_lo_start LO LUT start point.

RKNN dpu lut lo end

Address: Operational Base + offset (0x411C)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	lut_lo_end LO LUT end point.

RKNN dpu lut le slope scale

Address: Operational Base + offset (0x4120)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	lut_le_slope_oflow_scale LE LUT slope scale if over flow.
15:0	RW	0x0000	lut_le_slope_uflow_scale LE LUT slope scale if under flow.

RKNN dpu lut le slope shift

Address: Operational Base + offset (0x4124)

Bit	Attr	Reset Value	Description
31:10	RO	0x000000	reserved
9:5	RW	0x00	lut_le_slope_oflow_shift LE LUT slope shift if over flow.
4:0	RW	0x00	lut_le_slope_uflow_shift LE LUT slope shift if under flow.

RKNN dpu lut lo slope scale

Address: Operational Base + offset (0x4128)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	lut_lo_slope_oflow_scale LO LUT slope scale if over flow.
15:0	RW	0x0000	lut_lo_slope_uflow_scale LO LUT slope scale if under flow.

RKNN dpu lut lo slope shift

Address: Operational Base + offset (0x412C)

Bit	Attr	Reset Value	Description
31:10	RO	0x000000	reserved
9:5	RW	0x00	lut_lo_slope_oflow_shift LO LUT slope shift if over flow.
4:0	RW	0x00	lut_lo_slope_uflow_shift LO LUT slope shift if under flow.

RKNN dpu rdma s status

Address: Operational Base + offset (0x5000)

Bit	Attr	Reset Value	Description
31:18	RO	0x0000	reserved

Bit	Attr	Reset Value	Description
17:16	RO	0x0	status_1 Executer 1 status. 2'd0: Executer 1 is in idle state; 2'd1: Executer 1 is operating; 2'd2: Executer 1 is operating, executer 1 is waiting to operate; 2'd3: Reserved.
15:2	RO	0x0000	reserved
1:0	RO	0x0	status_0 Executer 0 status. 2'd0: Executer 0 is in idle state; 2'd1: Executer 0 is operating; 2'd2: Executer 0 is operating, executer 1 is waiting to operate; 2'd3: Reserved.

RKNN_dpu_rdma_s_pointer

Address: Operational Base + offset (0x5004)

Bit	Attr	Reset Value	Description
31:17	RO	0x0000	reserved
16	RO	0x0	executer Which register group to be used. 1'd0: Executer group 0; 1'd1: Executer group 1.
15:6	RO	0x000	reserved
5	W1C	0x0	executer_pp_clear Clear executer group pointer. Set this bit to 1 to clear pointer to 0.
4	W1C	0x0	pointer_pp_clear Clear register group pointer. Set this bit to 1 to clear pointer to 0.
3	RW	0x0	pointer_pp_mode Register group ping-pong mode. 1'd0: Pointer ping-pong by executer; eg. if current executer is 0, next pointer will toggle to 1; 1'd1: Pointer ping-pong by pointer; eg. if current pointer is 0, next pointer will toggle to 1.
2	RW	0x0	executer_pp_en Executer group ping-pong enable. 1'd0: Disable; 1'd1: Enable.
1	RW	0x0	pointer_pp_en Register group ping-pong enable. 1'd0: Disable; 1'd1: Enable.

Bit	Attr	Reset Value	Description
0	RW	0x0	pointer Which register group ready to be set. 1'd0: Register group 0; 1'd1: Register group 1.

RKNN_dpu_rdma_operation_enable

Address: Operational Base + offset (0x5008)

Bit	Attr	Reset Value	Description
31:1	RO	0x00000000	reserved
0	RW	0x0	op_en Set this register will trigger DPU_RDMA block operate. This register and after this are all shadowed for ping-pong operation. 1'd0: Disable; 1'd1: Enable.

RKNN_dpu_rdma_data_cube_width

Address: Operational Base + offset (0x500C)

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	reserved
12:0	RW	0x0000	width Input feature width (need to minus 1).

RKNN_dpu_rdma_data_cube_height

Address: Operational Base + offset (0x5010)

Bit	Attr	Reset Value	Description
31:29	RO	0x0	reserved
28:16	RW	0x0000	ew_line_notch_addr Line notch of EW.
15:13	RO	0x0	reserved
12:0	RW	0x0000	height Input feature height (need to minus 1).

RKNN_dpu_rdma_data_cube_channel

Address: Operational Base + offset (0x5014)

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	reserved
12:0	RW	0x0000	channel Input feature channel (need to minus 1).

RKNN_dpu_rdma_src_base_addr

Address: Operational Base + offset (0x5018)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_base_addr Fly mode source address.

RKNN_dpu_rdma_brdma_cfg

Address: Operational Base + offset (0x501C)

Bit	Attr	Reset Value	Description
31:5	RO	0x00000000	reserved
4:1	RW	0x0	brdma_data_use How many data type need to be read. [0]: If read ALU operand, set 1 to enable; [1]: If read CPEND operand, set 1 to enable; [2]: If read MUL operand, set 1 to enable; [3]: If read TRT operand, set 1 to enable.
0	RO	0x0	reserved

RKNN_dpu_rdma_bs_base_addr

Address: Operational Base + offset (0x5020)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	bs_base_addr Base address to read BS ALU, BS CPEND, BS MUL operand.

RKNN_dpu_rdma_nrdma_cfg

Address: Operational Base + offset (0x5028)

Bit	Attr	Reset Value	Description
31:5	RO	0x00000000	reserved
4:1	RW	0x0	nrdma_data_use How many data type need to be read. [0]: If read ALU operand, set 1 to enable; [1]: If read CPEND operand, set 1 to enable, (tie to 0, cause BN do not have CPEND); [2]: If read MUL operand, set 1 to enable; [3]: If read TRT operand, set 1 to enable.
0	RO	0x0	reserved

RKNN_dpu_rdma_bn_base_addr

Address: Operational Base + offset (0x502C)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	bn_base_addr Base address to read BN ALU, BN MUL operand.

RKNN_dpu_rdma_erdma_cfg

Address: Operational Base + offset (0x5034)

Bit	Attr	Reset Value	Description
31:30	RW	0x0	erdma_data_mode If the read data is per channel or per pixel 2'd0: Per channel; 2'd1: Per pixel; 2'd2: Per channel by pixel; 2'd3: Reserved.
29	RW	0x0	erdma_surf_mode Surface mode of the EW cube. 1'd0: 1 surface series; 1'd1: 2 surface series.
28	RW	0x0	erdma_nonalign If read the EW cube in non-align mode. 1'd0: Do not use non-align mode; 1'd1: Use non-align mode.
27:4	RO	0x000000	reserved
3:2	RW	0x0	erdma_data_size What the precision of the cube that the ERDMA read. 2'd0: 4bit; 2'd1: 8bit; 2'd2: 16bit; 2'd3: 32bit.
1	RW	0x0	ov4k_bypass Separate the burst command of over 4k to 2 independent burst commands. 1'd0: Enable this feature; 1'd1: Bypass this feature.
0	RW	0x0	erdma_disable If disable ERDMA. 1'd0: Do not disable ERDMA; 1'd1: Disable ERDMA.

RKNN_dpu_rdma_ew_base_addr

Address: Operational Base + offset (0x5038)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	ew_base_addr Base address to read EW operand.

RKNN_dpu_rdma_ew_surf_stride

Address: Operational Base + offset (0x5040)

Bit	Attr	Reset Value	Description
31:4	RW	0x00000000	ew_surf_stride The surface stride of the element wise feature map; If erdma_data_mode is per channel, it need set to be 1.
3:0	RO	0x0	reserved

RKNN dpu rdma feature mode cfg

Address: Operational Base + offset (0x5044)

Bit	Attr	Reset Value	Description
31:18	RO	0x0000	reserved
17:15	RW	0x0	in_precision Input data precision. 3'd0: Integer 8bit; 3'd1: Integer 16bit; 3'd2: Gloat point 16bit; 3'd3: Bfloat 16bit; 3'd4: Integer 32bit; 3'd5: Float point 32bit; 3'd6: Integer 4bit.
14:11	RW	0x0	burst_len Burst length. 4'd3: Burst4; 4'd7: Burst8; 4'd15: Burst16.
10:8	RW	0x0	comb_use [0]: If enable MRDMA and ERDMA to read the same data; [1]: Read the data to MRDMA; [2]: Read the data to ERDMA.
7:5	RW	0x0	proc_precision Process precision. 3'd0: Integer 8bit; 3'd1: Integer 16bit; 3'd2: Float point 16bit; 3'd3: Bfloat 16bit; 3'd4: Integer 32bit; 3'd5: Float point 32bit; 3'd6: Integer 4bit.
4	RW	0x0	mrdma_disable If disable MRDMA. 1'd0: Do not disable MRDMA; 1'd1: Disable MRDMA.
3	RW	0x0	mrdma_fp16tofp32_en If enable DPU input from fp16 to fp32.
2:1	RW	0x0	conv_mode Convolution mode. 2'd0: Dc; 2'd1: Reserved; 2'd2: Reserved; 2'd3: Depthwise.

Bit	Attr	Reset Value	Description
0	RW	0x0	flying_mode Flying mode enable. 1'd0: DPU core main data is from convolution output; 1'd1: DPU core main data is from MRDMA.

RKNN dpu rdma src dma cfg

Address: Operational Base + offset (0x5048)

Bit	Attr	Reset Value	Description
31:19	RW	0x0000	line_notch_addr How many pixels from the end of width end to the shape feature line end.
18:14	RO	0x00	reserved
13	RW	0x0	pooling_method Pooling method 1'd0: Average pooling (up sampling can use this mode); 1'd1: Min or max pooling.
12	RW	0x0	unpooling_en If enable un-pooling.
11:9	RW	0x0	kernel_stride_height Un-pooling kernel stride height (minus 1).
8:6	RW	0x0	kernel_stride_width Un-pooling kernel stride width (minus 1).
5:3	RW	0x0	kernel_height Un-pooling kernel height (minus 1).
2:0	RW	0x0	kernel_width Un-pooling kernel width (minus 1).

RKNN dpu rdma surf notch

Address: Operational Base + offset (0x504C)

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	surf_notch_addr How many pixels from the end of this process feature map to the end of the shape feature map.
3:0	RO	0x0	reserved

RKNN dpu rdma pad cfg

Address: Operational Base + offset (0x5064)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	pad_value Pad value.
15:7	RO	0x000	reserved
6:4	RW	0x0	pad_top Un-pooling top pad.
3	RO	0x0	reserved

Bit	Attr	Reset Value	Description
2:0	RW	0x0	pad_left Un-pooling left pad.

RKNN_dpu_rdma_weight

Address: Operational Base + offset (0x5068)

Bit	Attr	Reset Value	Description
31:24	RW	0x00	e_weight The arbiter weight for ERDMA.
23:16	RW	0x00	n_weight The arbiter weight for NRDMA.
15:8	RW	0x00	b_weight The arbiter weight for BRDMA.
7:0	RW	0x00	m_weight The arbiter weight for MRDMA.

RKNN_dpu_rdma_ew_surf_notch

Address: Operational Base + offset (0x506C)

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	ew_surf_notch Surface notch of EW.
3:0	RO	0x0	reserved

RKNN_ppu_s_status

Address: Operational Base + offset (0x6000)

Bit	Attr	Reset Value	Description
31:18	RO	0x0000	reserved
17:16	RO	0x0	status_1 Executer 1 status. 2'd0: Executer 1 is in idle state; 2'd1: Executer 1 is operating; 2'd2: Executer 1 is operating, executer 1 is waiting to operate; 2'd3: Reserved.
15:2	RO	0x0000	reserved
1:0	RO	0x0	status_0 Executer 0 status. 2'd0: Executer 0 is in idle state; 2'd1: Executer 0 is operating; 2'd2: Executer 0 is operating, executer 1 is waiting to operate; 2'd3: Reserved.

RKNN_ppu_s_pointer

Address: Operational Base + offset (0x6004)

Bit	Attr	Reset Value	Description
31:17	RO	0x0000	reserved

Bit	Attr	Reset Value	Description
16	RO	0x0	executer Which register group to be used. 1'd0: Executer group 0; 1'd1: Executer group 1.
15:6	RO	0x000	reserved
5	W1 C	0x0	executer_pp_clear Clear executer group pointer. Set this bit to 1 to clear pointer to 0.
4	W1 C	0x0	pointer_pp_clear Clear register group pointer. Set this bit to 1 to clear pointer to 0.
3	RW	0x0	pointer_pp_mode Register group ping-pong mode. 1'd0: Pointer ping-pong by executer; eg. if current executer is 0, next pointer will toggle to 1; 1'd1: Pointer ping-pong by pointer; eg. if current pointer is 0, next pointer will toggle to 1.
2	RW	0x0	executer_pp_en Executer group ping-pong enable. 1'd0: Disable; 1'd1: Enable.
1	RW	0x0	pointer_pp_en Register group ping-pong enable. 1'd0: Disable; 1'd1: Enable.
0	RW	0x0	pointer Which register group ready to be set. 1'd0: Register group 0; 1'd1: Register group 1.

RKNN ppu operation enable

Address: Operational Base + offset (0x6008)

Bit	Attr	Reset Value	Description
31:1	RO	0x00000000	reserved
0	RW	0x0	op_en Set this register will trigger PPU block operate. This register and after this are all shadowed for ping-pong operation. 1'd0: Disable; 1'd1: Enable.

RKNN ppu data cube in width

Address: Operational Base + offset (0x600C)

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	reserved

Bit	Attr	Reset Value	Description
12:0	RW	0x0000	cube_in_width Pooling cube width (need to minus 1).

RKNN ppu data cube in height

Address: Operational Base + offset (0x6010)

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	reserved
12:0	RW	0x0000	cube_in_height Pooling cube height (need to minus 1).

RKNN ppu data cube in channel

Address: Operational Base + offset (0x6014)

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	reserved
12:0	RW	0x0000	cube_in_channel Pooling cube channel (need to minus 1).

RKNN ppu data cube out width

Address: Operational Base + offset (0x6018)

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	reserved
12:0	RW	0x0000	cube_out_width Pooling output cube width (need to minus 1).

RKNN ppu data cube out height

Address: Operational Base + offset (0x601C)

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	reserved
12:0	RW	0x0000	cube_out_height Pooling output cube height (need to minus 1).

RKNN ppu data cube out channel

Address: Operational Base + offset (0x6020)

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	reserved
12:0	RW	0x0000	cube_out_channel Pooling output cube channel (need to minus 1).

RKNN ppu operation mode cfg

Address: Operational Base + offset (0x6024)

Bit	Attr	Reset Value	Description
31	RO	0x0	reserved

Bit	Attr	Reset Value	Description
30	RW	0x0	index_en If enable output the position of each kernel. 1'd0: Disable; 1'd1: Enable.
29	RO	0x0	reserved
28:16	RW	0x0000	notch_addr How many pixels from the end of the width end to the shape line end.
15:8	RO	0x00	reserved
7:5	RW	0x0	use_cnt Use_cnt.
4	RW	0x0	flying_mode Where the pooling cube from. 1'd0: DPU; 1'd1: Outside.
3:2	RO	0x0	reserved
1:0	RW	0x0	pooling_method Pooling method. 2'd0: Average pooling; 2'd1: Max pooling; 2'd2: Min pooling; 2'd3: Reserved.

RKNN_ppu_pooling_kernel_cfg

Address: Operational Base + offset (0x6034)

Bit	Attr	Reset Value	Description
31:24	RO	0x00	reserved
23:20	RW	0x0	kernel_stride_height Pooling kernel stride height (need to minus 1).
19:16	RW	0x0	kernel_stride_width Pooling kernel stride width (need to minus 1).
15:12	RO	0x0	reserved
11:8	RW	0x0	kernel_height Pooling kernel height (need to minus 1).
7:4	RO	0x0	reserved
3:0	RW	0x0	kernel_width Pooling kernel width (need to minus 1).

RKNN_ppu_recip_kernel_width

Address: Operational Base + offset (0x6038)

Bit	Attr	Reset Value	Description
31:17	RO	0x0000	reserved
16:0	RW	0x00000	recip_kernel_width The Reciprocal of the shape kernel width multiple 2^16.

RKNN ppu recip kernel height

Address: Operational Base + offset (0x603C)

Bit	Attr	Reset Value	Description
31:17	RO	0x0000	reserved
16:0	RW	0x00000	recip_kernel_height The Reciprocal of the shape kernel height multiple 2^{16} .

RKNN ppu pooling padding cfg

Address: Operational Base + offset (0x6040)

Bit	Attr	Reset Value	Description
31:15	RO	0x00000	reserved
14:12	RW	0x0	pad_bottom Pooling bottom pad.
11	RO	0x0	reserved
10:8	RW	0x0	pad_right Pooling right pad.
7	RO	0x0	reserved
6:4	RW	0x0	pad_top Pooling top pad.
3	RO	0x0	reserved
2:0	RW	0x0	pad_left Pooling left pad.

RKNN ppu padding value 1 cfg

Address: Operational Base + offset (0x6044)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	pad_value_0 Pad_value*1 [31:0].

RKNN ppu padding value 2 cfg

Address: Operational Base + offset (0x6048)

Bit	Attr	Reset Value	Description
31:3	RO	0x00000000	reserved
2:0	RW	0x0	pad_value_1 Pad_value*1 [34:32].

RKNN ppu dst base addr

Address: Operational Base + offset (0x6070)

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	dst_base_addr Base address the output cube goes.
3:0	RO	0x0	reserved

RKNN ppu dst surf stride

Address: Operational Base + offset (0x607C)

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	dst_surf_stride Output shape area.
3:0	RO	0x0	reserved

RKNN_ppu_data_format

Address: Operational Base + offset (0x6084)

Bit	Attr	Reset Value	Description
31:4	RW	0x0000000	index_add If index_en enable, this register is dst_surface_stride x number of cube surface (every 8bytes per surface), else it equals to dst_surface_stride.
3	RW	0x0	dpu_flyin If the data from DPU, and DPU data is from outside, this bit set to be 1.
2:0	RW	0x0	proc_precision Process precision.

RKNN_ppu_misc_ctrl

Address: Operational Base + offset (0x60DC)

Bit	Attr	Reset Value	Description
31:16	RW	0x0000	surf_len Surface count length.
15:9	RO	0x00	reserved
8	RW	0x0	mc_surf_out If enable multiple surfaces out. 1'd0: Disable; 1'd1: Enable.
7	RW	0x0	nonalign If enable non-align mode. 1'd0: Disable; 1'd1: Enable.
6:4	RO	0x0	reserved
3:0	RW	0x0	burst_len Burst length 4'd3: Burst4; 4'd7: Burst8; 4'd15: Burst16.

RKNN_ppu_rdma_s_status

Address: Operational Base + offset (0x7000)

Bit	Attr	Reset Value	Description
31:18	RO	0x0000	reserved

Bit	Attr	Reset Value	Description
17:16	RO	0x0	status_1 Executer 1 status. 2'd0: Executer 1 is in idle state; 2'd1: Executer 1 is operating; 2'd2: Executer 1 is operating, executer 1 is waiting to operate; 2'd3: Reserved.
15:2	RO	0x0000	reserved
1:0	RO	0x0	status_0 Executer 0 status. 2'd0: Executer 0 is in idle state; 2'd1: Executer 0 is operating; 2'd2: Executer 0 is operating, executer 1 is waiting to operate; 2'd3: Reserved.

RKNN_ppu_rdma_s_pointer

Address: Operational Base + offset (0x7004)

Bit	Attr	Reset Value	Description
31:17	RO	0x0000	reserved
16	RO	0x0	executer Which register group to be used. 1'd0: Executer group 0; 1'd1: Executer group 1.
15:6	RO	0x000	reserved
5	W1C	0x0	executer_pp_clear Clear executer group pointer. Set this bit to 1 to clear pointer to 0.
4	W1C	0x0	pointer_pp_clear Clear register group pointer. Set this bit to 1 to clear pointer to 0.
3	RW	0x0	pointer_pp_mode Register group ping-pong mode. 1'd0: Pointer ping-pong by executer; eg. if current executer is 0, next pointer will toggle to 1; 1'd1: Pointer ping-pong by pointer; eg. if current pointer is 0, next pointer will toggle to 1.
2	RW	0x0	executer_pp_en Executer group ping-pong enable. 1'd0: Disable; 1'd1: Enable.
1	RW	0x0	pointer_pp_en Register group ping-pong enable. 1'd0: Disable; 1'd1: Enable.

Bit	Attr	Reset Value	Description
0	RW	0x0	pointer Which register group ready to be set. 1'd0: Register group 0; 1'd1: Register group 1.

RKNN_ppu_rdma_operation_enable

Address: Operational Base + offset (0x7008)

Bit	Attr	Reset Value	Description
31:1	RO	0x00000000	reserved
0	RW	0x0	op_en Set this register will trigger PPU_RDMA block operate. This register and after this are all shadowed for ping-pong operation. 1'd0: Disable; 1'd1: Enable.

RKNN_ppu_rdma_cube_in_width

Address: Operational Base + offset (0x700C)

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	reserved
12:0	RW	0x0000	cube_in_width Pooling cube width (need to minus 1).

RKNN_ppu_rdma_cube_in_height

Address: Operational Base + offset (0x7010)

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	reserved
12:0	RW	0x0000	cube_in_height Pooling cube height (need to minus 1).

RKNN_ppu_rdma_cube_in_channel

Address: Operational Base + offset (0x7014)

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	reserved
12:0	RW	0x0000	cube_in_channel Pooling cube channel (need to minus 1).

RKNN_ppu_rdma_src_base_addr

Address: Operational Base + offset (0x701C)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	src_base_addr Base address of the pooling cube.

RKNN_ppu_rdma_src_line_stride

Address: Operational Base + offset (0x7024)

Bit	Attr	Reset Value	Description
31:4	RW	0x00000000	src_line_stride Pooling cube shape width.
3:0	RO	0x0	reserved

RKNN_ppu_rdma_src_surf_stride

Address: Operational Base + offset (0x7028)

Bit	Attr	Reset Value	Description
31:4	RW	0x00000000	src_surf_stride Pooling cube shape area.
3:0	RO	0x0	reserved

RKNN_ppu_rdma_data_format

Address: Operational Base + offset (0x7030)

Bit	Attr	Reset Value	Description
31:2	RO	0x00000000	reserved
1:0	RW	0x0	in_precision Input precision. 2'd0: 4bit; 2'd1: 8bit; 2'd2: 16bit; 2'd3: 32bit.

RKNN_ddma_cfg_outstanding

Address: Operational Base + offset (0x8000)

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	reserved
15:8	RW	0x00	wr_os_cnt Max numbers of write outstanding.
7:0	RW	0x00	rd_os_cnt Max numbers of read outstanding.

RKNN_ddma_rd_weight_0

Address: Operational Base + offset (0x8004)

Bit	Attr	Reset Value	Description
31:24	RW	0x00	rd_weight_pdp Weight of PPU read burst.
23:16	RW	0x00	rd_weight_dpu Weight of DPU read burst.
15:8	RW	0x00	rd_weight_kernel Weight of read weight burst.
7:0	RW	0x00	rd_weight_feature Weight of read feature burst.

RKNN_ddma_wr_weight_0

Address: Operational Base + offset (0x8008)

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	reserved
15:8	RW	0x00	wr_weight_pdp Write_weight_ppu.
7:0	RW	0x00	wr_weight_dpu Write_weight_dpu.

RKNN_ddma_cfg_id_error

Address: Operational Base + offset (0x800C)

Bit	Attr	Reset Value	Description
31:10	RO	0x000000	reserved
9:6	RW	0x0	wr_resp_id Error write id.
5	RO	0x0	reserved
4:0	RW	0x00	rd_resp_id Error read id.

RKNN_ddma_rd_weight_1

Address: Operational Base + offset (0x8010)

Bit	Attr	Reset Value	Description
31:8	RO	0x000000	reserved
7:0	RW	0x00	rd_weight_pc Weight of PC read burst.

RKNN_ddma_cfg_dma_fifo_clr

Address: Operational Base + offset (0x8014)

Bit	Attr	Reset Value	Description
31:1	RO	0x00000000	reserved
0	RW	0x0	dma_fifo_clr Clear DMA FIFO.

RKNN_ddma_cfg_dma_arb

Address: Operational Base + offset (0x8018)

Bit	Attr	Reset Value	Description
31:10	RO	0x000000	reserved
9	RW	0x0	wr_arbit_model Write_arbit_model.
8	RW	0x0	rd_arbit_model Read_arbit_model.
7	RO	0x0	reserved
6:4	RW	0x0	wr_fix_arb Write_fix_arb.
3	RO	0x0	reserved
2:0	RW	0x0	rd_fix_arb Read_fix_arb.

RKNN ddma cfg dma rd qos

Address: Operational Base + offset (0x8020)

Bit	Attr	Reset Value	Description
31:10	RO	0x000000	reserved
9:8	RW	0x0	rd_pc_qos Read_pc_qos.
7:6	RW	0x0	rd_ppu_qos Read_ppu_qos.
5:4	RW	0x0	rd_dpu_qos Read_dpu_qos.
3:2	RW	0x0	rd_kernel_qos Read_kernel_qos.
1:0	RW	0x0	rd_feature_qos Read feature_qos.

RKNN ddma cfg dma rd cfg

Address: Operational Base + offset (0x8024)

Bit	Attr	Reset Value	Description
31:13	RO	0x000000	reserved
12	RW	0x0	rd_arlock Read_arlock.
11:8	RW	0x0	rd_arcache Read_arcache.
7:5	RW	0x0	rd_arprot Read_arprot.
4:3	RW	0x0	rd_arburst Read_arburst.
2:0	RW	0x0	rd_arsize Read_arsize.

RKNN ddma cfg dma wr cfg

Address: Operational Base + offset (0x8028)

Bit	Attr	Reset Value	Description
31:13	RO	0x000000	reserved
12	RW	0x0	wr_awlock Write_awlock.
11:8	RW	0x0	wr_awcache Write awcache.
7:5	RW	0x0	wr_awprot Write_awprot.
4:3	RW	0x0	wr_awburst Write_awburst.
2:0	RW	0x0	wr_awsiz Write_awsiz.

RKNN_ddma_cfg_dma_wstrb

Address: Operational Base + offset (0x802C)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	wr_wstrb Write_wstrb.

RKNN_ddma_cfg_status

Address: Operational Base + offset (0x8030)

Bit	Attr	Reset Value	Description
31:9	RO	0x000000	reserved
8	RW	0x0	idel Idel.
7:0	RO	0x00	reserved

RKNN_sdma_cfg_outstanding

Address: Operational Base + offset (0x9000)

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	reserved
15:8	RW	0x00	wr_os_cnt Max numbers of write outstanding.
7:0	RW	0x00	rd_os_cnt Max numbers of read outstanding.

RKNN_sdma_rd_weight_0

Address: Operational Base + offset (0x9004)

Bit	Attr	Reset Value	Description
31:24	RW	0x00	rd_weight_pdp Weight of PPU read burst.
23:16	RW	0x00	rd_weight_dpu Weight of DPU read burst.
15:8	RW	0x00	rd_weight_kernel Weight of read weight burst.
7:0	RW	0x00	rd_weight_feature Weight of read feature burst.

RKNN_sdma_wr_weight_0

Address: Operational Base + offset (0x9008)

Bit	Attr	Reset Value	Description
31:16	RO	0x0000	reserved
15:8	RW	0x00	wr_weight_pdp Write_weight_ppu.
7:0	RW	0x00	wr_weight_dpu Write_weight_dpu.

RKNN_sdma_cfg_id_error

Address: Operational Base + offset (0x900C)

Bit	Attr	Reset Value	Description
31:10	RO	0x000000	reserved
9:6	RW	0x0	wr_resp_id Error write id.
5	RO	0x0	reserved
4:0	RW	0x00	rd_resp_id Error read id.

RKNN_sdma_rd_weight_1

Address: Operational Base + offset (0x9010)

Bit	Attr	Reset Value	Description
31:8	RO	0x000000	reserved
7:0	RW	0x00	rd_weight_pc Weight of PC read burst.

RKNN_sdma_cfg_dma_fifo_clr

Address: Operational Base + offset (0x9014)

Bit	Attr	Reset Value	Description
31:1	RO	0x00000000	reserved
0	RW	0x0	dma_fifo_clr Clear DMA FIFO.

RKNN_sdma_cfg_dma_arb

Address: Operational Base + offset (0x9018)

Bit	Attr	Reset Value	Description
31:10	RO	0x000000	reserved
9	RW	0x0	wr_arbit_model Write_arbit_model.
8	RW	0x0	rd_arbit_model Read_arbit_model.
7	RO	0x0	reserved
6:4	RW	0x0	wr_fix_arb Write_fix_arb.
3	RO	0x0	reserved
2:0	RW	0x0	rd_fix_arb Read_fix_arb.

RKNN_sdma_cfg_dma_rd_qos

Address: Operational Base + offset (0x9020)

Bit	Attr	Reset Value	Description
31:10	RO	0x000000	reserved
9:8	RW	0x0	rd_pc_qos Read_pc_qos.
7:6	RW	0x0	rd_ppu_qos Read_ppu_qos.

Bit	Attr	Reset Value	Description
5:4	RW	0x0	rd_dpu_qos Read_dpu_qos.
3:2	RW	0x0	rd_kernel_qos Read_kernel_qos.
1:0	RW	0x0	rd_feature_qos Read feature_qos.

RKNN sdma cfg dma rd cfg

Address: Operational Base + offset (0x9024)

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	reserved
12	RW	0x0	rd_arlock Read_arlock.
11:8	RW	0x0	rd_arcache Read_arcache.
7:5	RW	0x0	rd_arprot Read_arprot.
4:3	RW	0x0	rd_arburst Read_arburst.
2:0	RW	0x0	rd_arsize Read_arsize.

RKNN sdma cfg dma wr cfg

Address: Operational Base + offset (0x9028)

Bit	Attr	Reset Value	Description
31:13	RO	0x00000	reserved
12	RW	0x0	wr_awlock Write_awlock.
11:8	RW	0x0	wr_awcache Write awcache.
7:5	RW	0x0	wr_awprot Write_awprot.
4:3	RW	0x0	wr_awburst Write_awburst.
2:0	RW	0x0	wr_awsiz Write_awsiz.

RKNN sdma cfg dma wstrb

Address: Operational Base + offset (0x902C)

Bit	Attr	Reset Value	Description
31:0	RW	0x00000000	wr_wstrb Write_wstrb.

RKNN sdma cfg status

Address: Operational Base + offset (0x9030)

Bit	Attr	Reset Value	Description
31:9	RO	0x000000	reserved
8	RW	0x0	idel Idel.
7:0	RO	0x00	reserved

RKNN global operation enable

Address: Operational Base + offset (0xF008)

Bit	Attr	Reset Value	Description
31:7	RO	0x0000000	reserved
6	RW	0x0	ppu_rdma_op_en PPU_RDMA operation enable signal.
5	RW	0x0	ppu_op_en PPU operation enable signal.
4	RW	0x0	dpu_rdma_op_en DPU_RDMA operation enable signal.
3	RW	0x0	dpu_op_en DPU operation enable signal.
2	RW	0x0	core_op_en CORE operation enable signal.
1	RO	0x0	reserved
0	RW	0x0	cna_op_en CNA operation enable signal.

36.5 Application Notes

36.5.1 Ping-pong registers

In order to reduce the time of fetch registers, every Calculate Core and Control Core of RKNN has its owner ping-pong registers. Configure the group 0 when use the group 1, and configure the group 1 when use group 0. As a result of hiding the time of fetch registers. Writing S_POINTER of every block can enable this function.

36.5.2 Clock and Reset

1.5.2.1 Clock Domains

RKNN has two clock domains, one is AHB clock, the other is AXI clock. AHB clock, which is the clock for AHB interface, while AXI clock, which is the clock for AXI interface. AXI clock also used for core clock for every Calculate Core and Control Core. Clock frequency can be controlled by CRU, please refer to the relevant sections. Automatic localized clock gating is employed throughout the design in order to minimize the dynamic power consumption. Almost all of the flip-flops are clock gated in the design. Block level clock gating also implemented in every separate block. If a block and the interface to the block are both idle, then the clock of that block will be gated automatically. This feature can be disabled by software.

1.5.2.2 NPU Reset

Correspond to the clock domain, there are two reset signals. Aresetn, the reset signal for AXI interface and every Calculate Core and Control Core. Hresetn is the AHB interface reset pin and which is synchronized to the AHB clock domain.

All the two signals must be asserted for a minimum of 32 core clock cycles, using the slowest of the two clocks. Then two signals must be release at the same time.

36.5.3 NPU Interrupt Application

RKNN has 3 interrupt output signal and it remains asserted until the host processor clears the interrupt. Each bit of PC_INTERRUPT_STATUS represents one of the 17 possible events that the RKNN can signal to the host processor. By setting the bits of the interrupt enable register (PC_INTERRUPT_MASK) the programmer can control which of those events will generate an interrupt.

36.5.4 NPU operate flow

RKNN has two types of work mode: slave configured mode and pc work mode. Pc work mode need to initial register information to the system memory, then obey flowing flows as Figure 1-3 shows.

The information of registers initial to the system memory need to generate by the op you want to specify. And we describe the normal work flows by describe the slave configured mode.

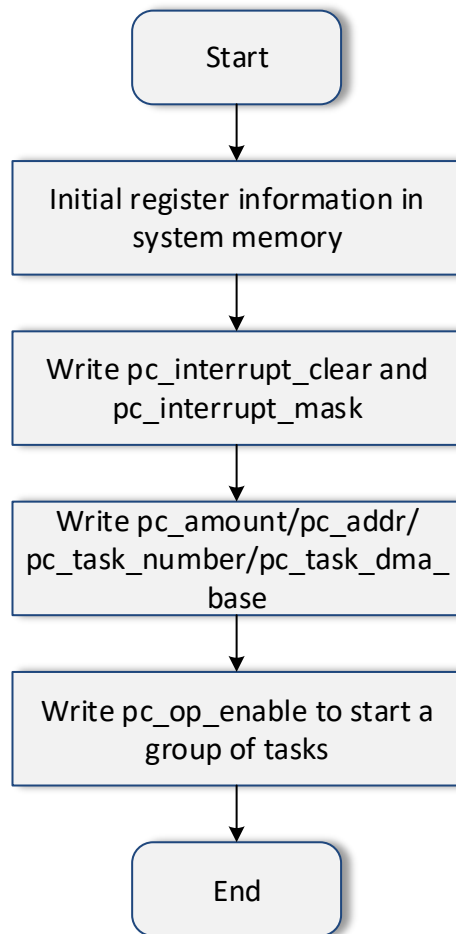


Fig. 36-3 PC flow

Convolution flow

RKNN supports variety of convolution and matrix multiplication. If use the MAC array to calculate, the work flow should obey the Figure 1-4, and if use DPU CORE calculate, work flow should obey the Figure 1-5.

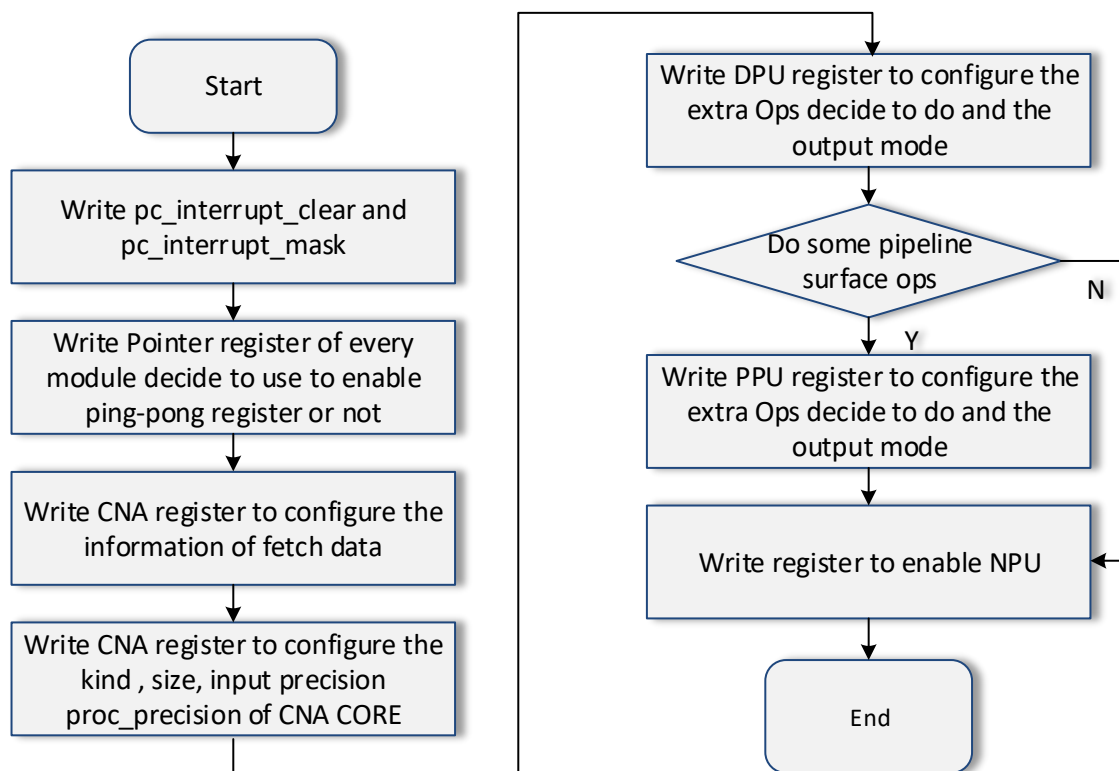


Fig. 36-4 Convolution flow 1

In step 1, do not set PPU mask bit if don't use PPU.

In step 3, you need to configure the size of feature map, and the input precision, process precision, grains, entries and weight size, including weight height, weight width, kernels and weight bytes etc.

In step 4, you need to configure process precision, conv_mode and auto gating registers of CMAC and CNA sequence controller.

In step 5, you can specify some operators to the outputs of convolution according to configure the DPU_CORE. DPU CORE has some fixed operators, also you can specify some programmed operators.

In step 6, you can do some extra operators to the outputs of DPU, pooling for example.

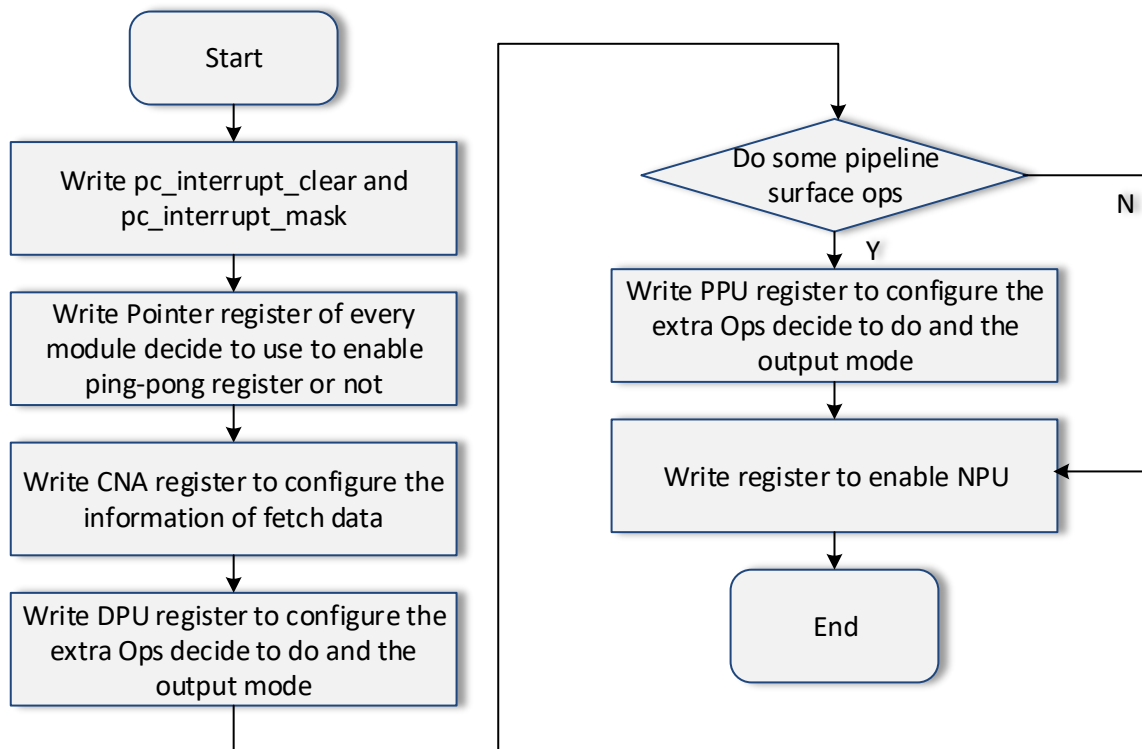


Fig. 36-5 Convolution flow 2

In step 3, you need to configure the size of feature map, and the input precision, process precision, grains, entries and weight size, including height, width, kernels and weight bytes etc. Here we supply zero skipping switch, if there's a lot of zero or some number else in the feature map, you can enable zero skipping by writing `cna_fc_con0`, `cna_fc_con1` and `cna_fc_con2`, as a result of without reading the weights correspond to the pixel in the feature map. In zero skipping mode, you can specify the feature map size for read different from the feature map size for calculate.

If enable zero skipping, the `conv_mode` of DPU must configure to be 3, and must bypass the `BS_CORE`. The `alu_src` must set to be 0, the `mul_src` must set to be 1, and the `alu_algo` must set be 3. We bypass `BS_CORE`, and use `BN_CORE` to calculate convolution, so the extra operators have to achieve by `EW_CORE`. But we use `NRDMA` to read operands to `EW_CORE` when `ew_src` is 1.

Pooling flow

You can use PPU in two ways, pipeline with DPU, and flying mode independently. The first one is described above, we describe the second flow below.

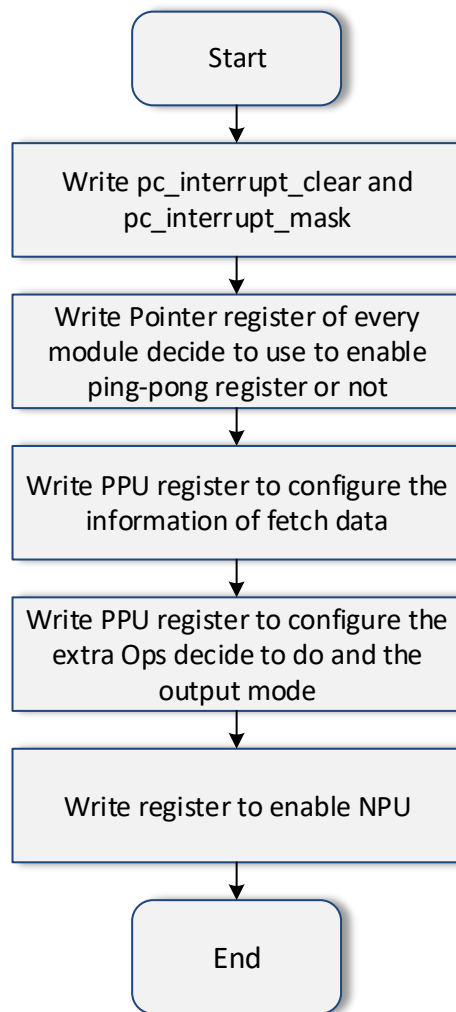


Fig. 36-6 Pooling flow

In step 4, we supply two type of output mode, 8byte align per pixel and non-align mode. In some case with uncomfortable feature map size, you can enable non-align mode by writing ppu_misc_ctrl_nonalign and ppu_misc_ctrl_surf_len.

Separate op flow

As for the operators that RKNN do not fixed, you can combine DPU and PPU to achieve it. Following we describe the DPU flying mode flow.

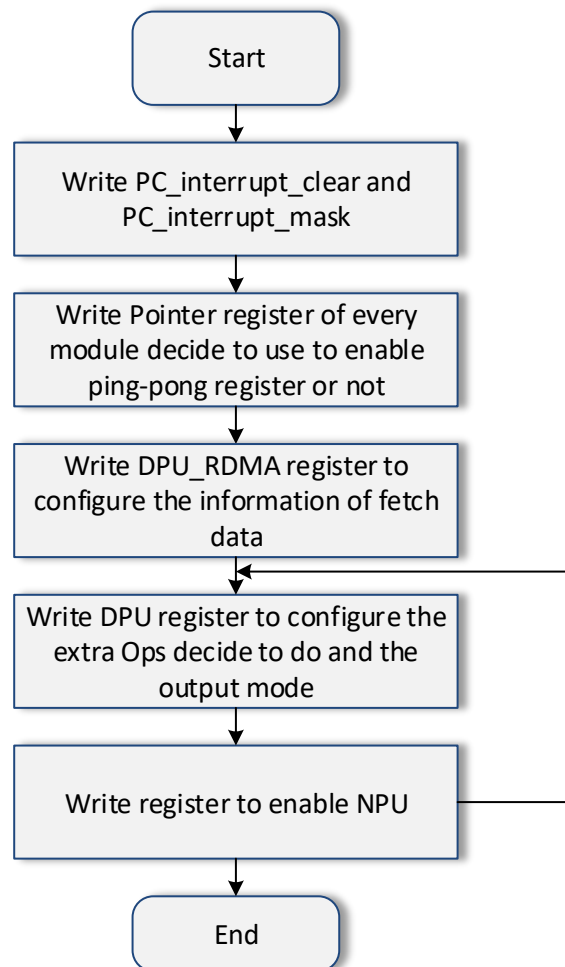


Fig. 36-7 DPU flying mode flow

In step 3, you can enable MRDMA, BRDMA, NRDMA, ERDMA to fetch data. RKNN supports 6 types of data input precision.

In step 4, you can configure BS_ALU_BYPASS, BS_MUL_BYPASS, BS_RELU_BYPASS, BS_MUL_PRELU, BS_MUL_SHIFT, BS_RELUX_EN etc. to specify the operators you want to achieve.

Chapter 37 Video Capture (VICAP)

37.1 Overview

The Video Capture, receives the data from Camera via DVP/MIPI, and transfers the data into system main memory by AXI bus.

- Support BT601 YCbCr 422 8bit input, RAW 8/10/12bit input
- Support BT656 YCbCr 422 8bit progressive/interlace input
- Support BT1120 YCbCr 422 16bit progressive/interlace input, single/dual-edge sampling
- Support 2/4 channels mixed BT656/BT1120 YCbCr 422 8/16bit progressive/interlace input
- Support YUYV sequence configurable
- Support the polarity of hsync and vsync configurable
- Support receiving six interfaces of MIPI CSI/DSI, up to four IDs for each interface
- Support five CSI data formats: RAW8/10/12/14, YUV422
- Support three modes of HDR: virtual channel mode, identification code mode, line counter mode
- Support window cropping
- Support RAW data through to ISP0/1
- Support four channels of 8/16/32 times down-sampling for RAW data
- Support virtual stride when write to DDR
- Support NV16/NV12/YUV400/YUYV output format for YUV data
- Support compact/non-compact output format for RAW data
- Support MMU
- Support soft reset, auto-reset when DMA error

37.2 Block Diagram

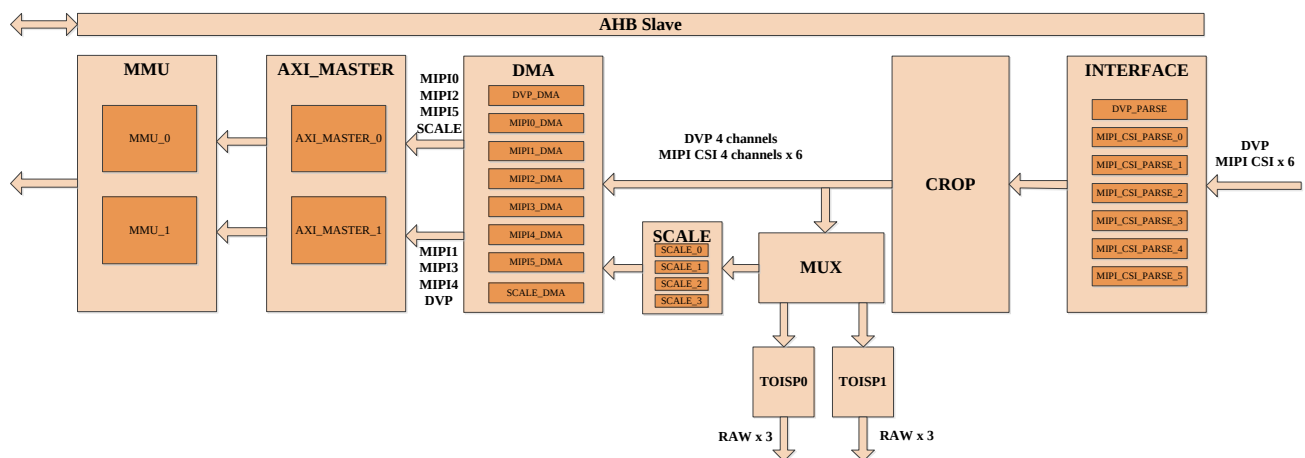


Fig. 37-1 VICAP Block Diagram

VICAP comprises with:

- AHB Slave
- AXI Master
- MMU
- INTERFACE
- CROP
- SCALE
- TOISP0/1
- DMA

37.3 Function Description

37.3.1 Interface

Translate the input video data(DVP/MIPI CSI) into the requisite data format

- DVP BT656/BT1120 format

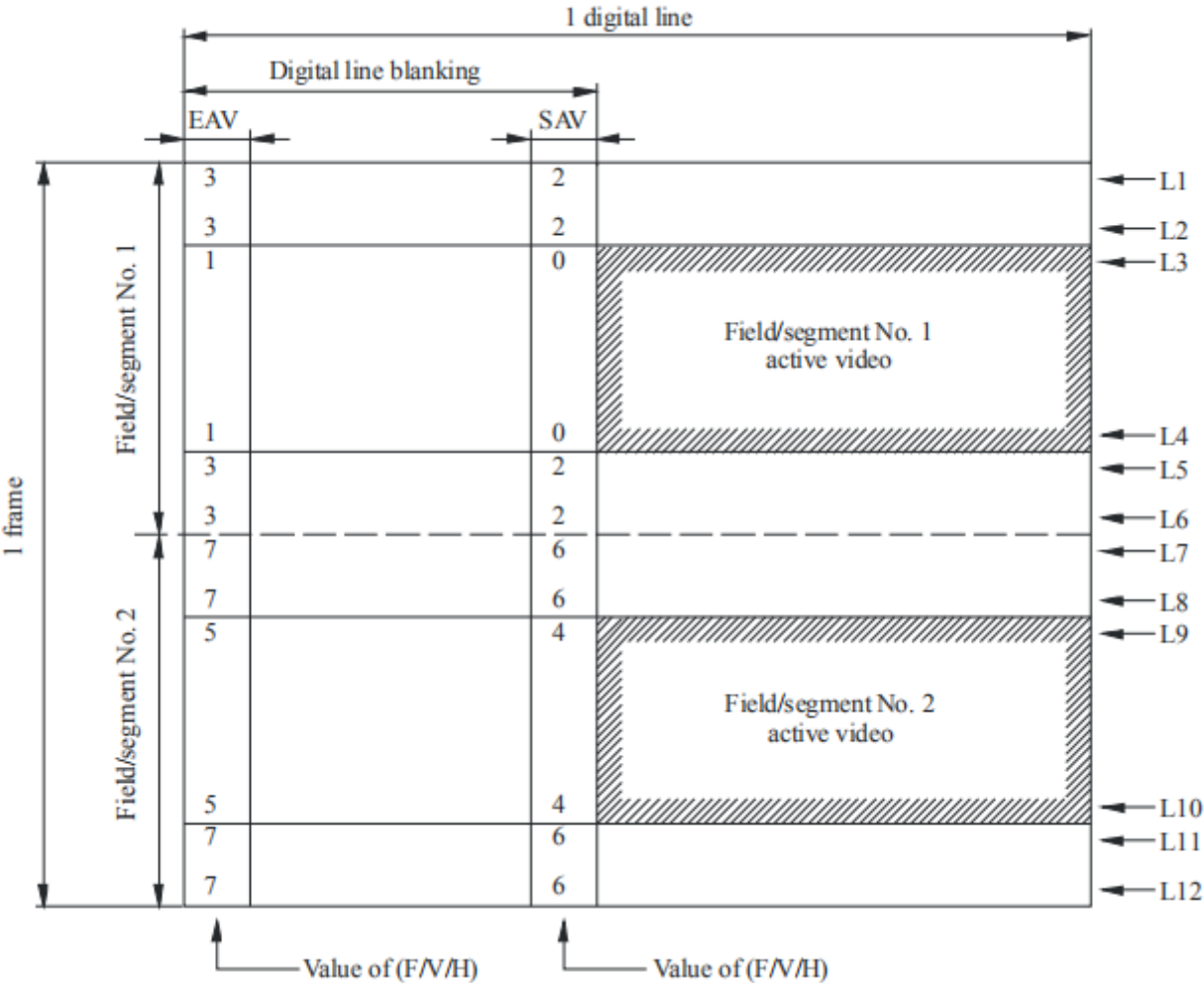


Fig. 37-2 BT656/BT1120 interlace timing relationship

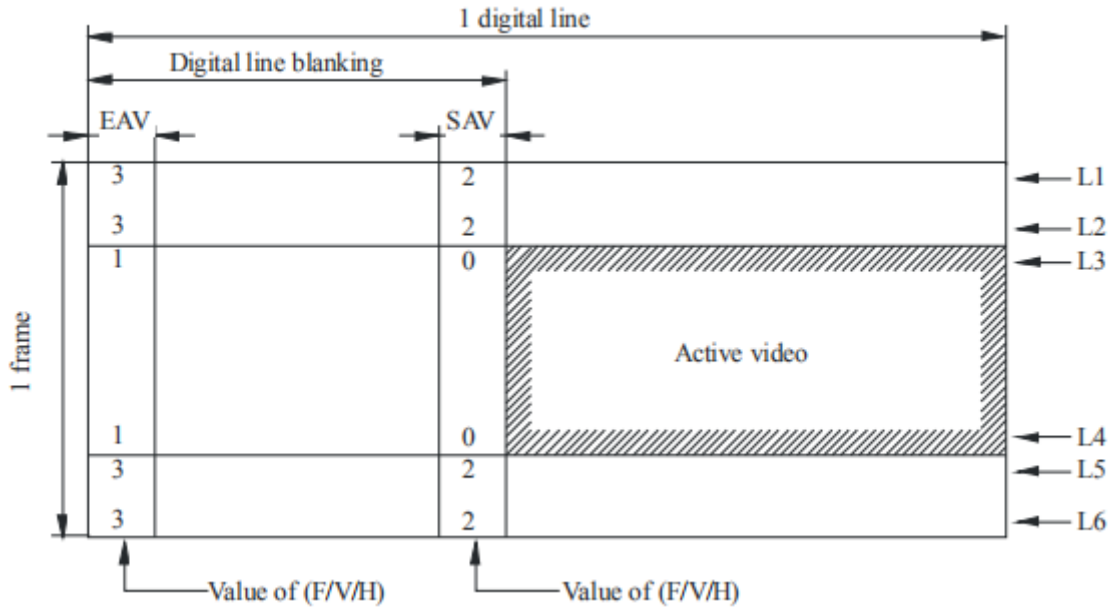


Fig. 37-3 BT656/BT1120 progressive timing relationship

- DVP 2/4 channels mixed BT656/BT1120 format

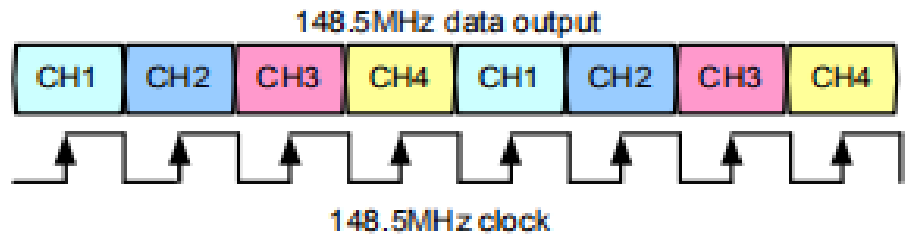


Fig. 37-4 4 channels mixed BT656/BT1120

- DVP BT601 format

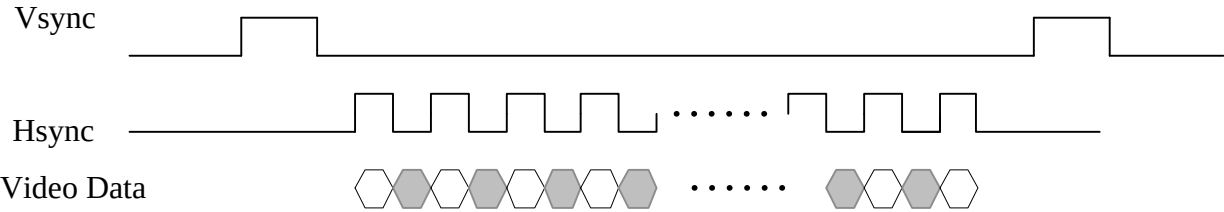


Fig. 37-5 BT656/BT1120 timing relationship

- MIPI CSI RAW8 format

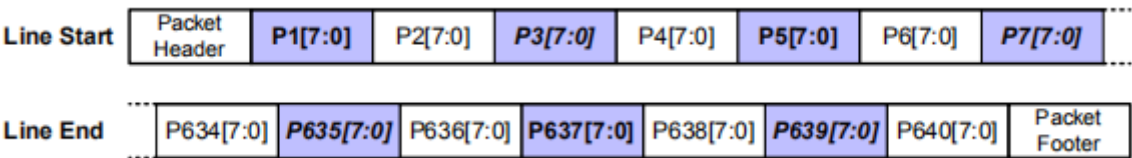


Fig. 37-6 MIPI CSI RAW8 format

- MIPI CSI RAW10 format

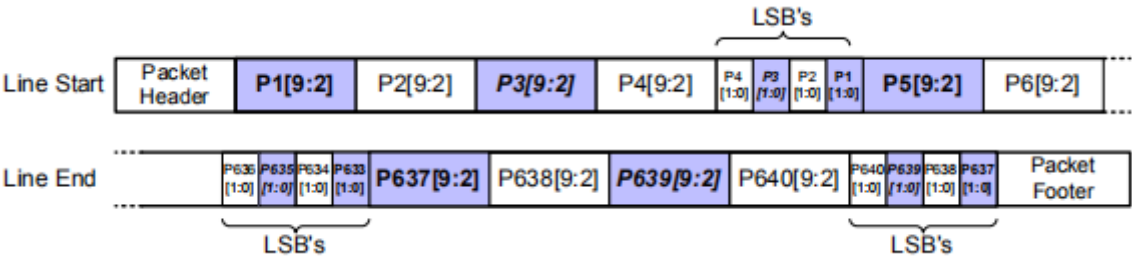


Fig. 37-7 MIPI CSI RAW10 format

- MIPI CSI RAW12 format

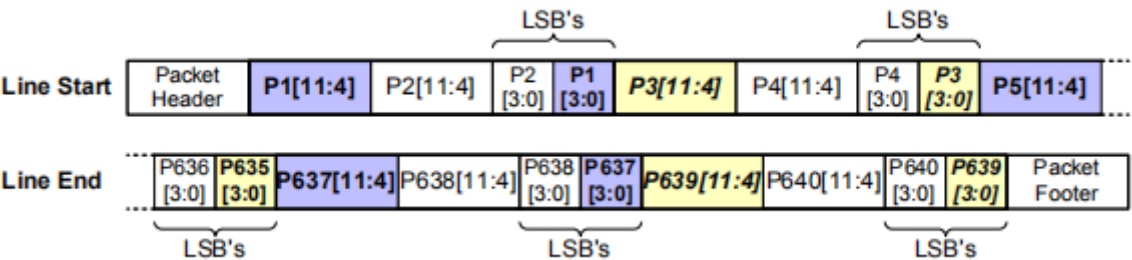


Fig. 37-8 MIPI CSI RAW12 format

- MIPI CSI RAW14 format

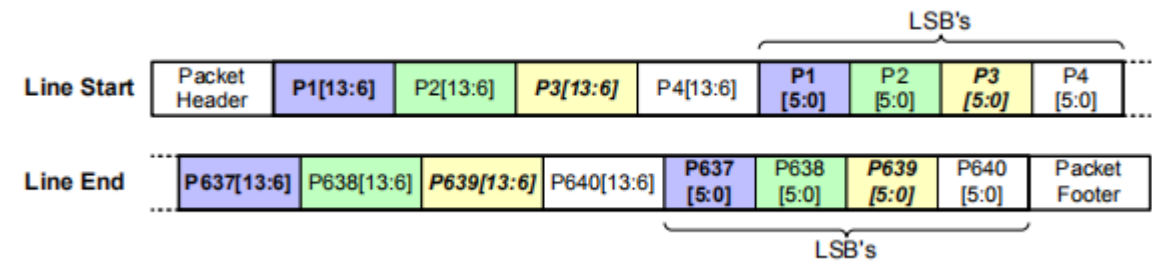


Fig. 37-9 MIPI CSI RAW14 format

- MIPI CSI YUV422 format

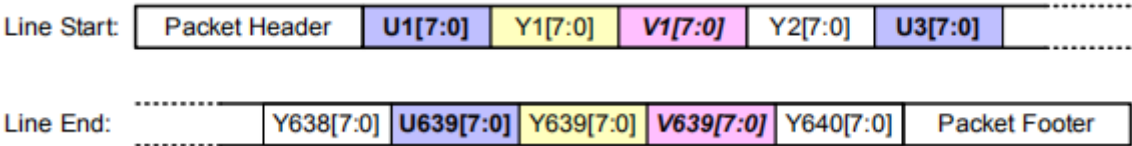


Fig. 37-10 MIPI CSI YUV422 format

37.3.2 Crop

Bypass or crop the source video data to a smaller size destination.

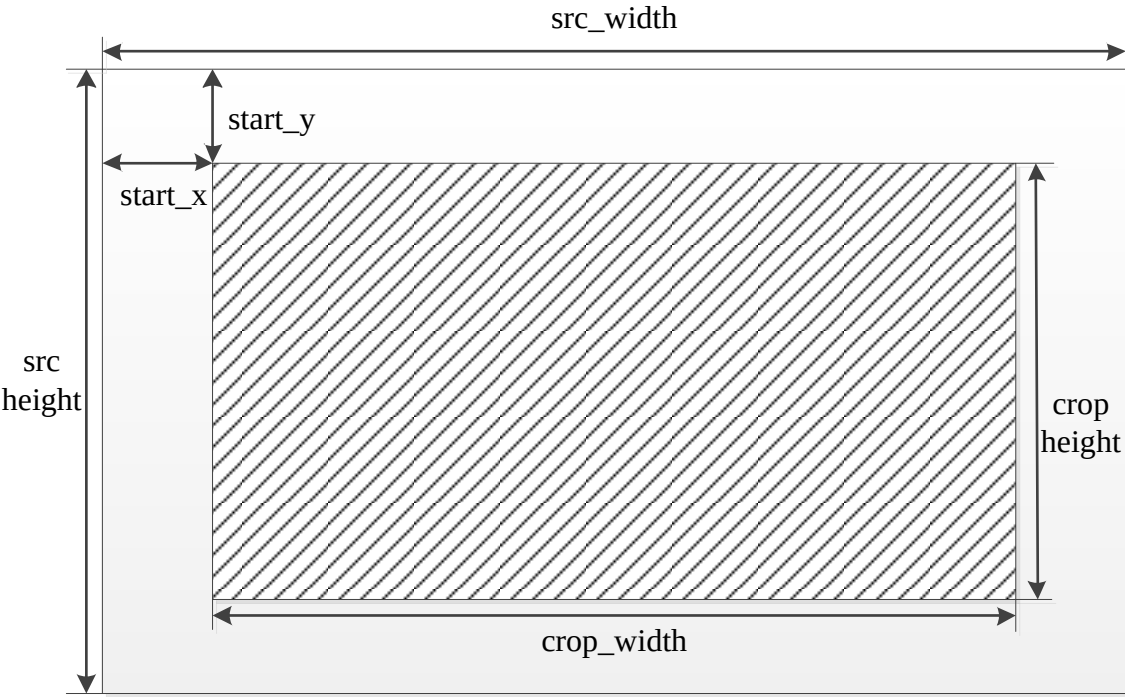


Fig. 37-11 Crop

37.3.3 Scale

Scale down for RAW data.

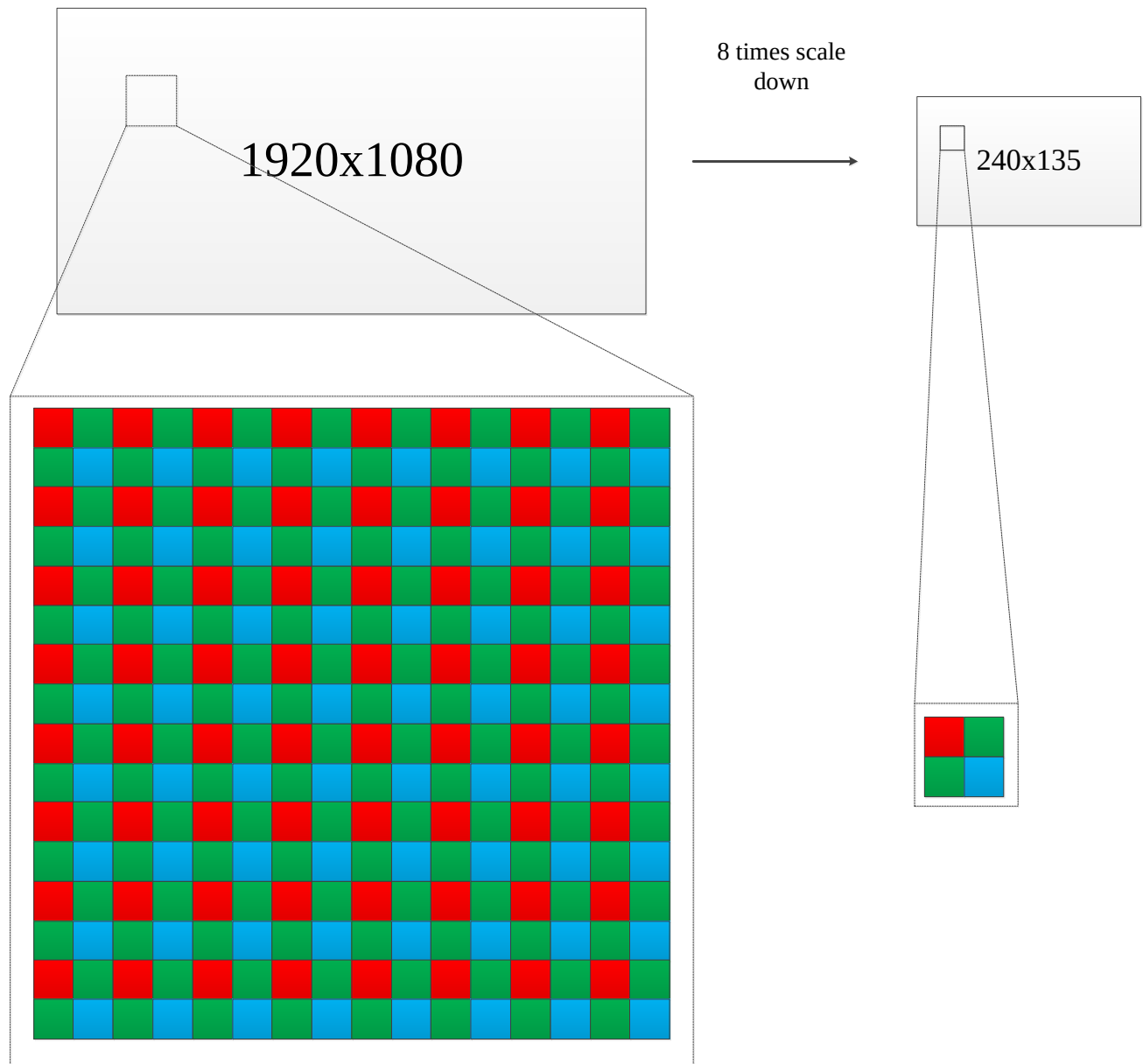


Fig. 37-12 8 times Scale down

37.3.4 Toisp

The parsed video data can go straight to ISP0/1 for real-time processing.

37.3.5 DMA

The DMA is used to transfer the data from crop module to the AXI master block which will send the data to the AXI bus.

- NV16

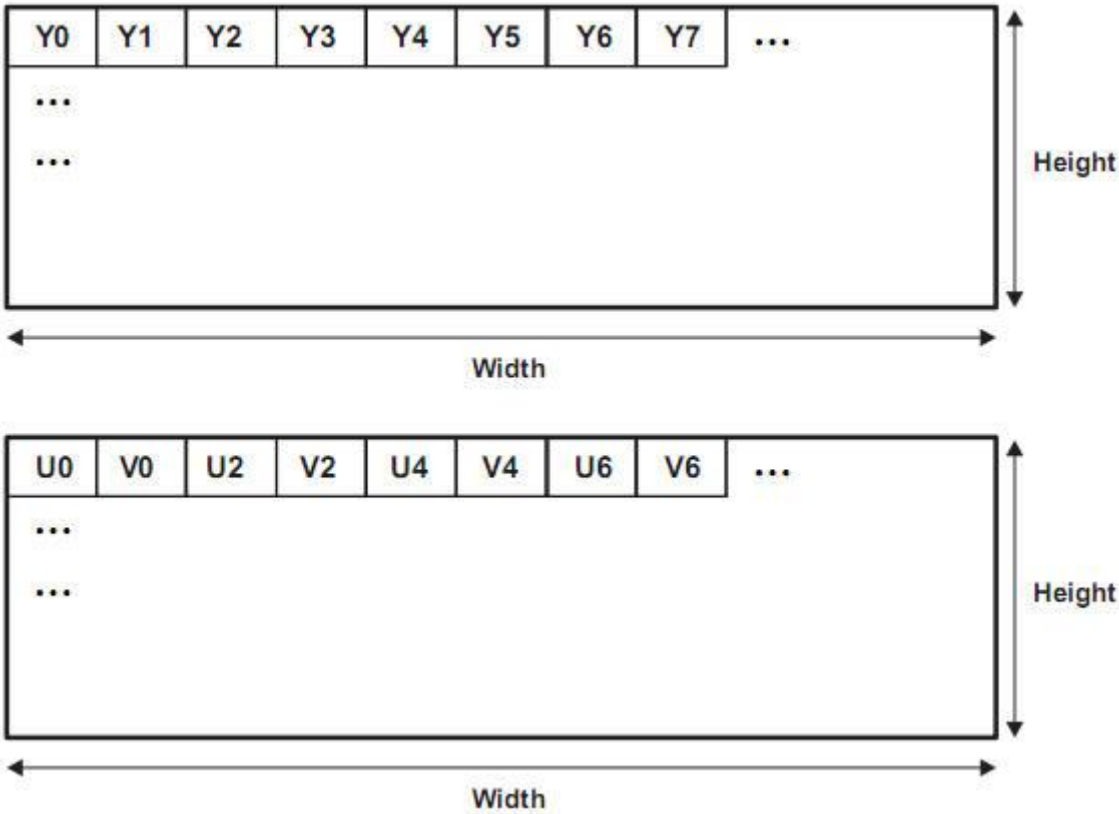


Fig. 37-13 NV16

● NV12

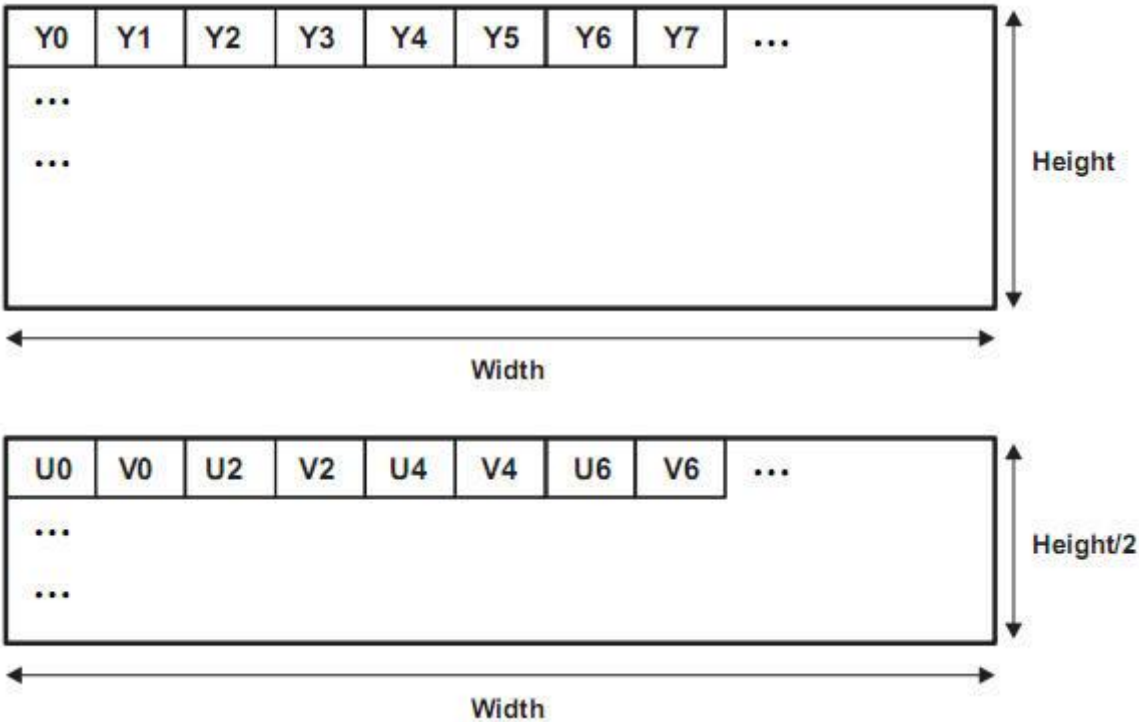


Fig. 37-14 NV12

● YUV400

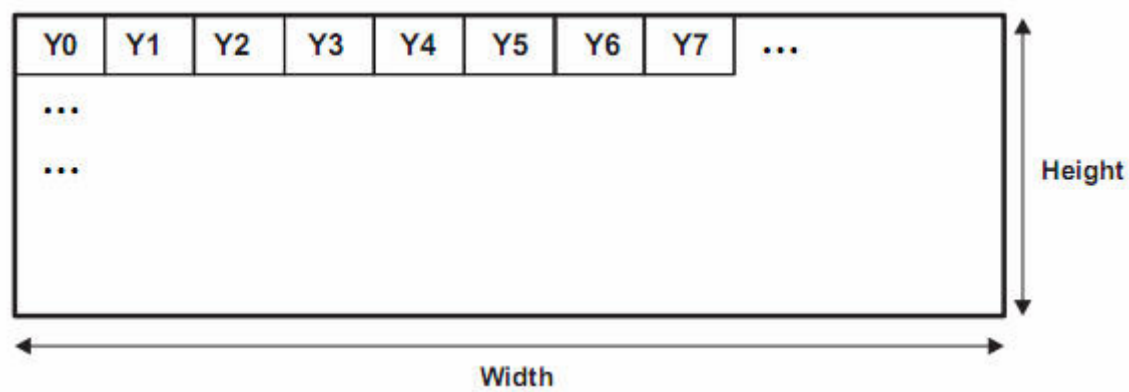


Fig. 37-15 YUV400

● YUYV

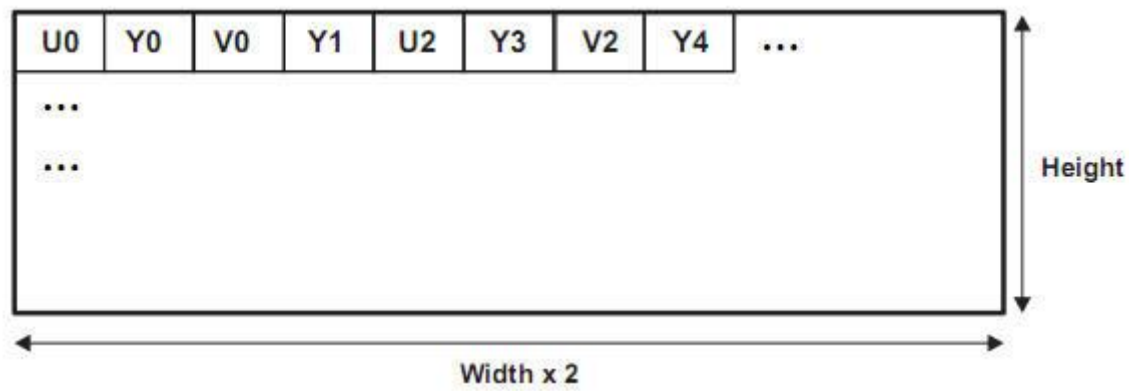


Fig. 37-16 YUYV

● Compact RAW

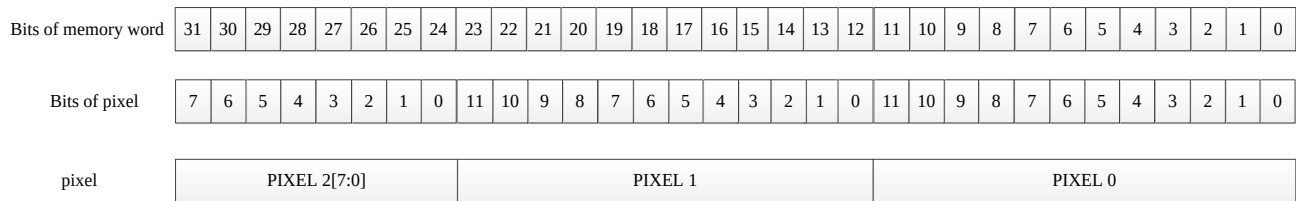


Fig. 37-17 Compact RAW12

● Noncompact RAW

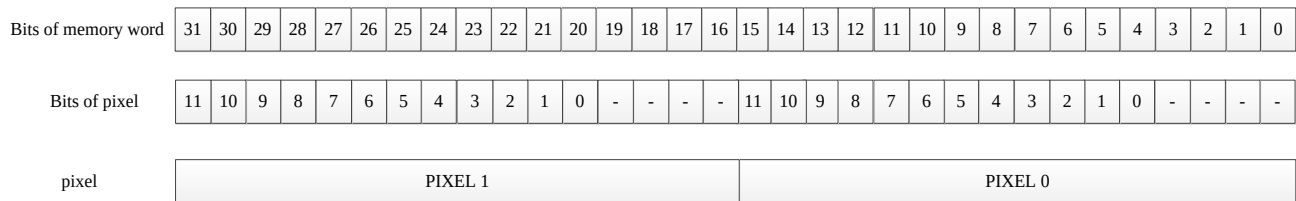


Fig. 37-18 noncompact RAW12(high align)

37.3.6 AXI Master

Transmit the data to chip memory via the AXI Master.

37.3.7 MMU

Map the virtual address to physical address.

37.3.8 AHB Slave

Host configure the registers via the AHB Slave.

37.4 VICAP Register Description

37.4.1 Internal Address Mapping

Slave address can be divided into different length for different usage, which is shown as follows.

Table 37-1 VICAP Address Mapping

Base Address[11:8]	Device	Address Length	Offset Address Range
4'b0000	Global/DVP	256 BYTE	0x0000 ~ 0x00ff
4'b0001	MIPI0	256 BYTE	0x0100 ~ 0x01ff
4'b0010	MIPI1	256 BYTE	0x0200 ~ 0x02ff
4'b0011	MIPI2	256 BYTE	0x0300 ~ 0x03ff
4'b0100	MIPI3	256 BYTE	0x0400 ~ 0x04ff
4'b0101	MIPI4	256 BYTE	0x0500 ~ 0x05ff
4'b0110	MIPI5	256 BYTE	0x0600 ~ 0x06ff
4'b0111	Scale/TOISP	256 BYTE	0x0700 ~ 0x07ff
4'b1000	MMU0	256 BYTE	0x0800 ~ 0x08ff
4'b1001	MMU1	256 BYTE	0x0900 ~ 0x09ff

37.4.2 Registers Summary

Name	Offset	Size	Reset Value	Description
<u>VICAP_GLB_CTRL</u>	0x0000	W	0x00010001	VICAP global control
<u>VICAP_GLB_INTEN</u>	0x0004	W	0x00000000	VICAP global interrupt enable.
<u>VICAP_GLB_INTST</u>	0x0008	W	0x00000000	VICAP global interrupt status.
<u>VICAP_DVP_CTRL</u>	0x0010	W	0x00000000	DVP path control
<u>VICAP_DVP_INTEN</u>	0x0014	W	0x00000000	DVP path interrupt status
<u>VICAP_DVP_INTSTAT</u>	0x0018	W	0x00000000	DVP path interrupt status
<u>VICAP_DVP_FORMAT</u>	0x001C	W	0x00000000	DVP path format
<u>VICAP_DVP_MULTI_ID</u>	0x0020	W	0x00000000	Channel ID for multi-ID mode
<u>VICAP_DVP_SAV_EAV</u>	0x0024	W	0xFEDCBA98	SAV/EAV for ACT/BLK
<u>VICAP_DVP_CROP_SIZE</u>	0x0028	W	0x01E002D0	The expected width and height of received image
<u>VICAP_DVP_CROP_START</u>	0x002C	W	0x00000000	The start point of DVP path cropping
<u>VICAP_DVP_FRM0_ADDR_Y_ID0</u>	0x0030	W	0x00000000	DVP path frame0 y address
<u>VICAP_DVP_FRM0_ADDR_UV_ID0</u>	0x0034	W	0x00000000	DVP path frame0 uv address
<u>VICAP_DVP_FRM1_ADDR_Y_ID0</u>	0x0038	W	0x00000000	DVP path frame1 y address
<u>VICAP_DVP_FRM1_ADDR_UV_ID0</u>	0x003C	W	0x00000000	DVP path frame1 uv address
<u>VICAP_DVP_FRM0_ADDR_Y_ID1</u>	0x0040	W	0x00000000	DVP path frame0 y address for ID1
<u>VICAP_DVP_FRM0_ADDR_UV_ID1</u>	0x0044	W	0x00000000	DVP path frame0 uv address for id1

Name	Offset	Size	Reset Value	Description
<u>VICAP DVP FRM1 ADDR Y ID1</u>	0x0048	W	0x00000000	DVP path frame1 y address for id1
<u>VICAP DVP FRM1 ADDR UV ID1</u>	0x004C	W	0x00000000	DVP path frame1 uv address for id1
<u>VICAP DVP FRM0 ADDR Y ID2</u>	0x0050	W	0x00000000	DVP path frame0 y address for id2
<u>VICAP DVP FRM0 ADDR UV ID2</u>	0x0054	W	0x00000000	DVP path frame0 uv address for id2
<u>VICAP DVP FRM1 ADDR Y ID2</u>	0x0058	W	0x00000000	DVP path frame1 y address for id2
<u>VICAP DVP FRM1 ADDR UV ID2</u>	0x005C	W	0x00000000	DVP path frame1 uv address for id2
<u>VICAP DVP FRM0 ADDR Y ID3</u>	0x0060	W	0x00000000	DVP path frame0 y address for id3
<u>VICAP DVP FRM0 ADDR UV ID3</u>	0x0064	W	0x00000000	DVP path frame0 uv address for id3
<u>VICAP DVP FRM1 ADDR Y ID3</u>	0x0068	W	0x00000000	DVP path frame1 y address for id3
<u>VICAP DVP FRM1 ADDR UV ID3</u>	0x006C	W	0x00000000	DVP path frame1 uv address for id3
<u>VICAP DVP VIR LINE WIDTH</u>	0x0070	W	0x00000000	DVP path virtual line width
<u>VICAP DVP LINE INT NUM ID0 1</u>	0x0074	W	0x00400040	DVP path id0/id1 line interrupt number
<u>VICAP DVP LINE INT NUM ID2 3</u>	0x0078	W	0x00400040	DVP path id2/id3 line interrupt number
<u>VICAP DVP LINE CNT ID0 1</u>	0x007C	W	0x00000000	DVP path id0/id1 line count
<u>VICAP DVP LINE CNT ID2 3</u>	0x0080	W	0x00000000	DVP path id2/id3 line count
<u>VICAP DVP PIX NUM ID0</u>	0x0084	W	0x00000000	DVP path ID0 PIXEL NUMBER
<u>VICAP DVP LINE NUM ID0</u>	0x0088	W	0x00000000	DVP path ID0 LINE NUMBER
<u>VICAP DVP PIX NUM ID1</u>	0x008C	W	0x00000000	DVP path ID1 PIXEL NUMBER
<u>VICAP DVP LINE NUM ID1</u>	0x0090	W	0x00000000	DVP path ID1 LINE NUMBER
<u>VICAP DVP PIX NUM ID2</u>	0x0094	W	0x00000000	DVP path ID2 PIXEL NUMBER
<u>VICAP DVP LINE NUM ID2</u>	0x0098	W	0x00000000	DVP path ID2 LINE NUMBER

Name	Offset	Size	Reset Value	Description
<u>VICAP DVP PIX NUM ID3</u>	0x009C	W	0x00000000	DVP path ID3 PIXEL NUMBER
<u>VICAP DVP LINE NUM ID3</u>	0x00A0	W	0x00000000	DVP path ID3 LINE NUMBER
<u>VICAP DVP SYNC HEADER</u>	0x00A4	W	0x00000000	DVP SYNC HEADER FOR BT656/BT1120
<u>VICAP MIPI0 ID0 CTRL0</u>	0x0100	W	0x00000000	MIPI0 path id0 control0
<u>VICAP MIPI0 ID0 CTRL1</u>	0x0104	W	0x00000000	MIPI0 path id0 control1
<u>VICAP MIPI0 ID1 CTRL0</u>	0x0108	W	0x00000000	MIPI0 path id1 control0
<u>VICAP MIPI0 ID1 CTRL1</u>	0x010C	W	0x00000000	MIPI0 path id1 control1
<u>VICAP MIPI0 ID2 CTRL0</u>	0x0110	W	0x00000000	MIPI0 path id2 control0
<u>VICAP MIPI0 ID2 CTRL1</u>	0x0114	W	0x00000000	MIPI0 path id2 control1
<u>VICAP MIPI0 ID3 CTRL0</u>	0x0118	W	0x00000000	MIPI0 path id3 control0
<u>VICAP MIPI0 ID3 CTRL1</u>	0x011C	W	0x00000000	MIPI0 path id3 control1
<u>VICAP MIPI0 CTRL</u>	0x0120	W	0x00000000	MIPI0 path control
<u>VICAP MIPI0 FRAME0 A DDR Y ID0</u>	0x0124	W	0x00000000	First address of even frame for ID0 Y/RAW/RGB path
<u>VICAP MIPI0 FRAME1 A DDR Y ID0</u>	0x0128	W	0x00000000	First address of odd frame for ID0 Y path
<u>VICAP MIPI0 FRAME0 A DDR UV ID0</u>	0x012C	W	0x00000000	First address of even frame for ID0 UV path
<u>VICAP MIPI0 FRAME1 A DDR UV ID0</u>	0x0130	W	0x00000000	First address of odd frame for ID0 UV path
<u>VICAP MIPI0 VLW ID0</u>	0x0134	W	0x00000000	Virtual line width for ID0
<u>VICAP MIPI0 FRAME0 A DDR Y ID1</u>	0x0138	W	0x00000000	First address of even frame for ID1 Y/RAW/RGB path
<u>VICAP MIPI0 FRAME1 A DDR Y ID1</u>	0x013C	W	0x00000000	First address of odd frame for ID1 Y/RAW/RGB path
<u>VICAP MIPI0 FRAME0 A DDR UV ID1</u>	0x0140	W	0x00000000	First address of even frame for ID1 UV path
<u>VICAP MIPI0 FRAME1 A DDR UV ID1</u>	0x0144	W	0x00000000	First address of odd frame for ID1 UV path
<u>VICAP MIPI0 VLW ID1</u>	0x0148	W	0x00000000	Virtual line width of even frame for ID1 path
<u>VICAP MIPI0 FRAME0 A DDR Y ID2</u>	0x014C	W	0x00000000	First address of even frame for ID2 Y/RAW/RGB path
<u>VICAP MIPI0 FRAME1 A DDR Y ID2</u>	0x0150	W	0x00000000	First address of odd frame for ID2 Y/RAW/RGB path
<u>VICAP MIPI0 FRAME0 A DDR UV ID2</u>	0x0154	W	0x00000000	First address of even frame for ID2 UV path
<u>VICAP MIPI0 FRAME1 A DDR UV ID2</u>	0x0158	W	0x00000000	First address of odd frame for ID2 UV path

Name	Offset	Size	Reset Value	Description
<u>VICAP_MIPI0_VLW_ID2</u>	0x015C	W	0x00000000	Virtual line width of even frame for ID2 path
<u>VICAP_MIPI0_FRAME0_A_DDR_Y_ID3</u>	0x0160	W	0x00000000	First address of even frame for ID3 Y/RAW/RGB path
<u>VICAP_MIPI0_FRAME1_A_DDR_Y_ID3</u>	0x0164	W	0x00000000	First address of odd frame for ID3 Y/RAW/RGB path
<u>VICAP_MIPI0_FRAME0_A_DDR_UV_ID3</u>	0x0168	W	0x00000000	First address of even frame for ID3 UV path
<u>VICAP_MIPI0_FRAME1_A_DDR_UV_ID3</u>	0x016C	W	0x00000000	First address of odd frame for ID3 UV path
<u>VICAP_MIPI0_VLW_ID3</u>	0x0170	W	0x00000000	Virtual line width of even frame for ID3 path
<u>VICAP_MIPI0_INTEN</u>	0x0174	W	0x00000000	MIPI0 path interrupt enable
<u>VICAP_MIPI0_INTSTAT</u>	0x0178	W	0x00000000	MIPI0 path interrupt status
<u>VICAP_MIPI0_LINE_INT_NUM_ID0_1</u>	0x017C	W	0x00400040	Line number of the MIPI0 path ID0/1 line interrupt
<u>VICAP_MIPI0_LINE_INT_NUM_ID2_3</u>	0x0180	W	0x00400040	Line number of the MIPI0 path ID2/3 line interrupt
<u>VICAP_MIPI0_LINE_CNT_ID0_1</u>	0x0184	W	0x00000000	Line count of the MIPI0 path ID0/1
<u>VICAP_MIPI0_LINE_CNT_ID2_3</u>	0x0188	W	0x00000000	Line count of the MIPI0 path ID2/3
<u>VICAP_MIPI0_ID0_CROP_START</u>	0x018C	W	0x00000000	The start point of MIPI0 ID0 cropping
<u>VICAP_MIPI0_ID1_CROP_START</u>	0x0190	W	0x00000000	The start point of MIPI0 ID1 cropping
<u>VICAP_MIPI0_ID2_CROP_START</u>	0x0194	W	0x00000000	The start point of MIPI0 ID2 cropping
<u>VICAP_MIPI0_ID3_CROP_START</u>	0x0198	W	0x00000000	The start point of MIPI0 ID3 cropping
<u>VICAP_MIPI0_FRAME_NUM_VC0</u>	0x019C	W	0x00000000	The frame number of virtual channel 0
<u>VICAP_MIPI0_FRAME_NUM_VC1</u>	0x01A0	W	0x00000000	The frame number of virtual channel 1
<u>VICAP_MIPI0_FRAME_NUM_VC2</u>	0x01A4	W	0x00000000	The frame number of virtual channel 2
<u>VICAP_MIPI0_FRAME_NUM_VC3</u>	0x01A8	W	0x00000000	The frame number of virtual channel 3
<u>VICAP_MIPI0_ID0_EFFECT_CODE</u>	0x01AC	W	0x00000000	The effect code of MIPI0 ID0
<u>VICAP_MIPI0_ID1_EFFECT_CODE</u>	0x01B0	W	0x00000000	The effect code of MIPI0 ID1

Name	Offset	Size	Reset Value	Description
<u>VICAP MIPI0 ID2 EFFEC T_CODE</u>	0x01B4	W	0x00000000	The effect code of MIPI0 ID2
<u>VICAP MIPI0 ID3 EFFEC T_CODE</u>	0x01B8	W	0x00000000	The effect code of MIPI0 ID3
<u>VICAP MIPI0 ON PAD V ALUE</u>	0x01BC	W	0x00000000	The ON padding value of MIPI0
<u>VICAP MIPI0 SIZE NUM ID0</u>	0x01C0	W	0x00000000	MIPI0 path ID0 SIZE NUMBER
<u>VICAP MIPI0 SIZE NUM ID1</u>	0x01C4	W	0x00000000	MIPI0 path ID1 SIZE NUMBER
<u>VICAP MIPI0 SIZE NUM ID2</u>	0x01C8	W	0x00000000	MIPI0 path ID2 SIZE NUMBER
<u>VICAP MIPI0 SIZE NUM ID3</u>	0x01CC	W	0x00000000	MIPI0 path ID3 SIZE NUMBER
<u>VICAP MIPI1 ID0 CTRL0</u>	0x0200	W	0x00000000	MIPI1 path id0 control0
<u>VICAP MIPI1 ID0 CTRL1</u>	0x0204	W	0x00000000	MIPI1 path id0 control1
<u>VICAP MIPI1 ID1 CTRL0</u>	0x0208	W	0x00000000	MIPI1 path id1 control0
<u>VICAP MIPI1 ID1 CTRL1</u>	0x020C	W	0x00000000	MIPI1 path id1 control1
<u>VICAP MIPI1 ID2 CTRL0</u>	0x0210	W	0x00000000	MIPI1 path id2 control0
<u>VICAP MIPI1 ID2 CTRL1</u>	0x0214	W	0x00000000	MIPI1 path id2 control1
<u>VICAP MIPI1 ID3 CTRL0</u>	0x0218	W	0x00000000	MIPI1 path id3 control0
<u>VICAP MIPI1 ID3 CTRL1</u>	0x021C	W	0x00000000	MIPI1 path id3 control1
<u>VICAP MIPI1 CTRL</u>	0x0220	W	0x00000000	MIPI1 path control
<u>VICAP MIPI1 FRAME0 A DDR Y ID0</u>	0x0224	W	0x00000000	First address of even frame for ID0 Y/RAW/RGB path
<u>VICAP MIPI1 FRAME1 A DDR Y ID0</u>	0x0228	W	0x00000000	First address of odd frame for ID0 Y path
<u>VICAP MIPI1 FRAME0 A DDR UV ID0</u>	0x022C	W	0x00000000	First address of even frame for ID0 UV path
<u>VICAP MIPI1 FRAME1 A DDR UV ID0</u>	0x0230	W	0x00000000	First address of odd frame for ID0 UV path
<u>VICAP MIPI1 VLW ID0</u>	0x0234	W	0x00000000	Virtual line width for ID0
<u>VICAP MIPI1 FRAME0 A DDR Y ID1</u>	0x0238	W	0x00000000	First address of even frame for ID1 Y/RAW/RGB path
<u>VICAP MIPI1 FRAME1 A DDR Y ID1</u>	0x023C	W	0x00000000	First address of odd frame for ID1 Y/RAW/RGB path
<u>VICAP MIPI1 FRAME0 A DDR UV ID1</u>	0x0240	W	0x00000000	First address of even frame for ID1 UV path
<u>VICAP MIPI1 FRAME1 A DDR UV ID1</u>	0x0244	W	0x00000000	First address of odd frame for ID1 UV path
<u>VICAP MIPI1 VLW ID1</u>	0x0248	W	0x00000000	Virtual line width of even frame for ID1 path

Name	Offset	Size	Reset Value	Description
<u>VICAP MIPI1 FRAME0 A DDR Y ID2</u>	0x024C	W	0x00000000	First address of even frame for ID2 Y/RAW/RGB path
<u>VICAP MIPI1 FRAME1 A DDR Y ID2</u>	0x0250	W	0x00000000	First address of odd frame for ID2 Y/RAW/RGB path
<u>VICAP MIPI1 FRAME0 A DDR UV ID2</u>	0x0254	W	0x00000000	First address of even frame for ID2 UV path
<u>VICAP MIPI1 FRAME1 A DDR UV ID2</u>	0x0258	W	0x00000000	First address of odd frame for ID2 UV path
<u>VICAP MIPI1 VLW ID2</u>	0x025C	W	0x00000000	Virtual line width of even frame for ID2 path
<u>VICAP MIPI1 FRAME0 A DDR Y ID3</u>	0x0260	W	0x00000000	First address of even frame for ID3 Y/RAW/RGB path
<u>VICAP MIPI1 FRAME1 A DDR Y ID3</u>	0x0264	W	0x00000000	First address of odd frame for ID3 Y/RAW/RGB path
<u>VICAP MIPI1 FRAME0 A DDR UV ID3</u>	0x0268	W	0x00000000	First address of even frame for ID3 UV path
<u>VICAP MIPI1 FRAME1 A DDR UV ID3</u>	0x026C	W	0x00000000	First address of odd frame for ID3 UV path
<u>VICAP MIPI1 VLW ID3</u>	0x0270	W	0x00000000	Virtual line width of even frame for ID3 path
<u>VICAP MIPI1 INTEN</u>	0x0274	W	0x00000000	MIPI1 path interrupt enable
<u>VICAP MIPI1 INTSTAT</u>	0x0278	W	0x00000000	MIPI1 path interrupt status
<u>VICAP MIPI1 LINE INT NUM ID0 1</u>	0x027C	W	0x00400040	Line number of the MIPI1 path ID0/1 line interrupt
<u>VICAP MIPI1 LINE INT NUM ID2 3</u>	0x0280	W	0x00400040	Line number of the MIPI1 path ID2/3 line interrupt
<u>VICAP MIPI1 LINE CNT ID0 1</u>	0x0284	W	0x00000000	Line count of the MIPI1 path ID0/1
<u>VICAP MIPI1 LINE CNT ID2 3</u>	0x0288	W	0x00000000	Line count of the MIPI1 path ID2/3
<u>VICAP MIPI1 ID0 CROP START</u>	0x028C	W	0x00000000	The start point of MIPI1 ID0 cropping
<u>VICAP MIPI1 ID1 CROP START</u>	0x0290	W	0x00000000	The start point of MIPI1 ID1 cropping
<u>VICAP MIPI1 ID2 CROP START</u>	0x0294	W	0x00000000	The start point of MIPI1 ID2 cropping
<u>VICAP MIPI1 ID3 CROP START</u>	0x0298	W	0x00000000	The start point of MIPI1 ID3 cropping
<u>VICAP MIPI1 FRAME NUM VC0</u>	0x029C	W	0x00000000	The frame number of virtual channel 0
<u>VICAP MIPI1 FRAME NUM VC1</u>	0x02A0	W	0x00000000	The frame number of virtual channel 1

Name	Offset	Size	Reset Value	Description
<u>VICAP MIPI1 FRAME NUM_VC2</u>	0x02A4	W	0x00000000	The frame number of virtual channel 2
<u>VICAP MIPI1 FRAME NUM_VC3</u>	0x02A8	W	0x00000000	The frame number of virtual channel 3
<u>VICAP MIPI1 ID0 EFFECT CODE</u>	0x02AC	W	0x00000000	The effect code of MIPI1 ID0
<u>VICAP MIPI1 ID1 EFFECT CODE</u>	0x02B0	W	0x00000000	The effect code of MIPI1 ID1
<u>VICAP MIPI1 ID2 EFFECT CODE</u>	0x02B4	W	0x00000000	The effect code of MIPI2 ID2
<u>VICAP MIPI1 ID3 EFFECT CODE</u>	0x02B8	W	0x00000000	The effect code of MIPI1 ID3
<u>VICAP MIPI1 ON PAD VALUE</u>	0x02BC	W	0x00000000	The ON padding value of MIPI1
<u>VICAP MIPI1 SIZE NUMBER ID0</u>	0x02C0	W	0x00000000	MIPI1 path ID0 SIZE NUMBER
<u>VICAP MIPI1 SIZE NUMBER ID1</u>	0x02C4	W	0x00000000	MIPI1 path ID1 SIZE NUMBER
<u>VICAP MIPI1 SIZE NUMBER ID2</u>	0x02C8	W	0x00000000	MIPI1 path ID2 SIZE NUMBER
<u>VICAP MIPI1 SIZE NUMBER ID3</u>	0x02CC	W	0x00000000	MIPI1 path ID3 SIZE NUMBER
<u>VICAP MIPI2 ID0 CTRL0</u>	0x0300	W	0x00000000	MIPI2 path id0 control0
<u>VICAP MIPI2 ID0 CTRL1</u>	0x0304	W	0x00000000	MIPI2 path id0 control1
<u>VICAP MIPI2 ID1 CTRL0</u>	0x0308	W	0x00000000	MIPI2 path id1 control0
<u>VICAP MIPI2 ID1 CTRL1</u>	0x030C	W	0x00000000	MIPI2 path id1 control1
<u>VICAP MIPI2 ID2 CTRL0</u>	0x0310	W	0x00000000	MIPI2 path id2 control0
<u>VICAP MIPI2 ID2 CTRL1</u>	0x0314	W	0x00000000	MIPI2 path id2 control1
<u>VICAP MIPI2 ID3 CTRL0</u>	0x0318	W	0x00000000	MIPI2 path id3 control0
<u>VICAP MIPI2 ID3 CTRL1</u>	0x031C	W	0x00000000	MIPI2 path id3 control1
<u>VICAP MIPI2 CTRL</u>	0x0320	W	0x00000000	MIPI2 path control
<u>VICAP MIPI2 FRAME0 ADDRESS DDR Y ID0</u>	0x0324	W	0x00000000	First address of even frame for ID0 Y/RAW/RGB path
<u>VICAP MIPI2 FRAME1 ADDRESS DDR Y ID0</u>	0x0328	W	0x00000000	First address of odd frame for ID0 Y path
<u>VICAP MIPI2 FRAME0 ADDRESS DDR UV ID0</u>	0x032C	W	0x00000000	First address of even frame for ID0 UV path
<u>VICAP MIPI2 FRAME1 ADDRESS DDR UV ID0</u>	0x0330	W	0x00000000	First address of odd frame for ID0 UV path
<u>VICAP MIPI2 VLW ID0</u>	0x0334	W	0x00000000	Virtual line width for ID0
<u>VICAP MIPI2 FRAME0 ADDRESS DDR Y ID1</u>	0x0338	W	0x00000000	First address of even frame for ID1 Y/RAW/RGB path

Name	Offset	Size	Reset Value	Description
<u>VICAP MIPI2 FRAME1 A DDR Y ID1</u>	0x033C	W	0x00000000	First address of odd frame for ID1 Y/RAW/RGB path
<u>VICAP MIPI2 FRAME0 A DDR UV ID1</u>	0x0340	W	0x00000000	First address of even frame for ID1 UV path
<u>VICAP MIPI2 FRAME1 A DDR UV ID1</u>	0x0344	W	0x00000000	First address of odd frame for ID1 UV path
<u>VICAP MIPI2 VLW ID1</u>	0x0348	W	0x00000000	Virtual line width of even frame for ID1 path
<u>VICAP MIPI2 FRAME0 A DDR Y ID2</u>	0x034C	W	0x00000000	First address of even frame for ID2 Y/RAW/RGB path
<u>VICAP MIPI2 FRAME1 A DDR Y ID2</u>	0x0350	W	0x00000000	First address of odd frame for ID2 Y/RAW/RGB path
<u>VICAP MIPI2 FRAME0 A DDR UV ID2</u>	0x0354	W	0x00000000	First address of even frame for ID2 UV path
<u>VICAP MIPI2 FRAME1 A DDR UV ID2</u>	0x0358	W	0x00000000	First address of odd frame for ID2 UV path
<u>VICAP MIPI2 VLW ID2</u>	0x035C	W	0x00000000	Virtual line width of even frame for ID2 path
<u>VICAP MIPI2 FRAME0 A DDR Y ID3</u>	0x0360	W	0x00000000	First address of even frame for ID3 Y/RAW/RGB path
<u>VICAP MIPI2 FRAME1 A DDR Y ID3</u>	0x0364	W	0x00000000	First address of odd frame for ID3 Y/RAW/RGB path
<u>VICAP MIPI2 FRAME0 A DDR UV ID3</u>	0x0368	W	0x00000000	First address of even frame for ID3 UV path
<u>VICAP MIPI2 FRAME1 A DDR UV ID3</u>	0x036C	W	0x00000000	First address of odd frame for ID3 UV path
<u>VICAP MIPI2 VLW ID3</u>	0x0370	W	0x00000000	Virtual line width of even frame for ID3 path
<u>VICAP MIPI2 INTEN</u>	0x0374	W	0x00000000	MIPI2 path interrupt enable
<u>VICAP MIPI2 INTSTAT</u>	0x0378	W	0x00000000	MIPI2 path interrupt status
<u>VICAP MIPI2 LINE INT NUM ID0 1</u>	0x037C	W	0x00400040	Line number of the MIPI2 path ID0/1 line interrupt
<u>VICAP MIPI2 LINE INT NUM ID2 3</u>	0x0380	W	0x00400040	Line number of the MIPI2 path ID2/3 line interrupt
<u>VICAP MIPI2 LINE CNT ID0 1</u>	0x0384	W	0x00000000	Line count of the MIPI2 path ID0/1
<u>VICAP MIPI2 LINE CNT ID2 3</u>	0x0388	W	0x00000000	Line count of the MIPI2 path ID2/3
<u>VICAP MIPI2 ID0 CROP START</u>	0x038C	W	0x00000000	The start point of MIPI2 ID0 cropping
<u>VICAP MIPI2 ID1 CROP START</u>	0x0390	W	0x00000000	The start point of MIPI2 ID1 cropping

Name	Offset	Size	Reset Value	Description
<u>VICAP MIPI2 ID2 CROP START</u>	0x0394	W	0x00000000	The start point of MIPI2 ID2 cropping
<u>VICAP MIPI2 ID3 CROP START</u>	0x0398	W	0x00000000	The start point of MIPI2 ID3 cropping
<u>VICAP MIPI2 FRAME NUM_VC0</u>	0x039C	W	0x00000000	The frame number of virtual channel 0
<u>VICAP MIPI2 FRAME NUM_VC1</u>	0x03A0	W	0x00000000	The frame number of virtual channel 1
<u>VICAP MIPI2 FRAME NUM_VC2</u>	0x03A4	W	0x00000000	The frame number of virtual channel 2
<u>VICAP MIPI2 FRAME NUM_VC3</u>	0x03A8	W	0x00000000	The frame number of virtual channel 3
<u>VICAP MIPI2 ID0 EFFECT CODE</u>	0x03AC	W	0x00000000	The effect code of MIPI2 ID0
<u>VICAP MIPI2 ID1 EFFECT CODE</u>	0x03B0	W	0x00000000	The effect code of MIPI2 ID1
<u>VICAP MIPI2 ID2 EFFECT CODE</u>	0x03B4	W	0x00000000	The effect code of MIPI2 ID2
<u>VICAP MIPI2 ID3 EFFECT CODE</u>	0x03B8	W	0x00000000	The effect code of MIPI2 ID3
<u>VICAP MIPI2 ON PAD VALUE</u>	0x03BC	W	0x00000000	The ON padding value of MIPI2
<u>VICAP MIPI2 SIZE NUMBER ID0</u>	0x03C0	W	0x00000000	MIPI2 path ID0 SIZE NUMBER
<u>VICAP MIPI2 SIZE NUMBER ID1</u>	0x03C4	W	0x00000000	MIPI2 path ID1 SIZE NUMBER
<u>VICAP MIPI2 SIZE NUMBER ID2</u>	0x03C8	W	0x00000000	MIPI2 path ID2 SIZE NUMBER
<u>VICAP MIPI2 SIZE NUMBER ID3</u>	0x03CC	W	0x00000000	MIPI2 path ID3 SIZE NUMBER
<u>VICAP MIPI3 ID0 CTRL0</u>	0x0400	W	0x00000000	MIPI3 path id0 control0
<u>VICAP MIPI3 ID0 CTRL1</u>	0x0404	W	0x00000000	MIPI3 path id0 control1
<u>VICAP MIPI3 ID1 CTRL0</u>	0x0408	W	0x00000000	MIPI3 path id1 control0
<u>VICAP MIPI3 ID1 CTRL1</u>	0x040C	W	0x00000000	MIPI3 path id1 control1
<u>VICAP MIPI3 ID2 CTRL0</u>	0x0410	W	0x00000000	MIPI3 path id2 control0
<u>VICAP MIPI3 ID2 CTRL1</u>	0x0414	W	0x00000000	MIPI3 path id2 control1
<u>VICAP MIPI3 ID3 CTRL0</u>	0x0418	W	0x00000000	MIPI3 path id3 control0
<u>VICAP MIPI3 ID3 CTRL1</u>	0x041C	W	0x00000000	MIPI3 path id3 control1
<u>VICAP MIPI3 CTRL</u>	0x0420	W	0x00000000	MIPI3 path control
<u>VICAP MIPI3 FRAME0 ADDRESS DDR Y ID0</u>	0x0424	W	0x00000000	First address of even frame for ID0 Y/RAW/RGB path
<u>VICAP MIPI3 FRAME1 ADDRESS DDR Y ID0</u>	0x0428	W	0x00000000	First address of odd frame for ID0 Y path

Name	Offset	Size	Reset Value	Description
<u>VICAP MIPI3 FRAME0 A DDR UV ID0</u>	0x042C	W	0x00000000	First address of even frame for ID0 UV path
<u>VICAP MIPI3 FRAME1 A DDR UV ID0</u>	0x0430	W	0x00000000	First address of odd frame for ID0 UV path
<u>VICAP MIPI3 VLW ID0</u>	0x0434	W	0x00000000	Virtual line width for ID0
<u>VICAP MIPI3 FRAME0 A DDR Y ID1</u>	0x0438	W	0x00000000	First address of even frame for ID1 Y/RAW/RGB path
<u>VICAP MIPI3 FRAME1 A DDR Y ID1</u>	0x043C	W	0x00000000	First address of odd frame for ID1 Y/RAW/RGB path
<u>VICAP MIPI3 FRAME0 A DDR UV ID1</u>	0x0440	W	0x00000000	First address of even frame for ID1 UV path
<u>VICAP MIPI3 FRAME1 A DDR UV ID1</u>	0x0444	W	0x00000000	First address of odd frame for ID1 UV path
<u>VICAP MIPI3 VLW ID1</u>	0x0448	W	0x00000000	Virtual line width of even frame for ID1 path
<u>VICAP MIPI3 FRAME0 A DDR Y ID2</u>	0x044C	W	0x00000000	First address of even frame for ID2 Y/RAW/RGB path
<u>VICAP MIPI3 FRAME1 A DDR Y ID2</u>	0x0450	W	0x00000000	First address of odd frame for ID2 Y/RAW/RGB path
<u>VICAP MIPI3 FRAME0 A DDR UV ID2</u>	0x0454	W	0x00000000	First address of even frame for ID2 UV path
<u>VICAP MIPI3 FRAME1 A DDR UV ID2</u>	0x0458	W	0x00000000	First address of odd frame for ID2 UV path
<u>VICAP MIPI3 VLW ID2</u>	0x045C	W	0x00000000	Virtual line width of even frame for ID2 path
<u>VICAP MIPI3 FRAME0 A DDR Y ID3</u>	0x0460	W	0x00000000	First address of even frame for ID3 Y/RAW/RGB path
<u>VICAP MIPI3 FRAME1 A DDR Y ID3</u>	0x0464	W	0x00000000	First address of odd frame for ID3 Y/RAW/RGB path
<u>VICAP MIPI3 FRAME0 A DDR UV ID3</u>	0x0468	W	0x00000000	First address of even frame for ID3 UV path
<u>VICAP MIPI3 FRAME1 A DDR UV ID3</u>	0x046C	W	0x00000000	First address of odd frame for ID3 UV path
<u>VICAP MIPI3 VLW ID3</u>	0x0470	W	0x00000000	Virtual line width of even frame for ID3 path
<u>VICAP MIPI3 INTEN</u>	0x0474	W	0x00000000	MIPI3 path interrupt enable
<u>VICAP MIPI3 INTSTAT</u>	0x0478	W	0x00000000	MIPI3 path interrupt status
<u>VICAP MIPI3 LINE INT NUM ID0 1</u>	0x047C	W	0x00400040	Line number of the MIPI3 path ID0/1 line interrupt
<u>VICAP MIPI3 LINE INT NUM ID2 3</u>	0x0480	W	0x00400040	Line number of the MIPI3 path ID2/3 line interrupt
<u>VICAP MIPI3 LINE CNT ID0 1</u>	0x0484	W	0x00000000	Line count of the MIPI3 path ID0/1

Name	Offset	Size	Reset Value	Description
<u>VICAP MIPI3 LINE CNT ID2_3</u>	0x0488	W	0x00000000	Line count of the MIPI3 path ID2/3
<u>VICAP MIPI3 ID0 CROP START</u>	0x048C	W	0x00000000	The start point of MIPI3 ID0 cropping
<u>VICAP MIPI3 ID1 CROP START</u>	0x0490	W	0x00000000	The start point of MIPI3 ID1 cropping
<u>VICAP MIPI3 ID2 CROP START</u>	0x0494	W	0x00000000	The start point of MIPI3 ID2 cropping
<u>VICAP MIPI3 ID3 CROP START</u>	0x0498	W	0x00000000	The start point of MIPI3 ID3 cropping
<u>VICAP MIPI3 FRAME NUM_VC0</u>	0x049C	W	0x00000000	The frame number of virtual channel 0
<u>VICAP MIPI3 FRAME NUM_VC1</u>	0x04A0	W	0x00000000	The frame number of virtual channel 1
<u>VICAP MIPI3 FRAME NUM_VC2</u>	0x04A4	W	0x00000000	The frame number of virtual channel 2
<u>VICAP MIPI3 FRAME NUM_VC3</u>	0x04A8	W	0x00000000	The frame number of virtual channel 3
<u>VICAP MIPI3 ID0 EFFECT CODE</u>	0x04AC	W	0x00000000	The effect code of MIPI3 ID0
<u>VICAP MIPI3 ID1 EFFECT CODE</u>	0x04B0	W	0x00000000	The effect code of MIPI3 ID1
<u>VICAP MIPI3 ID2 EFFECT CODE</u>	0x04B4	W	0x00000000	The effect code of MIPI3 ID2
<u>VICAP MIPI3 ID3 EFFECT CODE</u>	0x04B8	W	0x00000000	The effect code of MIPI3 ID3
<u>VICAP MIPI3 ON PAD VALUE</u>	0x04BC	W	0x00000000	The ON padding value of MIPI3
<u>VICAP MIPI3 SIZE NUM ID0</u>	0x04C0	W	0x00000000	MIPI3 path ID0 SIZE NUMBER
<u>VICAP MIPI3 SIZE NUM ID1</u>	0x04C4	W	0x00000000	MIPI3 path ID1 SIZE NUMBER
<u>VICAP MIPI3 SIZE NUM ID2</u>	0x04C8	W	0x00000000	MIPI3 path ID2 SIZE NUMBER
<u>VICAP MIPI3 SIZE NUM ID3</u>	0x04CC	W	0x00000000	MIPI3 path ID3 SIZE NUMBER
<u>VICAP MIPI4 ID0 CTRL0</u>	0x0500	W	0x00000000	MIPI4 path id0 control0
<u>VICAP MIPI4 ID0 CTRL1</u>	0x0504	W	0x00000000	MIPI4 path id0 control1
<u>VICAP MIPI4 ID1 CTRL0</u>	0x0508	W	0x00000000	MIPI4 path id1 control0
<u>VICAP MIPI4 ID1 CTRL1</u>	0x050C	W	0x00000000	MIPI4 path id1 control1
<u>VICAP MIPI4 ID2 CTRL0</u>	0x0510	W	0x00000000	MIPI4 path id2 control0
<u>VICAP MIPI4 ID2 CTRL1</u>	0x0514	W	0x00000000	MIPI4 path id2 control1
<u>VICAP MIPI4 ID3 CTRL0</u>	0x0518	W	0x00000000	MIPI4 path id3 control0

Name	Offset	Size	Reset Value	Description
<u>VICAP MIPI4 ID3 CTRL1</u>	0x051C	W	0x00000000	MIPI4 path id3 control1
<u>VICAP MIPI4 CTRL</u>	0x0520	W	0x00000000	MIPI4 path control
<u>VICAP MIPI4 FRAME0 A DDR Y ID0</u>	0x0524	W	0x00000000	First address of even frame for ID0 Y/RAW/RGB path
<u>VICAP MIPI4 FRAME1 A DDR Y ID0</u>	0x0528	W	0x00000000	First address of odd frame for ID0 Y path
<u>VICAP MIPI4 FRAME0 A DDR UV ID0</u>	0x052C	W	0x00000000	First address of even frame for ID0 UV path
<u>VICAP MIPI4 FRAME1 A DDR UV ID0</u>	0x0530	W	0x00000000	First address of odd frame for ID0 UV path
<u>VICAP MIPI4 VLW ID0</u>	0x0534	W	0x00000000	Virtual line width for ID0
<u>VICAP MIPI4 FRAME0 A DDR Y ID1</u>	0x0538	W	0x00000000	First address of even frame for ID1 Y/RAW/RGB path
<u>VICAP MIPI4 FRAME1 A DDR Y ID1</u>	0x053C	W	0x00000000	First address of odd frame for ID1 Y/RAW/RGB path
<u>VICAP MIPI4 FRAME0 A DDR UV ID1</u>	0x0540	W	0x00000000	First address of even frame for ID1 UV path
<u>VICAP MIPI4 FRAME1 A DDR UV ID1</u>	0x0544	W	0x00000000	First address of odd frame for ID1 UV path
<u>VICAP MIPI4 VLW ID1</u>	0x0548	W	0x00000000	Virtual line width of even frame for ID1 path
<u>VICAP MIPI4 FRAME0 A DDR Y ID2</u>	0x054C	W	0x00000000	First address of even frame for ID2 Y/RAW/RGB path
<u>VICAP MIPI4 FRAME1 A DDR Y ID2</u>	0x0550	W	0x00000000	First address of odd frame for ID2 Y/RAW/RGB path
<u>VICAP MIPI4 FRAME0 A DDR UV ID2</u>	0x0554	W	0x00000000	First address of even frame for ID2 UV path
<u>VICAP MIPI4 FRAME1 A DDR UV ID2</u>	0x0558	W	0x00000000	First address of odd frame for ID2 UV path
<u>VICAP MIPI4 VLW ID2</u>	0x055C	W	0x00000000	Virtual line width of even frame for ID2 path
<u>VICAP MIPI4 FRAME0 A DDR Y ID3</u>	0x0560	W	0x00000000	First address of even frame for ID3 Y/RAW/RGB path
<u>VICAP MIPI4 FRAME1 A DDR Y ID3</u>	0x0564	W	0x00000000	First address of odd frame for ID3 Y/RAW/RGB path
<u>VICAP MIPI4 FRAME0 A DDR UV ID3</u>	0x0568	W	0x00000000	First address of even frame for ID3 UV path
<u>VICAP MIPI4 FRAME1 A DDR UV ID3</u>	0x056C	W	0x00000000	First address of odd frame for ID3 UV path
<u>VICAP MIPI4 VLW ID3</u>	0x0570	W	0x00000000	Virtual line width of even frame for ID3 path
<u>VICAP MIPI4 INTEN</u>	0x0574	W	0x00000000	MIPI4 path interrupt enable
<u>VICAP MIPI4 INTSTAT</u>	0x0578	W	0x00000000	MIPI4 path interrupt status

Name	Offset	Size	Reset Value	Description
<u>VICAP MIPI4 LINE INT NUM ID0 1</u>	0x057C	W	0x00400040	Line number of the MIPI4 path ID0/1 line interrupt
<u>VICAP MIPI4 LINE INT NUM ID2 3</u>	0x0580	W	0x00400040	Line number of the MIPI4 path ID2/3 line interrupt
<u>VICAP MIPI4 LINE CNT ID0 1</u>	0x0584	W	0x00000000	Line count of the MIPI4 path ID0/1
<u>VICAP MIPI4 LINE CNT ID2 3</u>	0x0588	W	0x00000000	Line count of the MIPI4 path ID2/3
<u>VICAP MIPI4 ID0 CROP START</u>	0x058C	W	0x00000000	The start point of MIPI4 ID0 cropping
<u>VICAP MIPI4 ID1 CROP START</u>	0x0590	W	0x00000000	The start point of MIPI4 ID1 cropping
<u>VICAP MIPI4 ID2 CROP START</u>	0x0594	W	0x00000000	The start point of MIPI4 ID2 cropping
<u>VICAP MIPI4 ID3 CROP START</u>	0x0598	W	0x00000000	The start point of MIPI4 ID3 cropping
<u>VICAP MIPI4 FRAME NUM VC0</u>	0x059C	W	0x00000000	The frame number of virtual channel 0
<u>VICAP MIPI4 FRAME NUM VC1</u>	0x05A0	W	0x00000000	The frame number of virtual channel 1
<u>VICAP MIPI4 FRAME NUM VC2</u>	0x05A4	W	0x00000000	The frame number of virtual channel 2
<u>VICAP MIPI4 FRAME NUM VC3</u>	0x05A8	W	0x00000000	The frame number of virtual channel 3
<u>VICAP MIPI4 ID0 EFFECT CODE</u>	0x05AC	W	0x00000000	The effect code of MIPI4 ID0
<u>VICAP MIPI4 ID1 EFFECT CODE</u>	0x05B0	W	0x00000000	The effect code of MIPI4 ID1
<u>VICAP MIPI4 ID2 EFFECT CODE</u>	0x05B4	W	0x00000000	The effect code of MIPI4 ID2
<u>VICAP MIPI4 ID3 EFFECT CODE</u>	0x05B8	W	0x00000000	The effect code of MIPI4 ID3
<u>VICAP MIPI4 ON PAD VALUE</u>	0x05BC	W	0x00000000	The ON padding value of MIPI4
<u>VICAP MIPI4 SIZE NUM ID0</u>	0x05C0	W	0x00000000	MIPI4 path ID0 SIZE NUMBER
<u>VICAP MIPI4 SIZE NUM ID1</u>	0x05C4	W	0x00000000	MIPI4 path ID1 SIZE NUMBER
<u>VICAP MIPI4 SIZE NUM ID2</u>	0x05C8	W	0x00000000	MIPI4 path ID2 SIZE NUMBER
<u>VICAP MIPI4 SIZE NUM ID3</u>	0x05CC	W	0x00000000	MIPI4 path ID3 SIZE NUMBER
<u>VICAP MIPI5 ID0 CTRL0</u>	0x0600	W	0x00000000	MIPI5 path id0 control0

Name	Offset	Size	Reset Value	Description
<u>VICAP MIPI5 ID0 CTRL1</u>	0x0604	W	0x00000000	MIPI5 path id0 control1
<u>VICAP MIPI5 ID1 CTRL0</u>	0x0608	W	0x00000000	MIPI5 path id1 control0
<u>VICAP MIPI5 ID1 CTRL1</u>	0x060C	W	0x00000000	MIPI5 path id1 control1
<u>VICAP MIPI5 ID2 CTRL0</u>	0x0610	W	0x00000000	MIPI5 path id2 control0
<u>VICAP MIPI5 ID2 CTRL1</u>	0x0614	W	0x00000000	MIPI5 path id2 control1
<u>VICAP MIPI5 ID3 CTRL0</u>	0x0618	W	0x00000000	MIPI5 path id3 control0
<u>VICAP MIPI5 ID3 CTRL1</u>	0x061C	W	0x00000000	MIPI5 path id3 control1
<u>VICAP MIPI5 CTRL</u>	0x0620	W	0x00000000	MIPI5 path control
<u>VICAP MIPI5 FRAME0 A DDR Y ID0</u>	0x0624	W	0x00000000	First address of even frame for ID0 Y/RAW/RGB path
<u>VICAP MIPI5 FRAME1 A DDR Y ID0</u>	0x0628	W	0x00000000	First address of odd frame for ID0 Y path
<u>VICAP MIPI5 FRAME0 A DDR UV ID0</u>	0x062C	W	0x00000000	First address of even frame for ID0 UV path
<u>VICAP MIPI5 FRAME1 A DDR UV ID0</u>	0x0630	W	0x00000000	First address of odd frame for ID0 UV path
<u>VICAP MIPI5 VLW ID0</u>	0x0634	W	0x00000000	Virtual line width for ID0
<u>VICAP MIPI5 FRAME0 A DDR Y ID1</u>	0x0638	W	0x00000000	First address of even frame for ID1 Y/RAW/RGB path
<u>VICAP MIPI5 FRAME1 A DDR Y ID1</u>	0x063C	W	0x00000000	First address of odd frame for ID1 Y/RAW/RGB path
<u>VICAP MIPI5 FRAME0 A DDR UV ID1</u>	0x0640	W	0x00000000	First address of even frame for ID1 UV path
<u>VICAP MIPI5 FRAME1 A DDR UV ID1</u>	0x0644	W	0x00000000	First address of odd frame for ID1 UV path
<u>VICAP MIPI5 VLW ID1</u>	0x0648	W	0x00000000	Virtual line width of even frame for ID1 path
<u>VICAP MIPI5 FRAME0 A DDR Y ID2</u>	0x064C	W	0x00000000	First address of even frame for ID2 Y/RAW/RGB path
<u>VICAP MIPI5 FRAME1 A DDR Y ID2</u>	0x0650	W	0x00000000	First address of odd frame for ID2 Y/RAW/RGB path
<u>VICAP MIPI5 FRAME0 A DDR UV ID2</u>	0x0654	W	0x00000000	First address of even frame for ID2 UV path
<u>VICAP MIPI5 FRAME1 A DDR UV ID2</u>	0x0658	W	0x00000000	First address of odd frame for ID2 UV path
<u>VICAP MIPI5 VLW ID2</u>	0x065C	W	0x00000000	Virtual line width of even frame for ID2 path
<u>VICAP MIPI5 FRAME0 A DDR Y ID3</u>	0x0660	W	0x00000000	First address of even frame for ID3 Y/RAW/RGB path
<u>VICAP MIPI5 FRAME1 A DDR Y ID3</u>	0x0664	W	0x00000000	First address of odd frame for ID3 Y/RAW/RGB path
<u>VICAP MIPI5 FRAME0 A DDR UV ID3</u>	0x0668	W	0x00000000	First address of even frame for ID3 UV path

Name	Offset	Size	Reset Value	Description
<u>VICAP MIPI5 FRAME1 A DDR UV ID3</u>	0x066C	W	0x00000000	First address of odd frame for ID3 UV path
<u>VICAP MIPI5 VLW ID3</u>	0x0670	W	0x00000000	Virtual line width of even frame for ID3 path
<u>VICAP MIPI5 INTEN</u>	0x0674	W	0x00000000	MIPI5 path interrupt enable
<u>VICAP MIPI5 INTSTAT</u>	0x0678	W	0x00000000	MIPI5 path interrupt status
<u>VICAP MIPI5 LINE INT NUM ID0 1</u>	0x067C	W	0x00400040	Line number of the MIPI5 path ID0/1 line interrupt
<u>VICAP MIPI5 LINE INT NUM ID2 3</u>	0x0680	W	0x00400040	Line number of the MIPI5 path ID2/3 line interrupt
<u>VICAP MIPI5 LINE CNT ID0 1</u>	0x0684	W	0x00000000	Line count of the MIPI5 path ID0/1
<u>VICAP MIPI5 LINE CNT ID2 3</u>	0x0688	W	0x00000000	Line count of the MIPI5 path ID2/3
<u>VICAP MIPI5 ID0 CROP START</u>	0x068C	W	0x00000000	The start point of MIPI5 ID0 cropping
<u>VICAP MIPI5 ID1 CROP START</u>	0x0690	W	0x00000000	The start point of MIPI5 ID1 cropping
<u>VICAP MIPI5 ID2 CROP START</u>	0x0694	W	0x00000000	The start point of MIPI5 ID2 cropping
<u>VICAP MIPI5 ID3 CROP START</u>	0x0698	W	0x00000000	The start point of MIPI5 ID3 cropping
<u>VICAP MIPI5 FRAME NUM VC0</u>	0x069C	W	0x00000000	The frame number of virtual channel 0
<u>VICAP MIPI5 FRAME NUM VC1</u>	0x06A0	W	0x00000000	The frame number of virtual channel 1
<u>VICAP MIPI5 FRAME NUM VC2</u>	0x06A4	W	0x00000000	The frame number of virtual channel 2
<u>VICAP MIPI5 FRAME NUM VC3</u>	0x06A8	W	0x00000000	The frame number of virtual channel 3
<u>VICAP MIPI5 ID0 EFFEC T CODE</u>	0x06AC	W	0x00000000	The effect code of MIPI5 ID0
<u>VICAP MIPI5 ID1 EFFEC T CODE</u>	0x06B0	W	0x00000000	The effect code of MIPI5 ID1
<u>VICAP MIPI5 ID2 EFFEC T CODE</u>	0x06B4	W	0x00000000	The effect code of MIPI5 ID2
<u>VICAP MIPI5 ID3 EFFEC T CODE</u>	0x06B8	W	0x00000000	The effect code of MIPI5 ID3
<u>VICAP MIPI5 ON PAD V ALUE</u>	0x06BC	W	0x00000000	The ON padding value of MIPI5
<u>VICAP MIPI5 SIZE NUM ID0</u>	0x06C0	W	0x00000000	MIPI5 path ID0 SIZE NUMBER

Name	Offset	Size	Reset Value	Description
<u>VICAP MIPI5 SIZE NUM ID1</u>	0x06C4	W	0x00000000	MIPI5 path ID1 SIZE NUMBER
<u>VICAP MIPI5 SIZE NUM ID2</u>	0x06C8	W	0x00000000	MIPI5 path ID2 SIZE NUMBER
<u>VICAP MIPI5 SIZE NUM ID3</u>	0x06CC	W	0x00000000	MIPI5 path ID3 SIZE NUMBER
<u>VICAP_SCL_CH_CTRL</u>	0x0700	W	0x00000000	Scale channel0 path control
<u>VICAP_SCL_CTRL</u>	0x0704	W	0x00000000	MIPI3 path control
<u>VICAP_SCL_FRAME0_ADDR_CH0</u>	0x0708	W	0x00000000	First address of even frame for CH0 path
<u>VICAP_SCL_FRAME1_ADDR_CH0</u>	0x070C	W	0x00000000	First address of odd frame for CH0 path
<u>VICAP_SCL_VLW_CH0</u>	0x0710	W	0x00000000	Virtual line width for CH0
<u>VICAP_SCL_FRAME0_ADDR_CH1</u>	0x0714	W	0x00000000	First address of even frame for CH1 path
<u>VICAP_SCL_FRAME1_ADDR_CH1</u>	0x0718	W	0x00000000	First address of odd frame for CH1 path
<u>VICAP_SCL_VLW_CH1</u>	0x071C	W	0x00000000	Virtual line width for CH0
<u>VICAP_SCL_FRAME0_ADDR_CH2</u>	0x0720	W	0x00000000	First address of even frame for CH2 path
<u>VICAP_SCL_FRAME1_ADDR_CH2</u>	0x0724	W	0x00000000	First address of odd frame for CH2 path
<u>VICAP_SCL_VLW_CH2</u>	0x0728	W	0x00000000	Virtual line width for CH2
<u>VICAP_SCL_FRAME0_ADDR_CH3</u>	0x072C	W	0x00000000	First address of even frame for CH3 path
<u>VICAP_SCL_FRAME1_ADDR_CH3</u>	0x0730	W	0x00000000	First address of odd frame for CH1 path
<u>VICAP_SCL_VLW_CH3</u>	0x0734	W	0x00000000	Virtual line width for CH3
<u>VICAP_SCL_CH0_BLACK_LEVEL</u>	0x0738	W	0x00000000	Scale channel0 black level
<u>VICAP_SCL_CH1_BLACK_LEVEL</u>	0x073C	W	0x00000000	Scale channel1 black level
<u>VICAP_SCL_CH2_BLACK_LEVEL</u>	0x0740	W	0x00000000	Scale channel2 black level
<u>VICAP_SCL_CH3_BLACK_LEVEL</u>	0x0744	W	0x00000000	Scale channel3 black level
<u>VICAP_TOISP0_CH_CTRL</u>	0x0780	W	0x00000000	TOISP0 path control
<u>VICAP_TOISP0_CROP_SIZE</u>	0x0784	W	0x01E002D0	The expected width and height of received image
<u>VICAP_TOISP0_CROP_START</u>	0x0788	W	0x00000000	The start point of toisp0 path cropping
<u>VICAP_TOISP1_CH_CTRL</u>	0x078C	W	0x00000000	TOISP1 path control